Probabilistic Data Science for Psychology

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Welcome

Welcome to the working version of *Probabilistic Data Science for Psychology (PDSP)*, a textbook on data-scientific methodology at the Institute of Psychology at Otto von Guericke University Magdeburg.

Figure 1 Probabilistic Data Science for Psychology

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Part I

Mathematical foundations

As the language of natural-scientific modeling, mathematics is also the language of probabilistic data science in psychology. In this role, mathematics mediates between intuitive scientific language and the variety of programming languages used to implement data-scientific analyses. Mathematical concept formation makes it possible to grasp concepts very precisely and in a generally understandable way, and to communicate them independently of the imprecision of everyday scientific language or the idiosyncrasies of different programming languages. Of course, mathematical language cannot assume this role detached from scientific-language intuition or practical application. In probabilistic data analysis, mathematics is therefore never an end in itself, but must always be understood as applied mathematics.

In this part, after some thoughts on the fundamental concepts of mathematics (Chapter 1), we present sets and functions (Chapter 2 and Chapter 4), the two pillars of modern mathematics, in compressed form. We supplement this presentation with some notational aspects in Chapter 3. Closely interwoven with the concept of a function are differential calculus and integral calculus. Here, too, the aim is not an exhaustive presentation, but in particular an explanation of some foundations that are central to the data-scientific concepts of optimization and to working with probability functions. We supplement these sections with some introductory thoughts on the analytical foundations of differential and integral calculus in Chapter 5. In dealing with the large data sets of contemporary science, matrix notation has proved useful. We first consider this subfield of linear algebra for single-column matrices and then for matrices in detail, again with the aim of providing notational and computational foundations for later sections.

Overall, this part therefore primarily serves as a brief introduction to foundations that will be taken up again in other parts, and in particular also as the introduction of a unified notation. Depending on need and interest, more in-depth self-study of the various contents is of course advisable. In the following literature notes, we provide an overview of sources and reading recommendations for the contents considered here.

Literature notes

Bärwolff (2017) and Arens et al. (2018) form the primary basis for many of the topics presented here and offer an excellent and comprehensive entry point into the material covered. At the international level, Spivak (2008) and Strang (2009) provide very readable introductions to differential calculus and linear algebra, respectively. A deeper understanding, especially of analytical concepts, is provided, for example, by Abbott (2015) and Chiossi (2021). Searle (1982) offers a compact presentation of those aspects of matrix calculus and linear algebra that are used mainly in probabilistic data analysis. Deisenroth et al. (2020) provides an overview of many of the topics treated here under the moniker of machine learning.

1 Language and logic

1.1 Mathematics is a language

Mathematics is the language of natural-scientific model building. For example, the expression

$$F = ma \tag{1.1}$$

corresponds, in the sense of Newton's second axiom, to a theory of the motion of objects under the action of forces (Newton (1687)). Likewise, the expression

$$\max_{q(z)} \int q(z) \ln \left(\frac{p(y,z)}{q(z)} \right) dz \tag{1.2}$$

corresponds, in the sense of variational inference, to contemporary theory about how the brain works (Friston (2005)). Mathematical symbolism serves in particular to communicate scientific insights precisely and aims to describe complex matters exactly and efficiently. As in reflective engagement with any form of language, the question "What is this supposed to mean?" therefore always remains the guiding question when working with mathematical content and symbolism.

As a language system, mathematics has a number of special features. First, its contents are often abstract. This is because mathematics strives for the broadest possible general comprehensibility and applicability. Mathematical approaches to the phenomena of the world are interested in making insights as easy as possible to transfer to other contexts. To make this possible, mathematics tries to work as precisely and understandably as possible, that is, in terms of precise concepts. It proceeds in a strictly hierarchical way, so that concepts introduced later often presuppose a good understanding of the underlying concepts introduced earlier.

The precision of mathematical language implies a high density of information. It is therefore rather sober and leaves out what is superfluous, so that, in the best case, *everything* in a mathematical text is relevant for communicating an idea. As recipients of mathematical texts, we perceive their information density through the high cognitive energy required when reading such a text. This high energy expenditure calls in particular for calm and slowness when reading with the aim of genuine understanding. The following quotation may serve as a guiding principle for working with mathematical texts: "A mathematical text cannot be read like a novel; one has to work through it" (Unger (2000)). After reading a short mathematical text, one should always ask critically whether one has really understood what has been read or whether further sources should be consulted to clarify the matter. It is also helpful, in the spirit of Richard Feynman's famous quotation "What I cannot create, I do not understand", to make one's own notes and to reconstruct mathematical language systems oneself.

If one wants to gain access to the world of natural-scientific model building, then it is helpful, when dealing with its mathematical expressions and symbolism, to use the same strategies as when learning a foreign language. In addition to immersing oneself in the corresponding language space, that is, constant exposure to mathematical modes of expression, this certainly includes first memorizing concepts and translating texts into everyday language. A deep and secure understanding of mathematical model building then arises in particular through the application of mathematical ways of thinking in written and spoken form.

1.2 Basic building blocks

In the following, we introduce three basic building blocks of mathematical communication that will accompany us throughout: the concepts of a *definition*, a *theorem*, and a *proof*.

Definition

A *definition* is a cornerstone of a mathematical system that is neither justified nor deductively derived within that system. Definitions can only be evaluated according to their usefulness within a mathematical system. A definition is best memorized first and questioned only once one has understood its use in application or is not convinced by it. Some relaxation and calm are generally helpful when dealing with definitions that appear complex at first sight. To indicate that we define a symbol as something, we use the notation ":=". For example, the expression "$a := 2$" defines the symbol a as the number two. In this text, definitions always end with the symbol $\bullet$.

Theorem

A *theorem* is a mathematical statement that can be recognized as correct (true) by means of a proof. That is, a theorem is always derived from definitions and/or other theorems. In this sense, theorems are the empirical results of mathematics. In applied, data-analytic mathematics, theorems are often helpful for calculations. It is therefore worth memorizing them, because they usually form the basis for data analysis and data interpretation. Equations often appear in theorems. These equations arise from the assumptions of the theorem. To mark equations, we use the equals sign "=". For example, in a given context, the expression "$a = 2$" says that, because of certain assumptions, the variable a has the value two. In this text, theorems always end with the symbol $\circ$.

Proof

A *proof* is a chain of argumentation that draws on known definitions and theorems to establish the correctness (truth) of a theorem. Short proofs often contribute to understanding a theorem; long proofs tend to do so less. Proofs are therefore, in particular, answers to the question why a mathematical statement holds ("Why is this so?"). Proofs are not memorized. When proofs are short, it is sensible to work through them, because they usually repeat content that is assumed to be known. When they are long, it is more sensible to skip them at first so as not to get lost in details and lose the main path through the mathematical structure. In this text, proofs always end with the symbol $\square$.

In addition to definitions, theorems, and proofs, mathematical texts contain other typical building blocks, such as *axioms*, *lemmas*, *corollaries*, and *conjectures*. We will not use these terms and therefore give only a brief overview of them.

- *Axioms* are unproved fundamental assumptions in the sense that they serve as starting points for building mathematical systems. The transition between definitions and axioms is often fluid. Because we will not work at a particularly deep mathematical level, we prefer the term definition in almost all cases.
- A *lemma* is an "auxiliary theorem", that is, a mathematical statement that is proved but is not as important as a theorem. Because, on the one hand, we focus on important content and, on the other hand, we do not want to discriminate among mathematical statements, we avoid this term and instead use the term theorem throughout.
- A *corollary* is a mathematical statement that follows from a theorem by a simple proof. Because the "simplicity" of mathematical proofs is a relative property, we avoid this term and instead also use the term theorem throughout.
- *Conjectures* are mathematical statements for which it is unknown whether they are provable or refutable. Because we work in the area of applied mathematics, we will not encounter conjectures.

1.3 Propositional logic

Having now become acquainted with some basic building blocks of mathematical language systems, we turn to *propositional logic*, a simple system that allows relationships between mathematical statements to be established and formalized. Propositional logic plays a central role, for example, in the definition of set operations, in optimization conditions for functions, and in many proofs. In data-analytic applications, propositional logic is the basis of Boolean logic in programming. In mathematical psychology, propositional logic is the basis of the representational theory of measurement.

We begin with the definition of the concept of a mathematical *statement*.

Definition 1.1 (Statement)**.** A *statement* is a sentence to which the property *true* or *false* can be assigned unambiguously.

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The adjective *true* can also be understood as *correct*. We abbreviate true by "t" and false by "f". In the field of real numbers, for example, the statement $1 + 1 = 2$ is true and the statement $1+1 = 3$ is false. Note that the binarity of the truth value of statements is a basic assumption of propositional logic and therefore to be understood formally and scientifically, not empirically. Truth values do not refer to definitions; definitions fix meanings.

A first way of working with statements is to negate them. This leads to the following definition.

Definition 1.2 (Negation)**.** Let A be a statement. Then the *negation of* A is the statement that is false when A is true and true when A is false. The negation of A is denoted by $\neg A$, read as "not A".

•

For example, the negation of the statement "The sun is shining" is the statement "The sun is not shining". The negation of the statement $1 + 1 = 2$ is the statement $1 + 1 \neq 2$, and the negation of the statement $x > 1$ is the statement $x \leq 1$. The definition of the negation of a statement A is represented in tabular form as follows:

Table 1.1 Truth table of negation

A	$\neg A$
t	f
f	t

Tables of this form are called *truth tables*. They are a popular tool in propositional logic, and we will use them often in the following.

If one wants to connect two statements logically, the concepts of *conjunction* and *disjunction* are available.

Definition 1.3 (Conjunction)**.** Let A and B be statements. Then the *conjunction of* A *and* B is the statement that is true if and only if both A and B are true. The conjunction of A and B is denoted by $A \wedge B$, read as "A and B".

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The definition of conjunction implies the following truth table:

Table 1.2 Truth table of conjunction

A	B	$A \wedge B$
t	t	t
t	f	f
f	t	f
f	f	f

As an example, let A be the statement $2 \geq 1$ and let B be the statement $2 > 1$. Since both A and B are true, the statement $(2 \geq 1) \wedge (2 > 1)$ is also true. As another example, let A be the statement $1 \geq 1$ and let B be the statement $1 > 1$. Here, only A is true and B is false. Thus the statement $(1 \geq 1) \wedge (1 > 1)$ is false.

Definition 1.4 (Disjunction)**.** Let A and B be statements. Then the *disjunction of* A *and* B is the statement that is true if and only if at least one of the two statements A and B is true. The disjunction of A and B is denoted by $A \vee B$, read as "A or B".

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The definition of disjunction implies the following truth table:

Table 1.3 Truth table of disjunction

A	B	$A \vee B$
t	t	t
t	f	t
f	t	t
f	f	f

$A \vee B$ is therefore also true when both A and B are true. The "or" considered here is thus more precisely an "and/or". For this reason, disjunction is also called a "non-exclusive or". As an example, let A be the statement $2 \geq 1$ and let B be the statement $2 > 1$. A is true and B is true. Thus the statement $(2 \geq 1) \vee (2 > 1)$ is true. Now again let A be the statement $1 \geq 1$ and let B be the statement $1 > 1$. Then A is true and B is false. Thus the statement $(1 \geq 1) \vee (1 > 1)$ is true.

One way of placing statements in a mechanical logical relationship is *implication*. It is defined as follows.

Definition 1.5 (Implication)**.** Let A and B be statements. Then the *implication*, denoted by $A \Rightarrow B$, is the statement that is false if and only if A is true and B is false. A is called the *assumption (premise)* and B the *conclusion* of the implication. $A \Rightarrow B$ is read as "from A follows B", "A implies B", or "if A, then B".

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The definition of implication can be represented by the following truth table:

Table 1.4 Truth table of implication

A	B	$A \Rightarrow B$
t	t	t
t	f	f
f	t	t
f	f	t

An intuitive understanding of the definition of implication in the sense of the truth table above is obtained most readily by reading it as an attempt to capture and formalize the intuitive idea of a consequence in the context of propositional logic. To understand this, it is best to read the truth states in Table 1.4 in the order truth state of A, truth state of $A \Rightarrow B$, and finally truth state of B. Read in this way, the truth table shows that if A is true and $A \Rightarrow B$ is true, then B is true. Thus, if one constructs a true implication based on a true statement, for example by transforming equations, it follows that B is also true. If this is not possible, that is, if A is true and $A \Rightarrow B$ is always false, then B is also false. This is how statements may be refuted. Finally, one sees that if A is false and $A \Rightarrow B$ is true, then B can be either true or false. Only from a true premise does a true implication always yield a true conclusion. In particular, the definition of implication therefore satisfies the principle "anything follows from falsehood (ex falso sequitur quodlibet)". Thus one cannot infer anything meaningful from false statements by means of implication.

In the context of implication, the concepts of *sufficient* and *necessary* conditions arise: If $A \Rightarrow B$ is true, one says that "A is *sufficient* for B" and that "B is *necessary* for A". This terminology is explained in the context of implication as follows: If $A \Rightarrow B$ is true, then if A is true, B is also true. The truth of A is therefore enough for the truth of B. Thus A is sufficient for B. Furthermore, if $A \Rightarrow B$ is true, then if B is false, A is also false. The truth of B is therefore necessary for the truth of A.

Finally, to illustrate Table 1.4, we give a concrete everyday example. Let A be the statement "The bridge is damaged", let B be the statement "The bridge is closed", and let $A \Rightarrow B$ be the implication "If the bridge is damaged, then the bridge is closed". We discuss the four cases of Table 1.4.

- A true, B true, $A \Rightarrow B$ true. If A is true, then the bridge is damaged, and if B is also true, then the bridge is also closed. In this case, the implication "If the bridge is damaged, then the bridge is closed" is therefore obviously true.
- A true, B false, $A \Rightarrow B$ false. If A is true, then the bridge is damaged, and if B is false, then the bridge is not closed. In this case, the implication "If the bridge is damaged, then the bridge is closed" is therefore obviously false.
- A false, B true, $A \Rightarrow B$ true. If A is false, then the bridge is not damaged, and if B is true, then the bridge is closed. Since the bridge is not damaged, however, no conclusion can be drawn regarding the implication "If the bridge is damaged, then the bridge is closed"; in particular, one cannot show that it would be false. Thus one assumes that the statement "If the bridge is damaged, then the bridge is closed" is true.
- A false, B false, $A \Rightarrow B$ true. If A is false, then the bridge is not damaged, and if B is false, then the bridge is not closed. Since the bridge is not damaged, however, no conclusion can be drawn regarding the implication "If the bridge is damaged, then the bridge is closed"; in particular, one cannot show that it would be false. Thus one again assumes that the statement "If the bridge is damaged, then the bridge is closed" is true.

A very frequently occurring relation between two statements is their *equivalence*.

Definition 1.6 (Equivalence)**.** Let A and B be statements. The *equivalence of A and B* is the statement that is true if and only if A and B are both true or if A and B are both false. The equivalence of A and B is denoted by $A \Leftrightarrow B$ and read as "A if and only if B" or "A is equivalent to B".

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The definition of equivalence implies the following truth table:

Table 1.5 Truth table of equivalence

A	B	$A \Leftrightarrow B$
t	t	t
t	f	f
f	t	f
f	f	t

The definition of the concept of *logical equivalence* allows, among other things, the equivalence of two statements to be established by means of implications.

Definition 1.7 (Logical equivalence)**.** Two statements are called *logically equivalent* if their truth tables are the same.

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As examples of logical equivalences that are frequently used in proof arguments, we show the statements of the following theorem.

Theorem 1.1 (Logical equivalences)**.** *Let A and B be two statements. Then the following statements are logically equivalent:*

(1) $A \Leftrightarrow B$ *and* $(A \Rightarrow B) \wedge (B \Rightarrow A)$
(2) $A \Rightarrow B$ *and* $(\neg B) \Rightarrow (\neg A)$

∘

Proof. By the definition of logical equivalence, we have to show that the truth tables of the statements under consideration are the same. We show first (1), then (2).

(1) We recall the truth table of $A \Leftrightarrow B$:

A	B	$A \Leftrightarrow B$
t	t	t
t	f	f
f	t	f
f	f	t

We further consider the truth table of $(A \Rightarrow B) \wedge (B \Rightarrow A)$:

A	B	$A \Rightarrow B$	$B \Rightarrow A$	$(A \Rightarrow B) \wedge (B \Rightarrow A)$
t	t	t	t	t
t	f	f	t	f
f	t	t	f	f
f	f	t	t	t

Comparing the truth table of $A \Leftrightarrow B$ with the first two and the last column of the truth table of $(A \Rightarrow B) \wedge (B \Rightarrow A)$ shows their equality.

(2) We recall the truth table of $A \Rightarrow B$:

A	B	$A \Rightarrow B$
t	t	t
t	f	f
f	t	t
f	f	t

We further consider the truth table of $(\neg B) \Rightarrow (\neg A)$:

A	B	$\neg B$	$\neg A$	$(\neg B) \Rightarrow (\neg A)$
t	t	f	f	t
t	f	t	f	f
f	t	f	t	t
f	f	t	t	t

Comparing the truth table of $A \Rightarrow B$ with the first two and the last column of the truth table of $(\neg B) \Rightarrow (\neg A)$ shows their equality. □

The first statement of Theorem 1.1 says that the statement "A and B are equivalent" is logically equivalent to the statement "from A follows B and from B follows A". This is the basis for many so-called *direct proofs* using equivalence transformations. The second statement of Theorem 1.1 says that the statement "from A follows B" is logically equivalent to the statement "from not B follows not A". This is the basis for the technique of *indirect proof*. We consider these proof techniques more closely in the following section. First, we summarize the meanings of the symbols introduced in this section once more in the table below.

Table 1.10 Symbol overview. A and B are statements, that is, A and B are either true or false.

Symbol	Meaning	Note
$\neg A$	Not A	True when A is false, and vice versa
$A \wedge B$	A and B	True only when both A and B are true
$A \vee B$	A and/or B	True when at least one of the statements is true
$A \Rightarrow B$	From A follows B	B is necessary for A, A is sufficient for B
$A \Leftrightarrow B$	A is equivalent to B	Both $A \Rightarrow B$ and $B \Rightarrow A$ hold

1.4 Equivalence transformations

Mathematical problems often lead to the case in which information about an unknown variable is represented implicitly by means of an equation or an inequality. To represent the information about the corresponding variable explicitly, that is, to determine its value in the case of equations or to determine the range of numbers in which the value of the variable lies in the case of inequalities, one uses *equivalence transformations*. Here, we consider only the equivalence transformations of equations and inequalities with respect to real variables that are familiar from school mathematics. Equivalence transformations of equations and inequalities have the property that the truth value of the statement formulated by an equation or inequality does not change when the corresponding transformation is applied. Both sides of the equation or inequality are always transformed. For the application of a mathematical operation that transforms an equation or inequality statement A into an equation or inequality statement B to be an equivalence transformation of the form $A \Leftrightarrow B$, both $A \Rightarrow B$ and $B \Rightarrow A$ must hold, as is well known. This implies that equivalence transformations are reversible (invertible) operations.

For equations, admissible equivalence transformations include in particular

- adding a number to both sides of the equation,
- subtracting a number from both sides of the equation,
- multiplying both sides of the equation by a nonzero number,
- dividing both sides of the equation by a nonzero number, and

- applying an invertible function to both sides of the equation.

Example

Anticipating Section 4.3, we consider the statement

$$2\exp(x) - 2 = 0. \tag{1.3}$$

Then

$$\begin{aligned}
& 2\exp(x) - 2 = 0 \\
\Leftrightarrow\ & 2\exp(x) - 2 + 2 = +2 \\
\Leftrightarrow\ & 2\exp(x) = 2 \\
\Leftrightarrow\ & \frac{1}{2} \cdot 2\exp(x) = \frac{1}{2} \cdot 2 \\
\Leftrightarrow\ & \exp(x) = 1 \\
\Leftrightarrow\ & \ln(\exp(x)) = \ln(1) \\
\Leftrightarrow\ & x = 0.
\end{aligned} \tag{1.4}$$

In summary, therefore,

$$2\exp(x) - 2 = 0 \Leftrightarrow x = 0. \tag{1.5}$$

For inequalities, admissible equivalence transformations include in particular

- adding a number to both sides of the inequality,
- subtracting a number from both sides of the inequality,
- multiplying both sides of the inequality by a nonzero number, with multiplication by a negative number reversing the inequality sign,
- dividing both sides of the inequality by a nonzero number, with division by a negative number reversing the inequality sign, and
- applying an invertible monotonically increasing function to both sides of the inequality; for monotonically decreasing functions, the inequality sign is reversed.

Example

We consider the statement

$$-5x - 2 \geq 8. \tag{1.6}$$

Then

$$\begin{aligned}
& -5x - 2 \geq 8 \\
\Leftrightarrow\ & -5x - 2 + 2 \geq 8 + 2 \\
\Leftrightarrow\ & -5x \geq 10 \\
\Leftrightarrow\ & -\frac{1}{5} \cdot 5x \geq \frac{1}{5} \cdot 10 \\
\Leftrightarrow\ & -x \geq 2 \\
\Leftrightarrow\ & x \leq -2.
\end{aligned} \tag{1.7}$$

In summary, therefore,

$$-5x - 2 \geq 8 \Leftrightarrow x \leq -2. \tag{1.8}$$

1.5 Proof techniques

In this section, we outline three fundamental proof techniques: *direct* and *indirect proofs*, and *proof by contradiction.* In what follows, especially the first technique will be used repeatedly to justify theorems.

We have:

- *Direct proofs* use equivalence transformations to show $A \Rightarrow B$.
- *Indirect proofs* use the logical equivalence of $A \Rightarrow B$ and $(\neg B) \Rightarrow (\neg A)$.
- *Proofs by contradiction* show that $(\neg B) \wedge A$ is false.

Equipped with this, we now prove the following theorem by means of a direct proof, an indirect proof, and a proof by contradiction (cf. Arens et al. (2018)).

Theorem 1.2 (Squares of positive numbers)**.** *Let a and b be two positive numbers. Then $a^2 < b^2 \Rightarrow a < b$.*

$\circ$

Proof. We first give a *direct proof.* Let $a^2 < b^2$ be the statement A and let $a < b$ be the statement B. Then

$$a^2 < b^2 \Leftrightarrow 0 < b^2 - a^2 \Leftrightarrow 0 < (b + a)(b - a) \Leftrightarrow 0 < (b - a) \Leftrightarrow a < b. \tag{1.9}$$

We now give an *indirect proof.* Let $a^2 \geq b^2$ be the statement $\neg A$. Furthermore, let $a \geq b$ be the statement $\neg B$. Then

$$a \geq b \Leftrightarrow a^2 \geq ab \wedge ab \geq b^2 \Leftrightarrow a^2 \geq b^2. \tag{1.10}$$

Finally, we give a *proof by contradiction.* To do so, we show that the assumption $(\neg B) \wedge A$ leads to a false statement. We have

$$a \geq b \wedge a^2 < b^2 \Leftrightarrow a^2 \geq ab \wedge a^2 < b^2 \Leftrightarrow ab \leq a^2 < b^2. \tag{1.11}$$

Furthermore,

$$a \geq b \wedge a^2 < b^2 \Leftrightarrow ab \geq b^2 \wedge a^2 < b^2 \Leftrightarrow a^2 < b^2 \leq ab. \tag{1.12}$$

Overall, this yields the false statement

$$ab \leq a^2 < b^2 \leq ab \Leftrightarrow ab < ab. \tag{1.13}$$

$\square$

2 Sets

2.1 Basic definitions

Sets collect mathematical objects and form the foundation of modern mathematics. We begin with the following definition.

Definition 2.1 (Sets)**.** Following Cantor (1895), a *set* M is defined as "a collection into a whole of definite, distinct objects m of our intuition or of our thought (which are called the elements of the set)". We write

$$m \in M \text{ or } m \notin M \tag{2.1}$$

to express that m is an element or not an element of M, respectively.

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There are at least the following ways of defining sets:

- Listing the elements in curly braces, for example $M := \{1, 2, 3\}$.
- Specifying the properties of the elements, for example $M := \{x \in \mathbb{N} | x < 4\}$.
- Equating the set with another uniquely defined set, for example $M := \mathbb{N}_3$.

The notation $\{x \in \mathbb{N} | x < 4\}$ is read as "$x \in \mathbb{N}$ such that $x < 4$", where the meaning of $\mathbb{N}$ will be explained below. It is important to recognize that sets are *unordered* mathematical objects, that is, the order in which the elements of a set are listed does not matter. For example, $\{1, 2, 3\}$, $\{1, 3, 2\}$, and $\{2, 3, 1\}$ denote the same set, namely the set of the first three natural numbers.

Basic relations between several sets are fixed in the next definition.

Definition 2.2 (Subsets and set equality)**.** Let M and N be two sets.

- A set M is called a *subset* of a set N if, for every element $m \in M$, we also have $m \in N$. If M is a subset of N, we write

$$M \subseteq N \tag{2.2}$$

  and call M a *subset* of N and N a *superset* of M.
- A set M is called a *proper subset* of a set N if, for every element $m \in M$, we also have $m \in N$, but there is at least one element $n \in N$ for which $n \notin M$. If M is a proper subset of N, we write

$$M \subset N. \tag{2.3}$$

- Two sets M and N are called *equal* if, for every element $m \in M$, we also have $m \in N$, and if, for every element $n \in N$, we also have $n \in M$. If the sets M and N are equal, we write
$$M = N. \tag{2.4}$$

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Example

For example, consider the sets $M := \{1\}$, $N := \{1, 2\}$, and $O := \{1, 2\}$. Then, by the definitions above, $M \subset N$, because $1 \in M$ and $1 \in N$, but $2 \in N$ and $2 \notin M$. Furthermore, $N \subseteq O$, because $1 \in N$ and $1 \in O$, as well as $2 \in N$ and $2 \in O$, and there is no element of O that is not in N. Likewise, $O \subseteq N$, because $1 \in O$ and $1 \in N$, as well as $2 \in O$ and $2 \in N$, and there is no element of N that is not in O. Finally, we even have $N = O$, because, for every element $n \in N$, we also have $n \in O$, and at the same time, for every element $o \in O$, we also have $o \in N$. We represent these relations schematically by means of *Venn-Euler diagrams* in Figure 2.1.

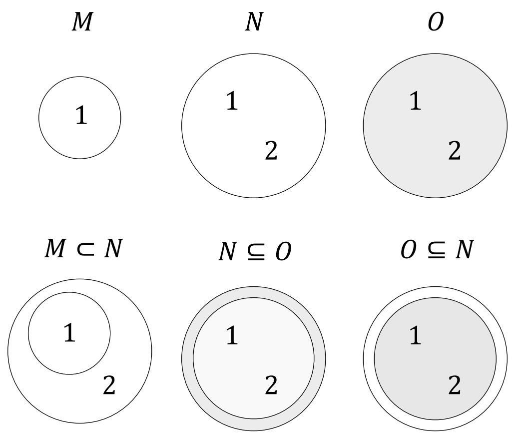

Figure 2.1 Venn-Euler diagrams of the subset properties of the sets $M := \{1\}$, $N := \{1, 2\}$, and $O := \{1, 2\}$.

An important property of a set is the number of elements it contains. This is called the *cardinality* of the set.

Definition 2.3 (Cardinality)**.** The number of elements of a set M is called its *cardinality* and is denoted by $|M|$.

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A special set is the set without elements.

Definition 2.4. A set with cardinality zero is called the *empty set* and is denoted by $\emptyset$.

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As examples, let $M := \{1, 2, 3\}$, $N = \{a, b, c, d\}$, and $O := \emptyset$. Then $|M| = 3$, $|N| = 4$, and $|O| = 0$.

For every set, one can consider the set of all subsets of this set. This leads to the important concept of the *power set*.

Definition 2.5 (Power set)**.** The set of all subsets of a set M is called the *power set of* M and is denoted by $\mathcal{P}(M)$.

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Note that the empty subset of M and M itself are also always elements of $\mathcal{P}(M)$. We first consider four examples of the concept of a power set.

- Let $M_0 := \emptyset$ be the empty set. Then

$$\mathcal{P}(M_0) = \{\emptyset\}. \tag{2.5}$$

- Let M_1 be the one-element set $M_1 := \{a\}$. Then

$$\mathcal{P}(M_1) = \{\emptyset, \{a\}\}. \tag{2.6}$$

- Let $M_2 := \{a, b\}$. Then M_2 has both one-element and two-element subsets, and

$$\mathcal{P}(M_2) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}. \tag{2.7}$$

- Finally, let $M_3 := \{a, b, c\}$. Then M_3 has, among others, one-element, two-element, and three-element subsets, and

$$\mathcal{P}(M_3) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}. \tag{2.8}$$

Theorem 2.1 (Cardinality of the power set)**.** *Given a set M with cardinality $|M| = n$, and let $\mathcal{P}(M)$ be its power set. Then $|\mathcal{P}(M)| = 2^n$.*

∘

Proof. To prove the statement of the theorem, we associate each element P of the power set of M uniquely with a binary sequence of length n, where the entry at the i-th position represents whether the i-th element of M is an element of P or not. For example, let $M := \{m_1, m_2, m_3\}$ and $P := \{m_2, m_3\}$. Then P corresponds to the binary sequence 011. The empty set $P := \emptyset$ corresponds to the binary sequence 000, and the original set $P = M$ corresponds to the binary sequence 111. Thus the question is how many unique binary sequences of length n there are. Since each element of the sequence has two possible states, there are n factors $2 \cdot 2 \cdots 2$, that is, 2^n. □

In the examples above, we have the cases

- $|M_0| = 0 \Rightarrow |\mathcal{P}(M_0)| = 2^0 = 1$,
- $|M_1| = 1 \Rightarrow |\mathcal{P}(M_1)| = 2^1 = 2$,
- $|M_2| = 2 \Rightarrow |\mathcal{P}(M_2)| = 2^2 = 4$,
- $|M_3| = 3 \Rightarrow |\mathcal{P}(M_3)| = 2^3 = 8$,

as can be verified by counting the elements of the corresponding power sets.

2.2 Operations

Two sets can be combined with one another in different ways. The result of such a combination is another set. We call the combination of two sets a *set operation* and give the following definitions.

Definition 2.6 (Set operations)**.** Let M and N be two sets.

- The *union of M and N* is defined as the set

$$M \cup N := \{x|x \in M \vee x \in N\}, \tag{2.9}$$

  where $\vee$ is to be understood according to Definition 1.4 as a *non-exclusive or*, that is, as and/or.

- The *intersection of M and N* is defined as the set

$$M \cap N := \{x|x \in M \wedge x \in N\}. \tag{2.10}$$

  If M and N satisfy $M \cap N = \emptyset$, then M and N are called *disjoint*.

- The *difference of M and N* is defined as the set

$$M \setminus N := \{x|x \in M \wedge x \notin N\}. \tag{2.11}$$

  The difference of M and N is also called, especially when $N \subseteq M$, the *complement of N with respect to M* and is denoted by N^c. In this context, M is also called the *basic set* or the *universe*.

- The *symmetric difference of M and N* is defined as the set

$$M\Delta N := \{x|(x \in M \vee x \in N) \wedge x \notin M \cap N\}. \tag{2.12}$$

  The symmetric difference can therefore be understood as an *exclusive or*.

•

Example

As an example, consider the sets $M := \{1, 2, 3\}$ and $N := \{2, 3, 4, 5\}$. Then

- $M \cup N = \{1, 2, 3, 4, 5\}$, because $1 \in M$, $2 \in M$, $3 \in M$, $4 \in N$, and $5 \in N$.
- $M \cap N = \{2, 3\}$, because only for 2 and 3 do we have $2 \in M, 3 \in M$ and also $2 \in N, 3 \in N$. For 1, we only have $1 \in M$, and for 4 and 5, we only have $4 \in N$ and $5 \in N$.
- $M \setminus N = \{1\}$, because $1 \in M$, but $1 \notin N$, and $2 \in M$, but also $2 \in N$.
- $N \setminus M = \{4, 5\}$, because $4 \in N$ and $5 \in N$, but $4 \notin M$ and $5 \notin M$. This shows in particular that the difference of M and N is *not* symmetric, that is, it is *not* necessarily the case that $M \setminus N$ equals $N \setminus M$.
- $M\Delta N = \{1, 4, 5\}$, because $1 \in M$, but $1 \notin \{2, 3\}$, $2 \in M$, but $2 \in \{2, 3\}$, $3 \in M$, but $3 \in \{2, 3\}$, $4 \in N$, but $4 \notin \{2, 3\}$, and $5 \in N$, but $5 \notin \{2, 3\}$.

Figure 2.2 visualizes the set operations considered in this example.

Finally, we introduce the concept of a partition of a set.

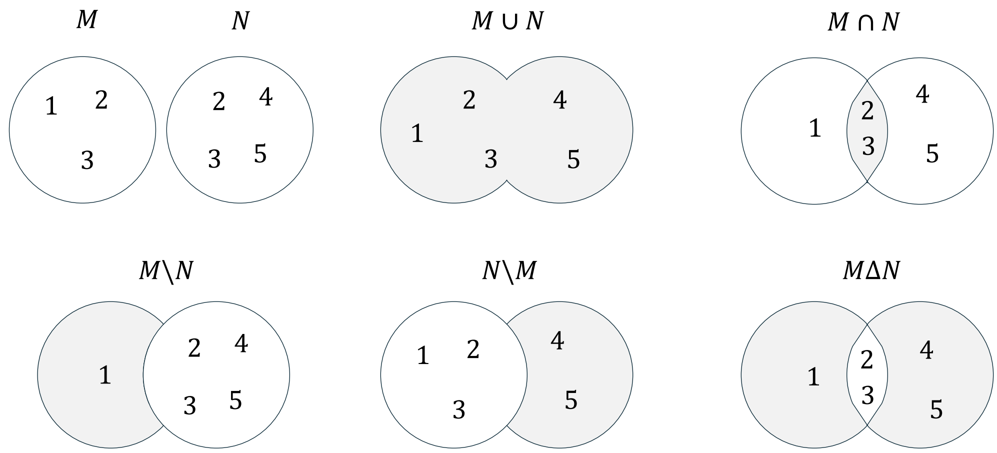

Figure 2.2 Venn-Euler diagrams of the set operations considered in the example for $M := \{1, 2, 3\}$ and $N := \{2, 3, 4, 5\}$. The gray shaded regions correspond to the resulting sets.

Definition 2.7 (Partition)**.** Let M be a set, and let $P := \{N_i\}$ be a set of sets N_i with $i = 1, ..., n$, such that

$$(M = \cup_{i=1}^{n} N_i) \wedge \left(N_i \cap N_j = \emptyset \text{ for } i, j = 1, ..., n \text{ and } i \neq j\right). \tag{2.13}$$

Then P is called a *partition of* M.

•

The partition of a set therefore corresponds to splitting the set into disjoint subsets. Partitions are generally not unique, that is, there are usually several ways to partition a given set.

As an example, consider the set $M := \{1, 2, 3, 4, 5, 6\}$. Then $P_1 := \{\{1\}, \{2, 3, 4, 5, 6\}\}$, $P_2 := \{\{1, 2, 3\}, \{4, 5, 6\}\}$, and $P_3 := \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$ are three possible partitions of M. Figure 2.3 visualizes the partitions considered in this example.

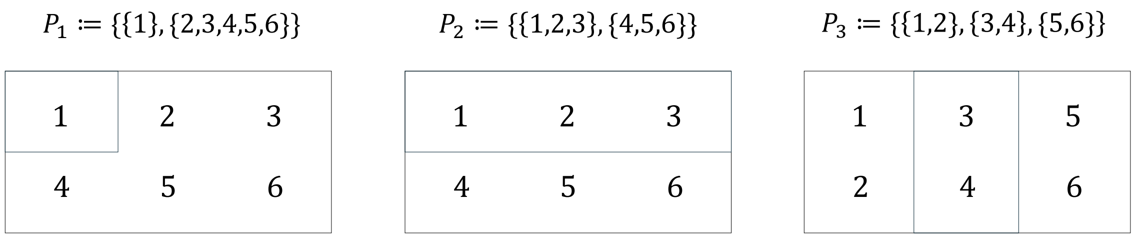

Figure 2.3 Diagrams of the partitions considered in the example for $M := \{1, 2, 3, 4, 5, 6\}$.

2.3 Special sets

Number sets

In natural science, one attempts to describe phenomena of the world that are intuitively identified as discrete or continuous by means of numbers. Depending on the type of phenomenon, different number sets are suitable for this purpose. Mathematics provides, among others, the number sets given in the following definition.

Definition 2.8 (Number sets)**.** The following denote:

- $\mathbb{N}$ $:= \{1, 2, 3, ...\}$ the *natural numbers*,
- $\mathbb{N}_n$ $:= \{1, 2, 3, ..., n\}$ the *natural numbers of order n*,
- $\mathbb{N}^0 := \mathbb{N} \cup \{0\}$ the *natural numbers* and zero,
- $\mathbb{Z}$ $:= \{..., -3, -2, -1, 0, 1, 2, 3...\}$ the *integers*,
- $\mathbb{Q}$ $:= \{\frac{p}{q} | p \in \mathbb{Z}, q \in \mathbb{N}\}$ the *rational numbers*,
- $\mathbb{R}$ the *real numbers*, and
- $\mathbb{C}$ $:= \{a + ib | a, b \in \mathbb{R}, i := \sqrt{-1}\}$ the *complex numbers*.

•

The natural numbers and the integers are suitable for quantifying discrete phenomena. The rational numbers and especially the real numbers are suitable for quantifying continuous phenomena. $\mathbb{R}$ includes the rational numbers and the so-called *irrational numbers* $\mathbb{R} \setminus \mathbb{Q}$. Rational numbers are numbers that can be expressed as fractions of integers and natural numbers. These include all integers as well as negative and positive decimal numbers such as $-\frac{9}{10} = -0.9$, $\frac{1}{3} = 0.\bar{3}$, and $\frac{196}{100} = 1.96$. Irrational numbers are numbers that cannot be expressed as rational numbers. Examples of irrational numbers are *Euler's number* $e \approx 2.71$, the *circle constant* $\pi \approx 3.14$, and the square root of 2, $\sqrt{2} \approx 1.41$.

The real numbers contain the natural numbers, the integers, and the rational numbers as subsets. Thus there are very many real numbers. Indeed, Cantor (1874) proved that there are more real numbers than natural numbers, even though there are infinitely many real numbers and infinitely many natural numbers. This property of the real numbers is called their *uncountability*. We will deepen this concept in Chapter 4. In particular,

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}. \tag{2.14}$$

Between any two real numbers there are infinitely many further real numbers. Positive infinity ∞ and negative infinity $-\infty$ are not numbers with which one could calculate in standard mathematics. They also do not belong to the number sets given in the definition above; thus $\infty \notin \mathbb{R}$ and $-\infty \notin \mathbb{R}$. Complex numbers are suitable for describing two-dimensional continuous phenomena. The values of the first dimension are represented in the real part a, and the values of the second dimension in the complex part b of a complex number. Complex numbers are used, for example, in the modeling of physical phenomena and in Fourier analysis.

Important subsets of the real numbers are the so-called *intervals*. We give the following definitions.

Intervals

Definition 2.9. Connected subsets of the real numbers are called *intervals*. For $a, b \in \mathbb{R}$, one distinguishes

- the *closed interval*
$$[a, b] := \{x \in \mathbb{R} | a \leq x \leq b\}, \tag{2.15}$$

- the *open interval*

$$]a, b[:= \{x \in \mathbb{R} | a < x < b\}, \tag{2.16}$$

- and the *half-open intervals*

$$]a, b] := \{x \in \mathbb{R} | a < x \leq b\} \text{ and } [a, b[:= \{x \in \mathbb{R} | a \leq x < b\}. \tag{2.17}$$

•

As examples, Figure 2.4 graphically represents the intervals $[1, 2]$, $]1, 2]$, $[1, 2[$, and $]1, 2[$. Here one imagines the real numbers as a continuous number line and must in each case note whether the left and right endpoints are part of the interval or not. As mentioned above, positive infinity (∞) and negative infinity $-\infty$ are not elements of $\mathbb{R}$. Thus one always writes $]-\infty, b]$ or $]-\infty, b[$ and $]a, \infty[$ or $[a, \infty[$, as well as $\mathbb{R} =]-\infty, \infty[$.

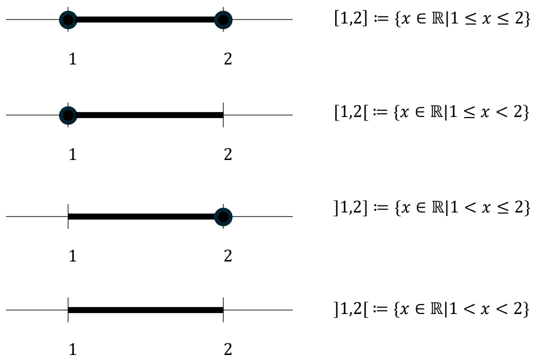

Figure 2.4 Representation of intervals on the number line. The black dot indicates that the corresponding number is part of the interval.

Cartesian products

Often one wants to quantitatively describe several independent properties of a phenomenon at the same time. For this purpose, the one-dimensional number sets defined above can be extended to multidimensional number sets by forming *Cartesian products*. The elements of Cartesian products are called *ordered tuples* or *vectors*.

Definition 2.10 (Cartesian products)**.** Let M and N be two sets. Then the *Cartesian product of the sets M and N* is the set of all ordered tuples (m, n) with $m \in M$ and $n \in N$, formally

$$M \times N := \{(m, n) | m \in M, n \in N\}. \tag{2.18}$$

The Cartesian product of a set M with itself is denoted by

$$M^2 := M \times M. \tag{2.19}$$

Furthermore, let $M_1, M_2, ..., M_n$ be sets. Then the *Cartesian product of the sets $M_1, ..., M_n$* is the set of all ordered n-tuples $(m_1, ..., m_n)$ with $m_i \in M_i$ for $i = 1, ..., n$, formally

$$\prod_{i=1}^{n} M_i := M_1 \times \cdots \times M_n := \{(m_1, ..., m_n) | m_i \in M_i \text{ for } i = 1, ..., n\}. \tag{2.20}$$

The n-fold Cartesian product of a set M with itself is denoted by

$$M^n := \prod_{i=1}^{n} M := \{(m_1, ..., m_n) | m_i \in M \text{ for } i = 1, ..., n\}. \tag{2.21}$$

•

In contrast to sets, the tuples introduced in Definition 2.10 are *ordered*. This means, for example, that for sets we have $\{1, 2\} = \{2, 1\}$, but for tuples we have $(1, 2) \neq (2, 1)$.

Example

Let $M := \{1, 2\}$ and $N := \{1, 2, 3\}$. Then the Cartesian product $M \times N$ is given by

$$M \times N := \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\} \tag{2.22}$$

and the Cartesian product $N \times M$ is given by

$$N \times M := \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}. \tag{2.23}$$

The Cartesian product is therefore generally not commutative, that is, it is not necessarily the case that $M \times N = N \times M$. One may construct the sets $M \times N \neq N \times M$ in this example from Table 2.1 and Table 2.2.

Table 2.1 Cartesian product $M \times N$

(m, n)	$n = 1$	$n = 2$	$n = 3$
$m = 1$	$(1, 1)$	$(1, 2)$	$(1, 3)$
$m = 2$	$(2, 1)$	$(2, 2)$	$(2, 3)$

Table 2.2 Cartesian product $N \times M$

(n, m)	$m = 1$	$m = 2$
$n = 1$	$(1, 1)$	$(1, 2)$
$n = 2$	$(2, 1)$	$(2, 2)$
$n = 3$	$(3, 1)$	$(3, 2)$

$\mathbb{R}$ to the Power of n

As described above, the real numbers are particularly suitable for describing continuous phenomena. For the simultaneous description of several aspects of a continuous phenomenon, the *set of real tuples of n-th order*, or $\mathbb{R}$ to the power of n for short, is correspondingly useful.

Definition 2.11 (Set of real tuples of n-th order)**.** The n-fold Cartesian product of the real numbers with themselves is denoted by

$$\mathbb{R}^n := \prod_{i=1}^{n} \mathbb{R} := \{x := (x_1, ..., x_n) | x_i \in \mathbb{R}\} \tag{2.24}$$

and is read as "$\mathbb{R}$ to the power of n". We write the elements of $\mathbb{R}^n$ as columns

$$x := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \tag{2.25}$$

and call them *n-dimensional vectors*. For distinction, we also call the elements of $\mathbb{R}^1 = \mathbb{R}$ *scalars*.

•

Examples

Familiar examples of $\mathbb{R}^n$ are $\mathbb{R}^1$ as the set of real numbers, $\mathbb{R}^2$ as the set of real tuples in the model of the two-dimensional plane, and $\mathbb{R}^3$ as the set of real triples in the model of three-dimensional space, as visualized in Figure 2.5.

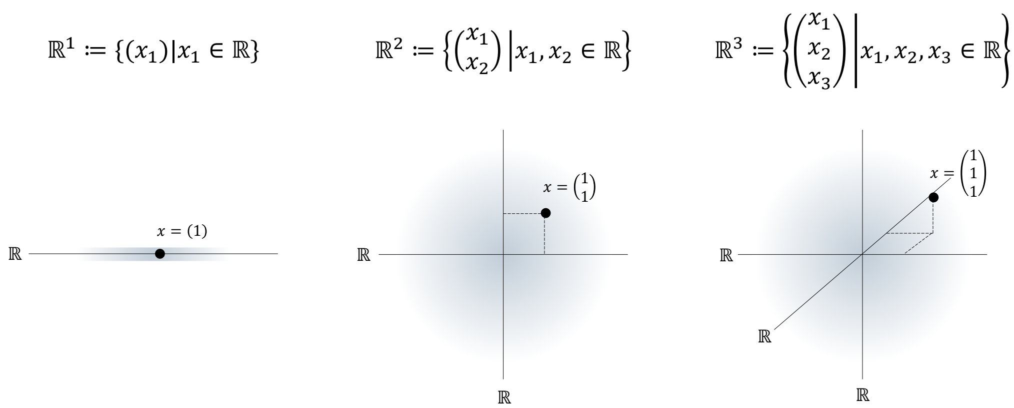

Figure 2.5 $\mathbb{R}^n$ for $n = 1, n = 2$, and $n = 3$. The black dots each represent an element of the corresponding set.

An example of an $x \in \mathbb{R}^4$ is

$$x = \begin{pmatrix} 0.16 \\ 1.76 \\ 0.23 \\ 7.11 \end{pmatrix}. \tag{2.26}$$

3 Sums, products, powers

This unit introduces notation for the elementary arithmetic operations.

3.1 Sums

Definition 3.1 (Summation symbol)**.** It denotes

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n. \tag{3.1}$$

Here,

- Σ denotes the Greek letter *S*igma, mnemonically standing for *s*um,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, \dots, x_n$ denote the *summands*.

•

For meaningful use of the summation symbol, it is essential to specify the beginning and the end of the summation by means of the subscript and the superscript. The precise name of the running index, by contrast, is irrelevant for the value of the sum; thus,

$$\sum_{i=1}^{n} x_i = \sum_{j=1}^{n} x_j. \tag{3.2}$$

Sometimes the running index is also given as an element of an *index set*. If, for example, the index set $I := \{1, 5, 7\}$ is defined, then

$$\sum_{i \in I} x_i := x_1 + x_5 + x_7. \tag{3.3}$$

In the following, we briefly consider a few examples of how the summation symbol is used.

- *Summation of predefined summands.* Let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. Then

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 2 + 10 - 4 = 8. \tag{3.4}$$

- *Summation of weighted predefined summands.* Again let $x_1 := 2$, $x_2 := 10$, and $x_3 := -4$. In addition, let the *weighting coefficients* $a_1 := \frac{1}{2}$, $a_2 := \frac{1}{5}$, and $a_3 := 2$ be defined. Then

$$\sum_{i=1}^{3} a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3 = \frac{1}{2} \cdot 2 + \frac{1}{5} \cdot 10 + 2 \cdot (-4) = 1 + 2 - 8 = -5. \quad (3.5)$$

  Expressions of the form $\sum_{i=1}^{n} a_i x_i$ are also called *linear combinations* of $x_1, ..., x_n$ with the *coefficients* or *weighting parameters* $a_1, ..., a_n$.
- *Summation of natural numbers.* We have

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5 = 15. \quad (3.6)$$

- *Summation of even natural numbers.* We have

$$\sum_{i=1}^{5} 2i = 2 \cdot 1 + 2 \cdot 2 + 2 \cdot 3 + 2 \cdot 4 + 2 \cdot 5 = 2 + 4 + 6 + 8 + 10 = 30. \quad (3.7)$$

- *Summation of odd natural numbers.* We have

$$\sum_{i=1}^{5} (2i-1) = 2 \cdot 1 - 1 + 2 \cdot 2 - 1 + 2 \cdot 3 - 1 + 2 \cdot 4 - 1 + 2 \cdot 5 - 1 = 1 + 3 + 5 + 7 + 9 = 25. \quad (3.8)$$

Working with the summation symbol can often be simplified by applying the following calculation rules.

Theorem 3.1 (Calculation rules for sums)**.**

(1) Sums of identical summands

$$\sum_{i=1}^{n} x = nx \quad (3.9)$$

(2) Associativity for sums of equal length

$$\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i = \sum_{i=1}^{n} (x_i + y_i) \quad (3.10)$$

(3) Distributivity under multiplication by a constant

$$\sum_{i=1}^{n} a x_i = a \sum_{i=1}^{n} x_i \quad (3.11)$$

(4) Splitting sums for $1 < m < n$

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{m} x_i + \sum_{i=m+1}^{n} x_i \quad (3.12)$$

(5) Reindexing

$$\sum_{i=0}^{n} x_i = \sum_{j=m}^{n+m} x_{j-m} \quad (3.13)$$

∘

Proof. These calculation rules can be verified by writing out the sums and applying the rules of addition and multiplication. Here, we show associativity for sums of equal length and distributivity under multiplication by a constant as examples. For the former, we have

$$\begin{aligned}\sum_{i=1}^n x_i + \sum_{i=1}^n y_i &= x_1 + x_2 + \cdots + x_n + y_1 + y_2 + \cdots + y_n \\ &= x_1 + y_1 + x_2 + y_2 + \cdots + x_n + y_n \\ &= \sum_{i=1}^n (x_i + y_i).\end{aligned} \tag{3.14}$$

For the latter, we have

$$\begin{aligned}\sum_{i=1}^n ax_i &= ax_1 + ax_2 + \cdots + ax_n \\ &= a(x_1 + x_2 + \cdots + x_n) \\ &= a\sum_{i=1}^n x_i.\end{aligned} \tag{3.15}$$

□

Examples

As a first example of applying the calculation rules recorded in Theorem 3.1, we consider the evaluation of a *mean* (sometimes also called an *average*). Let $x_1, x_2, ..., x_n$ be real numbers. The mean of these numbers is the sum of $x_1, x_2, ..., x_n$ divided by the number of numbers n. By statement (3) of Theorem 3.1, it is irrelevant whether the numbers are first added up and the resulting sum is then divided by n, or whether the individual numbers are each divided by n and the corresponding results are then added up. More precisely, by applying Theorem 3.1 (3) with $a = 1/n$, we obtain

$$\frac{1}{n}\sum_{i=1}^n x_i = \sum_{i=1}^n \frac{x_i}{n}. \tag{3.16}$$

For example, the mean of $x_1 := 1, x_2 := 4, x_3 := 2$, and $x_4 := 1$ is given by

$$\frac{1}{4}\sum_{i=1}^4 x_i = \frac{1}{4}(1+4+2+1) = \frac{8}{4} = 2 = \frac{8}{4} = \frac{1}{4} + \frac{4}{4} + \frac{2}{4} + \frac{1}{4} = \sum_{i=1}^4 \frac{x_i}{4}. \tag{3.17}$$

As a second example, we consider the reindexing rule recorded in Theorem 3.1 (5). Let $n := 3$ and $m := 2$, and let $x_0 := 2$, $x_1 := 3$, $x_2 := 5$, and $x_3 := 10$. Then, evidently,

$$\begin{aligned}\sum_{i=0}^3 x_i &= x_0 + x_1 + x_2 + x_3 \\ &= 2 + 3 + 5 + 10 \\ &= 20.\end{aligned} \tag{3.18}$$

But we also have

$$
\begin{aligned}
\sum_{j=m}^{n+m} x_{j-m} & = \sum_{j=2}^{3+2} x_{j-2} \\
& = \sum_{j=2}^{5} x_{j-2} \\
& = x_{2-2} + x_{3-2} + x_{4-2} + x_{5-2} \\
& = x_0 + x_1 + x_2 + x_3 \\
& = 2 + 3 + 5 + 10 \\
& = 20.
\end{aligned}
\tag{3.19}
$$

Double Sums

In applications, one often encounters expressions that carry out several summations one after another. For each of these sums, the definitions and calculation rules listed above apply. The meaning of double sums is best made clear by writing them out from the inside to the outside, while noting that the running index of the outer sum remains constant during each iteration of the inner sum.

The following examples may illustrate this.

(1)

$$
\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{3}(i+j) & = \sum_{i=1}^{2}(i+1+i+2+i+3) \\
& = (1+1+1+2+1+3) + (2+1+2+2+2+3) \\
& = 9 + 12 \\
& = 21
\end{aligned}
\tag{3.20}
$$

(2)

$$
\begin{aligned}
\sum_{i=1}^{2}\sum_{j=1}^{3}(x_i+y_j) & = \sum_{i=1}^{2}(x_i+y_1+x_i+y_2+x_i+y_3) \\
& = (x_1+y_1+x_1+y_2+x_1+y_3) + (x_2+y_1+x_2+y_2+x_2+y_3)
\end{aligned}
\tag{3.21}
$$

(3)

$$
\begin{aligned}
\sum_{i=1}^{3}\sum_{j=1}^{2}(x_i y_j) & = \sum_{i=1}^{3}(x_i y_1 + x_i y_2) \\
& = (x_1 y_1 + x_1 y_2) + (x_2 y_1 + x_2 y_2) + (x_3 y_1 + x_3 y_2)
\end{aligned}
\tag{3.22}
$$

3.2 Products

The product symbol provides notation for products that is analogous to the summation symbol.

Definition 3.2 (Product symbol)**.** It denotes

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot \dots \cdot x_n. \tag{3.23}$$

Here,

- $\prod$ denotes the Greek letter *P*i, mnemonically standing for *p*roduct,
- the subscript $i = 1$ denotes the *running index* and the *starting index*,
- the superscript n denotes the *terminal index*, and
- $x_1, x_2, ..., x_n$ denote the *product terms.*

•

Analogously to the summation symbol, the product symbol is meaningful only with a subscript and a superscript specifying the running and terminal index. The precise name of the running index is again irrelevant; thus,

$$\prod_{i=1}^{n} x_i = \prod_{j=1}^{n} x_j. \tag{3.24}$$

Here, too, the running index is in rare cases given as an element of an index set. If, for example, the index set $J := \mathbb{N}_2^0$ is defined, then

$$\prod_{j \in J} x_j := x_0 \cdot x_1 \cdot x_2. \tag{3.25}$$

One example of the use of the product symbol is the definition of the *factorial* of a natural number n by

$$n! := \prod_{i=1}^{n} i. \tag{3.26}$$

For example,

$$3! := \prod_{i=1}^{3} i = 1 \cdot 2 \cdot 3 = 6. \tag{3.27}$$

There are also a number of calculation rules for products that often simplify working with them. We list some in the following theorem. In doing so, we make anticipatory use of the notation of *powers* defined in Section 3.3.

Theorem 3.2 (Calculation rules for products)**.**

(1) Products of identical factors

$$\prod_{i=1}^{n} x = x^n \tag{3.28}$$

(2) Raising constants to powers

$$\prod_{i=1}^{n} a x_i = a^n \prod_{i=1}^{n} x_i \tag{3.29}$$

(3) Splitting products for $1 < m < n$

$$\prod_{i=1}^{n} x_i = \prod_{i=1}^{m} x_i \prod_{j=m+1}^{n} x_j \tag{3.30}$$

(4) Product of products

$$\prod_{i=1}^{n} x_i y_i = \prod_{i=1}^{n} x_i \prod_{i=1}^{n} y_i \tag{3.31}$$

∘

3.3 Powers

Products of numbers with themselves can be abbreviated using power notation.

Definition 3.3 (Power)**.** For $a \in \mathbb{R}$ and $n \in \mathbb{N}^0$, the *n-th power of a* is defined by

$$a^0 := 1 \text{ and } a^{n+1} := a^n \cdot a. \tag{3.32}$$

Furthermore, for $a \in \mathbb{R} \setminus \{0\}$ and $n \in \mathbb{N}^0$, the *negative n-th power of a* is defined by

$$a^{-n} := (a^n)^{-1} := \frac{1}{a^n}. \tag{3.33}$$

a is called the *base*, and n is called the *exponent*.

•

The way a^{n+1} is defined by referring back to the power a^n in the above definition is called *recursive*. The definition $a^0 := 1$ is called the *initial case* of the recursion; it is what makes the recursive definition of a^{n+1} possible in the first place. The definition $a^{n+1} := a^n \cdot a$ is also called the *recursion step*. The following calculation rules simplify computations with powers.

Theorem 3.3 (Calculation rules for powers)**.** *For $a, b \in \mathbb{R}$ and $n, m \in \mathbb{Z}$ with $a \neq 0$ for negative exponents, the following calculation rules hold:*

$$a^n a^m = a^{n+m} \tag{3.34}$$
$$(a^n)^m = a^{nm} \tag{3.35}$$
$$(ab)^n = a^n b^n \tag{3.36}$$

∘

Instead of a proof, we consider the following three examples.

(1)

$$2^2 \cdot 2^3 = (2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2^5 = 2^{2+3} \tag{3.37}$$

(2)

$$(3^2)^3 = (3 \cdot 3)^3 = (3 \cdot 3) \cdot (3 \cdot 3) \cdot (3 \cdot 3) = 3^6 = 3^{2 \cdot 3} \tag{3.38}$$

(3)
$$(2 \cdot 4)^2 = (2 \cdot 4) \cdot (2 \cdot 4) = (2 \cdot 2) \cdot (4 \cdot 4) = 2^2 \cdot 4^2 \tag{3.39}$$

Closely related to powers is the definition of the n-th root:

Definition 3.4 (n-th root)**.** For $a \in \mathbb{R}_{\geq 0}$ and $n \in \mathbb{N}$, the *n-th root of a* is defined as the nonnegative real number r with
$$r^n = a. \tag{3.40}$$

•

When computing with roots, power notation for roots is often helpful because it allows the calculation rules for powers to be applied directly.

Theorem 3.4 (Power notation for the n-th root)**.** *Let $a \in \mathbb{R}_{\geq 0}$, let $n \in \mathbb{N}$, and let r be the n-th root of a. Then*
$$r = a^{\frac{1}{n}} \tag{3.41}$$

∘

Proof. We have
$$\left(a^{\frac{1}{n}}\right)^n = a^{\frac{1}{n}} \cdot a^{\frac{1}{n}} \cdot \cdots \cdot a^{\frac{1}{n}} = a^{\sum_{i=1}^n \frac{1}{n}} = a^1 = a. \tag{3.42}$$
Thus, by Definition 3.4, $r = a^{\frac{1}{n}}$. □

Computing with square roots is greatly facilitated by the power notation
$$\sqrt{x} = x^{\frac{1}{2}} \tag{3.43}$$

For example,
$$\frac{2\pi}{\sqrt{2\pi}} = \frac{2\pi}{(2\pi)^{\frac{1}{2}}} = (2\pi)^1 \cdot (2\pi)^{-\frac{1}{2}} = (2\pi)^{1-\frac{1}{2}} = (2\pi)^{\frac{1}{2}} = \sqrt{2\pi}. \tag{3.44}$$

The following relation between square, root, and absolute value is used quite often.

Theorem 3.5 (Root and absolute value)**.** *Let $x \in \mathbb{R}$, and let*
$$|x| := \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases} \tag{3.45}$$
denote the absolute value of x. *Then*
$$\sqrt{x^2} = |x|. \tag{3.46}$$

∘

Proof. We consider the cases $x \geq 0$ and $x < 0$. First, let $x \geq 0$. Then
$$x \geq 0 \Rightarrow x^2 \geq 0 \Rightarrow \sqrt{x^2} = x = |x|. \tag{3.47}$$
Now let $x < 0$. Then
$$x < 0 \Rightarrow x^2 > 0 \Rightarrow \sqrt{x^2} = -x = |x|. \tag{3.48}$$
□

4 Functions

Alongside sets, functions are the second foundational pillar of modern mathematics. In this unit, we define the concept of a function, introduce first properties of functions, and give an overview of several elementary functions.

4.1 Definition and properties

Definition 4.1 (Function)**.** A *function* or *mapping* f is an assignment rule that assigns exactly one element of a set Z to each element of a set D. D is called the *domain of* f, and Z is called the *codomain of* f. We write

$$f : D \to Z, x \mapsto f(x), \tag{4.1}$$

where $f : D \to Z$ is read as "the function f maps all elements of the set D uniquely to elements in Z" and $x \mapsto f(x)$ is read as "x, which is an element of D, is mapped by the function f to $f(x)$, where $f(x)$ is an element of Z." The arrow $\to$ denotes the mapping between the sets D and Z, while the arrow $\mapsto$ denotes the mapping between an element of D and an element of Z.

•

It is essential to distinguish between the *function* f as an assignment rule and a *value of the function* $f(x)$ as an element of Z. x is the *argument* of the function (the function's *input*), and $f(x)$ is the value that the function f takes for the argument x (the function's *output*). Usually, the definition of a function specifies, after $f(x)$, the *functional form of* f, that is, a rule for forming the value $f(x)$ from x. For example, in the following definition of a function

$$f : \mathbb{R} \to \mathbb{R}_{\geq 0}, x \mapsto f(x) := x^2 \tag{4.2}$$

the definition of a power is used. Note that functions are always *unique* in the sense that, whenever the function is applied, they always assign one and the same $f(x) \in Z$ to each $x \in D$.

Figure 4.1 visualizes several aspects of the definition of a function. Figure 4.1 A first illustrates the central terms of the definition. Figure 4.1 B shows a first example of a function in which the function values assigned to the arguments are determined not by a computational rule but by direct definition. Figure 4.1 C and Figure 4.1 D use the same pictorial language to emphasize two aspects of the definition by means of counterexamples: the drawing in Figure 4.1 C is not the representation of a function, because by Definition 4.1 a function assigns *each element* of a set D exactly one element of a codomain Z. The sketched assignment rule does not assign any element in Z to the element $2 \in D$, and therefore it is not a function. Similarly, according to Definition 4.1, a function assigns exactly one element of a codomain Z to each element of a set D. The assignment rule

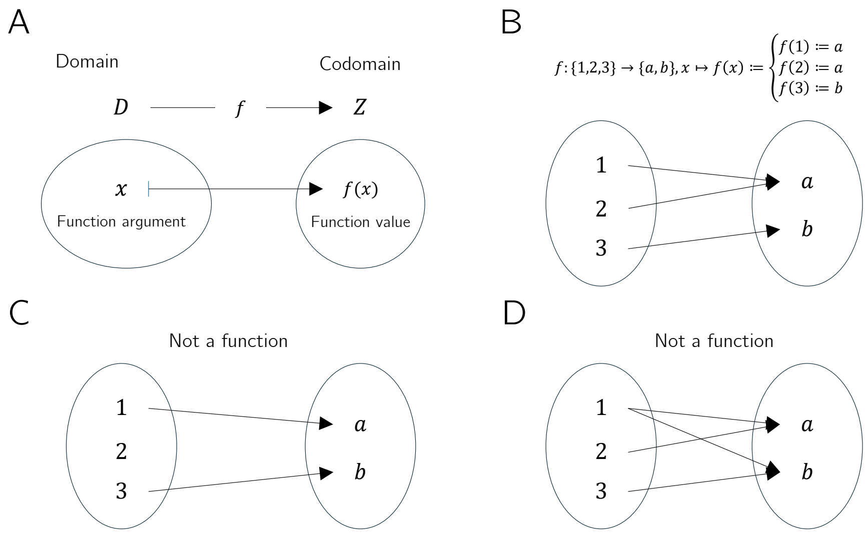

Figure 4.1 Aspects of the definition of a function

sketched in Figure 4.1 D assigns both the element $a \in Z$ and the element $b \in Z$ to $1 \in D$, and therefore it is incompatible with Definition 4.1.

Thus, functions put elements of sets into relation with one another. The sets of these elements receive special names.

Definition 4.2 (Image, range, preimage set, preimage)**.** Let $f : D \to Z, x \mapsto f(x)$ be a function and let $D' \subseteq D$ and $Z' \subseteq Z$. The set

$$f(D') := \{z \in Z | \text{there exists an } x \in D' \text{ with } z = f(x)\} \tag{4.3}$$

is called the *image of* D', and $f(D) \subseteq Z$ is called the *range* of f. Furthermore, the set

$$f^{-1}(Z') := \{x \in D | f(x) \in Z'\} \tag{4.4}$$

is called the *preimage set of* Z'. An $x \in D$ with $z = f(x) \in Z$ is called a *preimage of* z.

•

Note that the range $f(D)$ of f and the codomain Z of f need not be identical.

Example

To clarify the concepts introduced in Definition 4.2, we consider the function shown in Figure 4.2 A,

$$f : \{1, 2, 3, 4, 5\} \to \{a, b, c, d\}, x \mapsto f(x) := \begin{cases} f(1) := b \\ f(2) := d \\ f(3) := c \\ f(4) := c \\ f(5) := d \end{cases}. \tag{4.5}$$

According to Definition 4.2, an image is always defined with respect to a subset D' of the domain D. Thus, let $D' := \{2, 3\} \subset D$, as shown in Figure 4.2 B. Then the image of D' is the set of those $z \in Z$ for which there exists an $x \in D'$ such that $z = f(x)$. Here, the relevant $z \in Z$ for which such an $x \in D'$ exists are precisely $c, d \in Z$, because $f(2) = d$ and $f(3) = c$. Beyond these, there is no $z \in Z$ with $f(x) = z$ and $x \in \{2, 3\}$. By Definition 4.2, the range $f(D)$ is the subset of those $z \in Z$ for which there exists an $x \in D$ such that $z = f(x)$. This is true for all elements of Z except for $a \in Z$, because there is no $x \in D$ with $a = f(x)$. For the function considered here, the range of f is therefore given by $f(D) := \{b, c, d\}$.

Conversely, according to Definition 4.2, a preimage set is always defined with respect to a subset Z' of the codomain Z. Thus, let $Z' := \{c, d\} \subset Z$, as shown in Figure 4.2 C. Then the preimage set of Z' is the set of those $x \in D$ for which $f(x) \in Z'$. The elements of D whose function values under f are elements of Z' are precisely $\{2, 3, 4, 5\}$. By contrast, $1 \in D$ is not an element of this preimage set, because f maps $1 \in D$ to $b \in Z$ and $b \notin Z'$. Nevertheless, $1 \in D$ is of course a preimage of b.

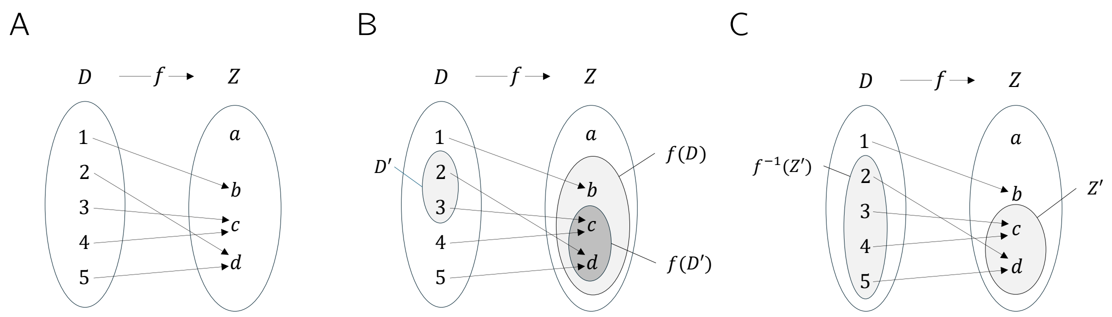

Figure 4.2 (A) Visualization of the function Equation 4.5. (B) Range and an image of f. (C) Preimage set of a set Z' under f

Basic properties of functions are named in the following definition.

Definition 4.3 (Injectivity, surjectivity, bijectivity)**.** Let $f : D \to Z, x \mapsto f(x)$ be a function. f is called *injective* if, for every image element $z \in f(D)$, there is exactly one preimage $x \in D$. Equivalently, f is injective if $x_1, x_2 \in D$ and $x_1 \neq x_2$ imply $f(x_1) \neq f(x_2)$. f is called *surjective* if $f(D) = Z$, that is, if every element of the codomain Z has a preimage in the domain D. Finally, f is called *bijective* if it is both injective and surjective. Bijective functions are also called *one-to-one correspondences* or *one-to-one mappings*.

•

Examples

We first illustrate Definition 4.3 using three (counter)examples in Figure 4.3. Figure 4.3 A shows the *non-injective* function

$$f : \{1, 2, 3\} \to \{a, b\}, x \mapsto f(x) := \begin{cases} f(1) := a \\ f(2) := a \\ f(3) := b \end{cases} . \tag{4.6}$$

The function is non-injective because the element a in the range of f has more than one preimage in the domain of f, namely the elements 1 and 2. Figure 4.3 B shows the *non-surjective* function

$$g : \{1,2,3\} \to \{a,b,c,d\}, x \mapsto g(x) := \begin{cases} g(1) := a \\ g(2) := b \\ g(3) := d \end{cases} . \tag{4.7}$$

The function is non-surjective because the element c in the codomain of g has no preimage in the domain of g. Finally, Figure 4.3 C shows the bijective function

$$h : \{1,2,3\} \to \{a,b,c\}, x \mapsto h(x) := \begin{cases} h(1) := a \\ h(2) := b \\ h(3) := c \end{cases} . \tag{4.8}$$

For *every* element in the codomain of h there is *exactly one* preimage, so the function is injective and surjective and therefore bijective.

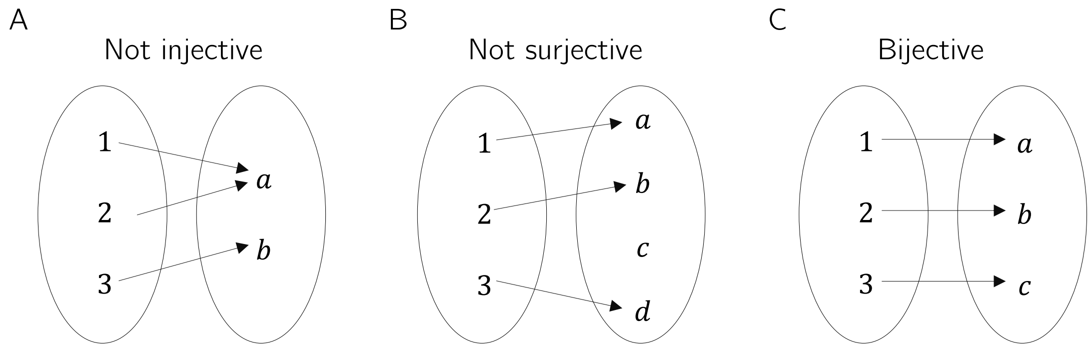

Figure 4.3 Injectivity, surjectivity, bijectivity.

As a further example, consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \tag{4.9}$$

This function is not injective because, for example, with $x_1 = 2 \neq -2 = x_2$ we have $f(x_1) = 2^2 = 4 = (-2)^2 = f(x_2)$. Moreover, f is not surjective because, for example, $-1 \in \mathbb{R}$ has no preimage under f. However, if the domain of f is restricted to the non-negative real numbers, that is, if we define the function

$$\tilde{f} : [0,\infty[\to [0,\infty[, x \mapsto \tilde{f}(x) := x^2, \tag{4.10}$$

then, in contrast to f, $\tilde{f}$ is injective and surjective, hence bijective.

On the uncountability of the real numbers

At this point, we use the concepts of surjectivity and bijectivity of functions to deepen the idea of the *uncountability of the real numbers* a little. This idea says that there are different kinds of infinities, which is certainly not easy to grasp intuitively at first. Moreover, some aspects of probability theory are formulated in ways that accommodate the uncountability of the real numbers, so even a rough understanding of this property can help motivate some of the subtleties of probability theory. Specifically, we briefly sketch *Cantor's diagonal argument* (Cantor (1891), Gray (1994)) for the uncountability of the real numbers. We begin by recording what, against the background of bijective mappings, is meant by a *finite* and a *countably infinite* set.

Definition 4.4. A set M is called *finite* if there exists a bijective mapping of the form

$$f : \mathbb{N}_n \to M \tag{4.11}$$

onto the elements of M. Furthermore, a set M is called *countably infinite* if there exists a bijective mapping of the form

$$f : \mathbb{N} \to M \tag{4.12}$$

onto the elements of M.

•

In the case of a finite set, each element of the set can thus be assigned a natural number with maximum value $n < \infty$ in the sense of a bijective mapping. In the case of a countably infinite set, each element of the set can at least be assigned a natural number in the sense of a bijective mapping. The elements of these sets can therefore be counted.

Cantor (1891) showed that this is not true in general for the real numbers: no bijective mapping between the real numbers and the natural numbers can be constructed, and hence there must be more real numbers than natural numbers. More specifically, one can argue that even the interval $[0, 1]$ already contains more real numbers than there are natural numbers, that is, no surjective function from $\mathbb{N}$ to $[0, 1]$ can be constructed.

Cantor's argument has the following form. Suppose that $f : \mathbb{N} \to [0, 1]$ is an injective function represented in tabular form, for example as in Table 4.1.

Table 4.1 Mapping $f : \mathbb{N} \to [0, 1]$. The dots $\cdots$ and $\vdots$ indicate that the table continues indefinitely to the right and downward.

n	$f(n)$										
1	0.	**3**	1	4	1	5	9	2	6	5	$\cdots$
2	0.	8	**9**	4	5	9	7	8	1	0	$\cdots$
3	0.	9	6	**2**	3	2	1	5	8	7	$\cdots$
4	0.	5	3	6	**8**	8	8	9	7	3	$\cdots$
5	0.	7	4	3	8	**1**	8	0	9	7	$\cdots$
$\vdots$	$\vdots$										

Now construct another number in $[0, 1]$ by adding 1 to each of the bold diagonal entries in Table 4.1 whenever the corresponding value is less than 9, and otherwise setting the value to 0. This is shown in Table 4.2.

Table 4.2 Construction of another real number in $[0, 1]$.

n	$f(n)$										
1	0.	**4**	1	4	1	5	9	2	6	5	$\cdots$
2	0.	8	**0**	4	5	9	7	8	1	0	$\cdots$
3	0.	9	6	**3**	3	2	1	5	8	7	$\cdots$
4	0.	5	3	6	**9**	8	8	9	7	3	$\cdots$
5	0.	7	4	3	8	**2**	8	0	9	7	$\cdots$
$\vdots$	$\vdots$										

The number thus constructed,

$$0.40392... \tag{4.13}$$

cannot occur in Table 4.1, because it differs from $f(1)$ in its first decimal place, from $f(2)$ in its second decimal place, from $f(3)$ in its third decimal place, ..., from $f(n)$ in its nth decimal place, and so on indefinitely. Thus there is at least one number in $[0,1]$ to which no natural number can be assigned. The injective mapping f is not surjective and therefore not bijective. Consequently, $[0,1]$ is not countably infinite and is accordingly called *uncountable*.

4.2 Types of functions

By composition, further functions can be formed from given functions.

Definition 4.5 (Composition of functions)**.** Let $f : D \to Z$ and $g : Z \to S$ be two functions, where the range of f is assumed to agree with the domain of g. Then

$$g \circ f : D \to S, x \mapsto (g \circ f)(x) := g(f(x)) \tag{4.14}$$

defines a function called the *composition of f and g*.

•

The notation for composed functions takes a little getting used to. It is important to recognize that $g \circ f$ denotes the composed function and $(g \circ f)(x)$ denotes an element in the codomain of the composed function. Intuitively, when evaluating $(g \circ f)(x)$, one first applies the function f to x and then applies the function g to the element $f(x)$ of Z. This is captured by the functional form $g(f(x))$. For simplicity, the composition of two functions is often denoted by a single letter; for example, one writes $h := g \circ f$ with $h(x) = g(f(x))$. A mild source of confusion can arise when elements in the codomain of f are denoted by y, so that the notation $y = f(x)$ and $h(x) = g(y)$ is used. However, this notation is sometimes needed for notational simplicity. We visualize Definition 4.5 in Figure 4.4.

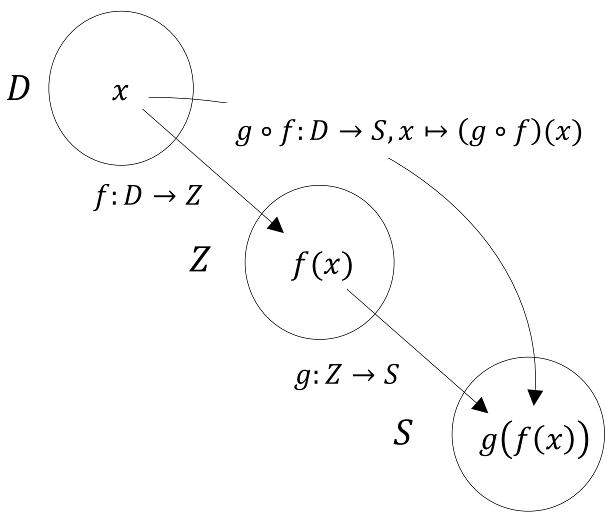

Figure 4.4 Composition of functions.

As an example of the composition of two functions, consider

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := -x^2 \tag{4.15}$$

and

$$g : \mathbb{R} \to \mathbb{R}, x \mapsto g(x) := \exp(x). \tag{4.16}$$

In this case, the composition of f and g is

$$g \circ f : \mathbb{R} \to \mathbb{R}, x \mapsto (g \circ f)(x) := g(f(x)) = \exp\left(-x^2\right). \tag{4.17}$$

A first application of function composition appears in the following definition.

Definition 4.6 (Inverse function)**.** Let $f : D \to Z, x \mapsto f(x)$ be a bijective function. Then the function f^{-1} with

$$f^{-1} \circ f : D \to D, x \mapsto (f^{-1} \circ f)(x) := f^{-1}(f(x)) = x \tag{4.18}$$

is called the *inverse function*, *inverse mapping*, or simply the *inverse of* f.

•

Inverse functions are always bijective. This follows because f is bijective and therefore each $x \in D$ is assigned exactly one $f(x) = z \in Z$. Hence, each $z \in Z$ is also assigned exactly one $x \in D$, namely $f^{-1}(f(x)) = x$. Intuitively, the inverse function of f undoes the effect of f on an element x. We visualize Definition 4.6 in Figure 4.5 A.

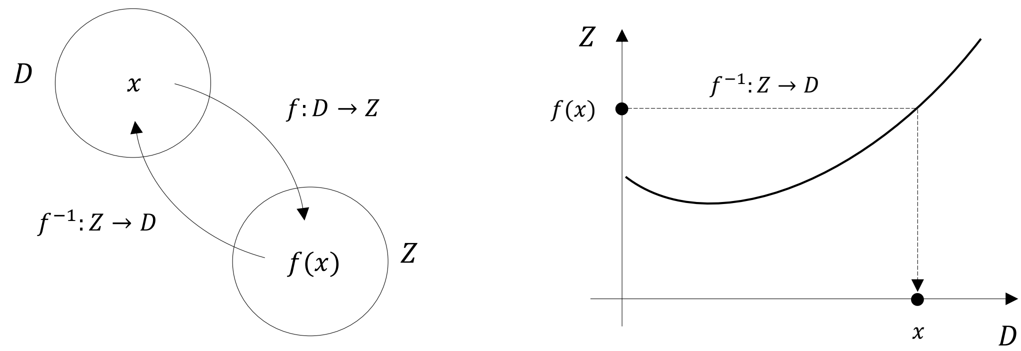

Figure 4.5 Inverse function.

If one considers the graph of a function in a Cartesian coordinate system, then applying a function to a value on the x-axis leads to a value on the y-axis. Applying the inverse function correspondingly leads from a value on the y-axis to a value on the x-axis. We visualize this in Figure 4.5 B. For example, consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := 2x =: y. \tag{4.19}$$

Then the inverse function of f is given by

$$f^{-1} : \mathbb{R} \to \mathbb{R}, y \mapsto f^{-1}(y) := \frac{1}{2}y, \tag{4.20}$$

because for every $x \in \mathbb{R}$,

$$(f^{-1} \circ f)(x) := f^{-1}(f(x)) = f^{-1}(2x) = \frac{1}{2} \cdot 2x = x. \tag{4.21}$$

An important class of functions consists of *linear maps*.

Definition 4.7 (Linear map)**.** A map $f : D \to Z, x \mapsto f(x)$ is called a *linear map* if, for $x, y \in D$ and a scalar c,

$$f(x+y) = f(x) + f(y) \qquad \text{(additivity)}$$

and

$$f(cx) = cf(x) \qquad \text{(homogeneity)}$$

hold. A map for which the above properties do not hold is called a *nonlinear map*.

•

Linear maps are often known as "straight lines." The general definition of linear maps is not completely congruent with this intuition. In particular, linear maps are only those functions that map zero to zero. We show this in the following theorem.

Theorem 4.1 (Linear map of zero)**.** *Let $f : D \to Z$ be a linear map. Then*

$$f(0) = 0. \tag{4.22}$$

∘

Proof. First note that by additivity of f,

$$f(0) = f(0+0) = f(0) + f(0). \tag{4.23}$$

Adding $-f(0)$ to both sides of the above equation gives

$$f(0) - f(0) = f(0) + f(0) - f(0) \Leftrightarrow f(0) = 0 \tag{4.24}$$

and this proves the claim. □

We illustrate the concept of a linear map with two examples.

- For $a \in \mathbb{R}$, the map

  $$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := ax \tag{4.25}$$

  is a linear map because

  $$f(x+y) = a(x+y) = ax+ay = f(x)+f(y) \text{ and } f(cx) = acx = cax = cf(x). \tag{4.26}$$

- For $a, b \in \mathbb{R}$, by contrast, the map

  $$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := ax + b \tag{4.27}$$

  is nonlinear because, for example, for $a := b := 1$,

  $$f(x+y) = 1(x+y) + 1 = x+y+1 \neq x+1+y+1 = f(x) + f(y). \tag{4.28}$$

A map of the form $f(x) := ax + b$ is called a *linear-affine map* or *linear-affine function*. Somewhat imprecisely, functions of the form $f(x) := ax + b$ are sometimes also called *linear functions*.

Besides the types of functions discussed so far, there are many further classes of functions. In the following definition, we classify functions by the dimensionality of their domains and codomains. This type of function classification is often helpful for gaining an initial overview of a mathematical model.

Definition 4.8 (Function types)**.** We distinguish

- *univariate real-valued functions* of the form
$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x), \tag{4.29}$$
- *multivariate real-valued functions* of the form
$$f : \mathbb{R}^n \to \mathbb{R}, x \mapsto f(x) = f(x_1, ..., x_n), \tag{4.30}$$
- and *multivariate vector-valued functions* of the form
$$f : \mathbb{R}^n \to \mathbb{R}^m, x \mapsto f(x) = \begin{pmatrix} f_1(x_1, ..., x_n) \\ \vdots \\ f_m(x_1, ..., x_n) \end{pmatrix}, \tag{4.31}$$
where $f_i, i = 1, ..., m$ are called the *components* or *component functions* of f.

•

In physics, multivariate real-valued functions are called *scalar fields*, and multivariate vector-valued functions are called *vector fields*. In some applications, *matrix-variate matrix-valued functions* also occur.

4.3 Elementary functions

By *elementary functions* we mean a small set of univariate real-valued functions that frequently appear as building blocks of more complex functions. These are the *polynomial functions*, the *exponential function*, the *logarithm function*, and the *gamma function*. In the following, we give the definitions of these functions and state their main properties as theorems, and then present their graphs. For proofs of the properties introduced here, we refer to the advanced literature.

Definition 4.9 (Polynomial functions)**.** A function of the form
$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \sum_{i=0}^{k} a_i x^i = a_0 + a_1 x^1 + a_2 x^2 + \cdots + a_k x^k \tag{4.32}$$
is called a *polynomial function* of degree k with coefficients $a_0, a_1, ..., a_k \in \mathbb{R}$. Typical polynomial functions are listed in Table 4.3.

Table 4.3 Polynomial functions.

Name	Form	Coefficients
Constant function	$f(x) = a$	$a_0 := a,\ a_i := 0, i > 0$
Identity function	$f(x) = x$	$a_0 := 0,\ a_1 := 1,\ a_i := 0, i > 1$
Linear-affine function	$f(x) = ax + b$	$a_0 := b,\ a_1 := a,\ a_i := 0, i > 1$
Square function	$f(x) = x^2$	$a_0 := 0,\ a_1 := 0,\ a_2 := 1,\ a_i := 0, i > 2$

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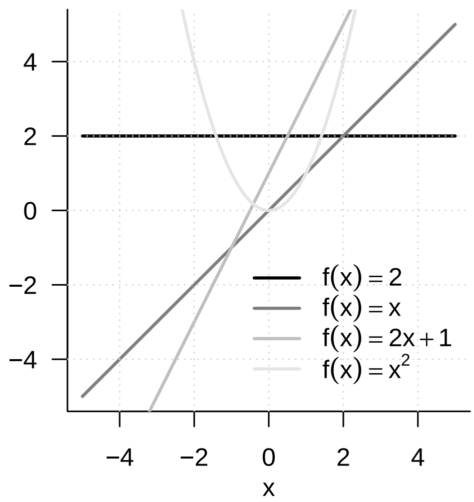

Figure 4.6 Selected polynomial functions

Figure 4.6 shows the graphs of the polynomial functions listed in Definition 4.9.

An important pair of functions consists of the exponential function and the logarithm function.

Theorem 4.2 (Exponential function and its properties)**.** *The* exponential function *is defined as*

$$\exp : \mathbb{R} \to \mathbb{R}, x \mapsto \exp(x) := e^x := \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \tag{4.33}$$

Among others, the exponential function has the properties listed in Table 4.4.

Table 4.4 Properties of the exponential function.

Property	*Meaning*
Range	$x \in]-\infty, 0[\Rightarrow \exp(x) \in]0, 1[$
	$x \in]0, \infty[\quad \Rightarrow \exp(x) \in]1, \infty[$
Monotonicity	$x < y \Rightarrow \exp(x) < \exp(y)$
Special values	$\exp(0) = 1$ *and* $\exp(1) = e$
Summation property	$\exp(x + y) = \exp(x)\exp(y)$
Subtraction property	$\exp(x - y) = \frac{\exp(x)}{\exp(y)}$

∘

In particular, the exponential function only takes positive values and intersects the y-axis at $x = 0$. The number $\exp(1) := e \approx 2.71...$ is called *Euler's number*. Finally, the special values of the exponential function imply

$$\exp(x)\exp(-x) = \exp(x - x) = \exp(0) = 1. \tag{4.34}$$

Theorem 4.3 (Logarithm function and its properties)**.** *The* logarithm function *is defined as the inverse function of the exponential function,*

$$\ln :]0, \infty[\to \mathbb{R}, x \mapsto \ln(x) \text{ with } \ln(\exp(x)) = x \text{ for all } x \in \mathbb{R}. \tag{4.35}$$

Among others, the logarithm function has the properties listed in Table 4.5.

Table 4.5 Properties of the logarithm function.

Property	*Meaning*
Range	$x \in]0, 1[\Rightarrow \ln(x) \in]-\infty, 0[$
	$x \in]1, \infty[\Rightarrow \ln(x) \in]0, \infty[$
Monotonicity	$x < y \Rightarrow \ln(x) < \ln(y)$
Special values	$\ln(1) = 0$ *and* $\ln(e) = 1$
Product property	$\ln(xy) = \ln(x) + \ln(y)$
Power property	$\ln(x^c) = c\ln(x)$
Division property	$\ln\left(\frac{1}{x}\right) = -\ln(x)$

∘

In contrast to the exponential function, the logarithm function takes both negative and positive values. The logarithm function intersects the x-axis at $x = 1$. The product property and the power property are central when calculating with the logarithm function. Intuitively, they can be remembered as: "The logarithm function turns products into sums and powers into products." The graphs of the exponential and logarithm functions are shown in Figure 4.7.

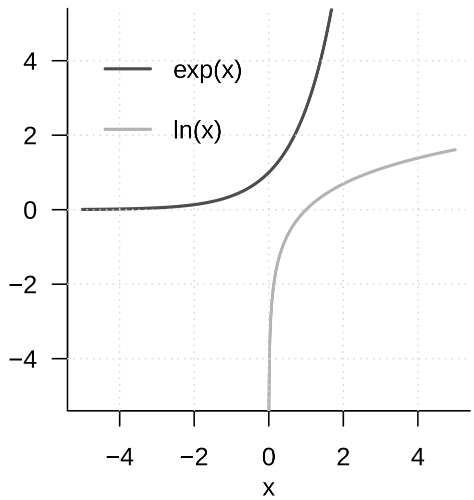

Figure 4.7 Exponential function and logarithm function

A frequent companion in probability theory is the *gamma function*.

Definition 4.10 (Gamma function)**.** The *gamma function* is defined by

$$\Gamma : \mathbb{R} \to \mathbb{R}, x \mapsto \Gamma(x) := \int_0^\infty \xi^{x-1} \exp(-\xi)\, d\xi \tag{4.36}$$

Among others, the gamma function has the properties listed in Table 4.6.

Table 4.6 Properties of the gamma function.

Property	Meaning
Special values	$\Gamma(1) = 1$
	$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
	$\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$
Recursion property	For $x > 0$, $\Gamma(x+1) = x\Gamma(x)$

•

A section of the graph of the gamma function is shown in Figure 4.8.

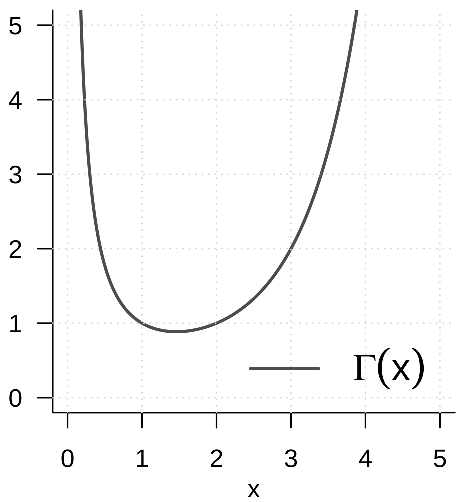

Figure 4.8 Gamma function

5 Sequences, limits, continuity

The topics treated in this chapter are not central to probabilistic data science but form basic building blocks of real analysis. Because modern probability theory is closely intertwined with analytic approaches, however, they help us understand certain results of probability theory. One example of such a result is the *central limit theorem*. The central limit theorem, in turn, forms the basis for the widely used normal-distribution assumption in probabilistic data science. The foundations considered here therefore indirectly increase our understanding of data-scientific principles. In brief, the central limit theorem is a statement about the *limit function* of a *sequence of functions*, more precisely about a sequence of random variables. Knowledge of the nature of *sequences*, *sequences of functions*, and their *limits* therefore allows an informed entry into the study of the central limit theorem. Furthermore, the topics treated in this chapter provide at least a first entry point for understanding continuity and smoothness of functions, which are important basic concepts in nonlinear optimization, for example when determining parameter estimators of probabilistic models.

5.1 Sequences

We begin with the definition of the concept of a *real sequence*.

Definition 5.1 (Real sequence)**.** A *real sequence* is a function of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n). \tag{5.1}$$

The function values $f(n)$ of a real sequence are usually denoted by x_n and called *sequence terms*. Common notations for sequences are

$$(x_1, x_2, ...) \text{ or } (x_n)_{n=1}^{\infty} \text{ or } (x_n)_{n\in\mathbb{N}} \text{ or } (x_n). \tag{5.2}$$

•

Note that a real sequence, because there are infinitely many natural numbers, always has infinitely many sequence terms. This should be kept in mind especially when using the notation $(x_1, x_2, ...)$. We consider two standard examples of real sequences.

Examples of Real Sequences

(1) Real sequences of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := \left(\frac{1}{n}\right)^{\frac{p}{q}} \text{ with } p, q \in \mathbb{N} \tag{5.3}$$

are called *harmonic sequences*. For $p := q := 1$, a harmonic sequence has the sequence-term form

$$\left(\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \cdots\right). \tag{5.4}$$

(2) Real sequences of the form

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := q^n \text{ with } q \in]-1,1[\tag{5.5}$$

are called *geometric sequences.* For $q := \frac{1}{2}$, a geometric sequence has the sequence-term form

$$\begin{aligned}\left(\left(\frac{1}{2}\right)^1, \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3, ...\right) &= \left(\frac{1^1}{2^1}, \frac{1^2}{2^2}, \frac{1^3}{2^3}, ...\right) \\ &= \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, ...\right).\end{aligned} \tag{5.6}$$

In addition to real sequences, that is, sequences of real numbers, one can also consider sequences of other mathematical objects. An important type of sequence is the *sequence of functions.*

Definition 5.2 (Sequence of functions)**.** Let ϕ be a set of univariate real-valued functions with domain $D \subseteq \mathbb{R}$. Then a sequence of functions is a function of the form

$$F : \mathbb{N} \to \phi, n \mapsto F(n). \tag{5.7}$$

The function values $F(n)$ of a sequence of functions are usually denoted by f_n and called *sequence terms.* Common notations for sequences of functions are

$$(f_1, f_2, ...) \text{ or } (f_n)_{n=1}^{\infty} \text{ or } (f_n)_{n \in \mathbb{N}} \text{ or } (f_n). \tag{5.8}$$

•

The definition of a sequence of functions is evidently analogous to the definition of a real sequence. The difference between a real sequence and a sequence of functions is that the sequence terms of a real sequence are real numbers, whereas the sequence terms of a sequence of functions are univariate real-valued functions. Here, too, we discuss two standard examples.

Examples of Sequences of Functions

(1) We consider the set ϕ of univariate real-valued functions of the form

$$\phi := \{f_n | f_n : [0,1] \to \mathbb{R}, x \mapsto f_n(x) := x^n \text{ for } n \in \mathbb{N}\}. \tag{5.9}$$

Then

$$F : \mathbb{N} \to \phi, n \mapsto F(n) := f_n \tag{5.10}$$

defines a sequence of functions. For the function values of the sequence terms of F, we have

$$f_1(x) := x^1, f_2(x) := x^2, f_3(x) := x^3, ... \tag{5.11}$$

(2) We consider the set ϕ of univariate real-valued functions of the form

$$\phi := \{f_n | f_n : [-a,a] \to \mathbb{R}, x \mapsto f_n(x) := \sum_{k=0}^{n} \frac{x^k}{k!} \text{ for } n \in \mathbb{N}\}. \tag{5.12}$$

Then

$$F : \mathbb{N} \to \phi, n \mapsto F(n) := f_n \tag{5.13}$$

defines a sequence of functions. For the function values of the sequence terms of F, we have

$$f_1(x) := \sum_{k=0}^{1} \frac{x^k}{k!}, f_2(x) := \sum_{k=0}^{2} \frac{x^k}{k!}, f_3(x) := \sum_{k=0}^{3} \frac{x^k}{k!}, ... \tag{5.14}$$

5.2 Limits

When considering the terms of a sequence, one can ask which values a sequence might take when the sequence index n becomes very large, that is, tends to infinity. If, in this case, the sequence terms take very similar values (and do not themselves become infinitely large), one is led to the concept of a *limit* for real sequences and a *limit function* for sequences of functions.

Definition 5.3 (Limit of a real sequence)**.** $x \in \mathbb{R}$ is called the limit of a real sequence $(x_n)_{n=1}^{\infty}$ if, for every $\epsilon > 0$, there exists an $m \in \mathbb{N}$ such that

$$|x_n - x| < \epsilon \text{ for all } n \geq m. \tag{5.15}$$

A sequence that has a limit is called a *convergent sequence*; a sequence that has no limit is called a *divergent sequence*. To express that $x \in \mathbb{R}$ is the limit of the sequence $(x_n)_{n=1}^{\infty}$, one also writes

$$\lim_{n\to\infty} x_n = x \text{ or } x_n \to x \text{ for } n \to \infty \text{ or } x_n \xrightarrow{n\to\infty} x. \tag{5.16}$$

•

According to Definition 5.3, the limit of a sequence may exist, but it need not. For example, the sequence

$$f : \mathbb{N} \to \mathbb{R}, n \mapsto f(n) := n \tag{5.17}$$

has no limit, because here both n and $f(n)$ become infinitely large. For a convergent sequence, that is, a sequence whose limit exists, Definition 5.3 says that, for an arbitrarily small but positive ϵ, an $m \in \mathbb{N}$ can be specified such that, for all sequence terms x_n with $n \geq m$, the distance from the limit in the positive or negative direction is smaller than ϵ. Thus the sequence terms with $n \geq m$ lie arbitrarily close to the limit x. It may often be the case that a smaller ϵ requires a larger m.

Examples

The examples of real sequences considered above, by contrast, have limits.

(1) For the generalized harmonic sequences, with $p, q \in \mathbb{N}$,

$$\lim_{n\to\infty} \left(\frac{1}{n}\right)^{\frac{p}{q}} = 0. \tag{5.18}$$

(2) For the geometric sequences, with $q \in]-1, 1[$,

$$\lim_{n\to\infty} q^n = 0. \tag{5.19}$$

The harmonic and geometric sequences are therefore also called *null sequences*. For general proofs of Equation 5.18 and Equation 5.19, we refer to the advanced literature. In fact, these proofs are not trivial and touch on the basic assumptions about the nature of the real numbers. In Figure 5.1, we visualize the first ten sequence terms as well as the limits of the harmonic sequence for $p := q := 1$ and of the geometric sequence for $q := 1/2$. The sequence terms shown as points evidently lie increasingly closer to the limit shown by a gray line as n increases.

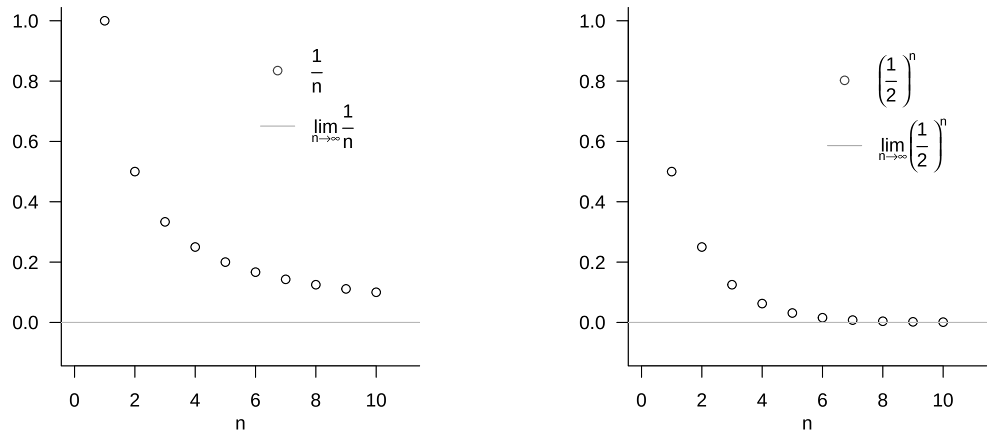

Figure 5.1 Examples of limits of real sequences.

For the special case of the harmonic sequence with $p = q = 1$, we briefly present the argument for the limit

$$\lim_{n\to\infty} \frac{1}{n} = 0. \tag{5.20}$$

According to Definition 5.3, Equation 5.20 holds if and only if, for an arbitrary $\epsilon > 0$ (and in particular for an arbitrarily small one), an $m \in \mathbb{N}$ can be specified such that, for all $n \geq m$,

$$\left|\frac{1}{n} - 0\right| < \epsilon \Leftrightarrow \frac{1}{n} < \epsilon.$$

Thus, using the ceiling function $\lceil \cdot \rceil$, if for an arbitrary $\epsilon > 0$ we set

$$m := \left\lceil \frac{1}{\epsilon} \right\rceil + 1,$$

for example, for $\epsilon := 0.01$ correspondingly $m := \lceil \frac{1}{0.01} \rceil + 1 = 101$, then for all $n \geq m$ we have $\frac{1}{n} < \epsilon$. Since $\epsilon > 0$ was chosen arbitrarily, this construction is possible for all $\epsilon > 0$.

For sequences of functions, one possible extension of the concepts of convergence and limit is the following.

Definition 5.4 (Pointwise convergence and limit function of a sequence of functions)**.** Let $F = (f_n)_{n\in\mathbb{N}}$ be a sequence of functions of univariate real-valued functions with domain D. F is called *pointwise convergent* if the real sequence $(f_n(x))_{n\in\mathbb{N}}$ is a convergent sequence for every $x \in D$, that is, if it has a limit. The function that assigns to each $x \in D$ this limit of $(f_n(x))_{n\in\mathbb{N}}$ is then called the *limit function of the sequence of functions* F and has the form

$$f : D \to \mathbb{R}, x \mapsto f(x) := \lim_{n\to\infty} f_n(x). \tag{5.21}$$

•

Note that the limits of convergent real sequences are real numbers, whereas the limit functions of pointwise convergent sequences of functions are functions. In addition to pointwise convergence of sequences of functions, there is the more powerful concept of *uniform convergence* of sequences of functions, for which we refer to the advanced literature. As examples, we consider the limit functions of the sequences of functions discussed above; for proofs, we again refer to the advanced literature.

Examples

(1) We consider the sequence of functions

$$F : \mathbb{N} \to \phi, n \mapsto F(n) \tag{5.22}$$

with

$$\phi := \{f_n | f_n : [0,1] \to \mathbb{R}, x \mapsto f_n(x) := x^n \text{ for } n \in \mathbb{N}\}. \tag{5.23}$$

Then F is pointwise convergent with limit function

$$f : [0,1] \to \mathbb{R}, x \mapsto f(x) := \begin{cases} 0, & \text{for } x \in [0,1[\\ 1, & \text{for } x = 1 \end{cases} \tag{5.24}$$

because $f_n(x) := x^n$ is a geometric sequence, and thus a null sequence, for $x \in [0,1[$, while $f_n(x) := x^n$ is a constant sequence for $x = 1$, all of whose sequence terms have distance 0 from 1. The sequence of functions F therefore converges to a function that is equal to zero on the entire interval $[0,1]$, except at the point 1. This function evidently has a jump.

(2) We consider the sequence of functions

$$F : \mathbb{N} \to \phi, n \mapsto F(n) \tag{5.25}$$

with

$$\phi := \{f_n | f_n : [-a,a] \to \mathbb{R}, x \mapsto f_n(x) := \sum_{k=0}^{n} \frac{x^k}{k!} \text{ for } n \in \mathbb{N}\}. \tag{5.26}$$

Then F is pointwise convergent with limit function

$$f : [-a,a] \to \mathbb{R}, x \mapsto f(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!} =: \exp(x). \tag{5.27}$$

The sequence of functions F therefore converges to the exponential function on $[-a,a]$. Conversely, the exponential function is defined precisely by

$$\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}. \tag{5.28}$$

5.3 Continuity

In this section, we approach the concept of the *continuity* of a function. Intuitively, a function is continuous if it has no jumps or, equivalently, if small changes in its arguments always lead only to small changes in its function values (and therefore precisely to no jumps). To define continuity, we first need the concept of the *limit of a function.*

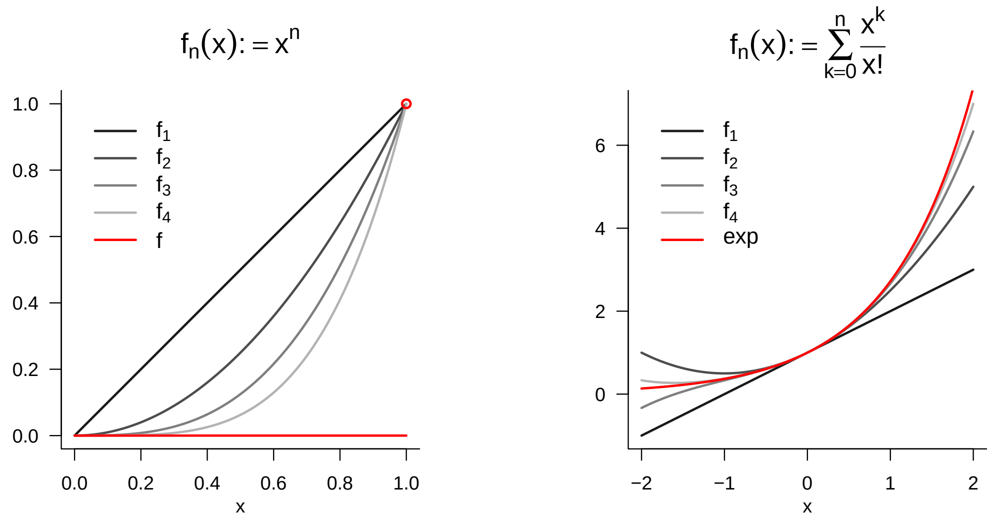

Figure 5.2 Examples of limits of sequences of functions.

Definition 5.5 (Limit of a function)**.**

For $D \subseteq \mathbb{R}$ and $Z \subseteq \mathbb{R}$, let $f : D \to Z, x \mapsto f(x)$ be a function, and let $a, b \in \mathbb{R}$. b is called the *limit of the function* f *as* x *tends to* a if

(1) there exists a real sequence $(x_n)_{n=1}^{\infty}$ with sequence terms in D and limit a, that is, $\lim_{n\to\infty} x_n = a$, and

(2) for every such sequence, b is the limit of the sequence of function values $f(x_n)$ of the sequence terms of $(x_n)_{n=1}^{\infty}$, that is, $\lim_{n\to\infty} f(x_n) = b$.

If b is the limit of the function f as x tends to a, one also writes $\lim_{x\to a} f(x) = b$.

•

In Figure 5.3, we visualize the limit of the exponential function at $a = 1$ by representing sequence terms $x_n \to 1$ as points parallel to the x-axis and the corresponding sequence terms $f(x_n)$ on the function graph. Evidently, $\lim_{x\to 1} \exp(x) = e \approx 2.71$.

We can now define the concept of continuity of a function.

Definition 5.6 (Continuity of a function)**.** A function $f : D \to Z$ with $D \subseteq \mathbb{R}$ and $Z \subseteq \mathbb{R}$ is called *continuous at* $a \in D$ if

$$\lim_{x\to a} f(x) = f(a). \tag{5.29}$$

If f is continuous at every $x \in D$, then f *is continuous on* D.

•

Note that, for a function continuous at a, it follows that

$$\lim_{x\to a} f(x) = f\left(\lim_{x\to a} x\right). \tag{5.30}$$

For continuous functions, taking limits and evaluating the function can therefore be interchanged.

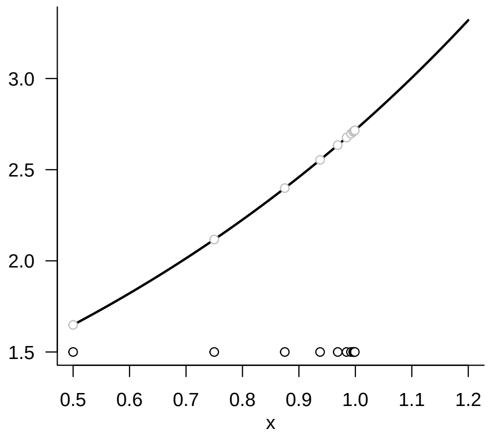

Figure 5.3 Example of the limit of a function. The sequence of x-values is defined here as $x_n := 1-(1/2)^n$ for $n = 1, 2, \ldots$.

6 Differential calculus

Differential calculus is concerned with the change of functions. On the one hand, it provides the basis for the mathematical modelling of dynamic systems by means of *differential equations*, that is, the description of functions in terms of their rates of change. On the other hand, differential calculus provides the basis of *optimization*, that is, of determining extrema of functions. In Section 6.1 we first introduce the concept of the derivative and elementary calculation rules associated with it. In Section 6.2 we then turn to the question of how derivatives can be used to determine extrema of functions.

6.1 Definitions and calculation rules

We begin with the following definition.

Definition 6.1 (Differentiability and derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval and let

$$f : I \to \mathbb{R}, x \mapsto f(x) \tag{6.1}$$

be a univariate real-valued function. f is called *differentiable* at $a \in I$ if the limit

$$f'(a) := \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{6.2}$$

exists. $f'(a)$ is then called the *derivative of f at the point a*. If f is differentiable for all $x \in I$, then f is called *differentiable*, and the function

$$f' : I \to \mathbb{R}, x \mapsto f'(x) \tag{6.3}$$

is called the *derivative of f*.

•

For $h > 0$, the expression

$$\frac{f(a+h) - f(a)}{h} \tag{6.4}$$

is called the *Newton difference quotient*. As shown in Figure 6.1, the Newton difference quotient measures the change $f(a+h) - f(a)$ of f on the y-axis per distance h on the x-axis. If, for example, $f(a)$ and $f(a+h)$ represent the position of an object at a time a and at a later time $a + h$, then $f(a + h) - f(a)$ is the distance travelled by this object in the time h, that is, its average velocity over the time interval h. For $h \to 0$, the Newton difference quotient then measures the instantaneous rate of change of f at a, in this example the velocity of the object at time a.

From a mathematical point of view, it is important to distinguish between the symbols $f'(a)$ and f' in the definition of the derivative. As usual, $f'(a)$ denotes the value of a

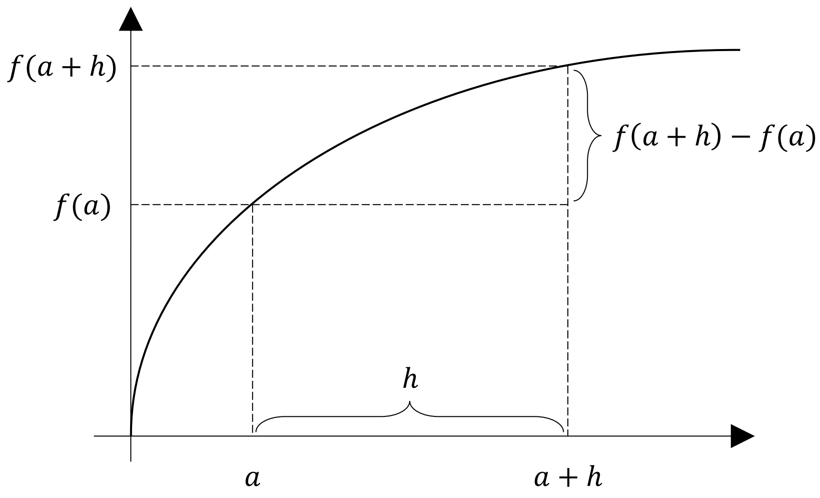

Figure 6.1 Elements of the Newton difference quotient.

function, that is, a number. In contrast, f' denotes a function, namely the function whose values are given by $f'(a)$ for all $a \in \mathbb{R}$.

Several historically established notations for derivatives exist in the literature; all of them represent the same concept of the derivative.

Definition 6.2 (Notation for derivatives of univariate real-valued functions)**.** Let f be a univariate real-valued function. Equivalent notations for the derivative of f and the derivative of f at a point x are

- the *Lagrange notation* f' and $f'(x)$,
- the *Leibniz notation* $\frac{df}{dx}$ and $\frac{df(x)}{dx}$,
- the *Newton notation* $\dot{f}$ and $\dot{f}(x)$,
- the *Euler notation* Df and $Df(x)$.

•

In the following, for univariate real-valued functions we will mainly use the Lagrange notation f' and $f'(x)$ as labels. In calculations, we also use an adapted form of Leibniz notation and understand the expression $\frac{d}{dx}f(x)$ as the instruction to compute the derivative of f. Newton notation is used mainly when the function argument represents time and is then usually denoted by t for *time*. Accordingly, $\dot{f}(t)$ denotes the rate of change of f at time t. Euler notation is particularly useful in the context of multivariate real-valued or vector-valued functions.

We assume a number of derivatives of elementary functions to be known; these are summarized in Table 6.1. For proofs, we refer to the advanced literature.

Table 6.1 Derivatives of elementary functions

Name	Definition	Derivative
Polynomial function	$f(x) := \sum_{i=0}^{k} a_i x^i$	$f'(x) = \sum_{i=1}^{k} i a_i x^{i-1}$
Constant function	$f(x) := a$	$f'(x) = 0$
Identity function	$f(x) := x$	$f'(x) = 1$
Linear-affine function	$f(x) := ax + b$	$f'(x) = a$
Square function	$f(x) := x^2$	$f'(x) = 2x$
Exponential function	$f(x) := \exp(x)$	$f'(x) = \exp(x)$
Logarithm function	$f(x) := \ln(x)$	$f'(x) = \frac{1}{x}$

In Figure 6.2 we visualize the identity function, a linear function, and the square function together with their respective derivatives. In Figure 6.3 we visualize the exponential and logarithm functions together with their respective derivatives.

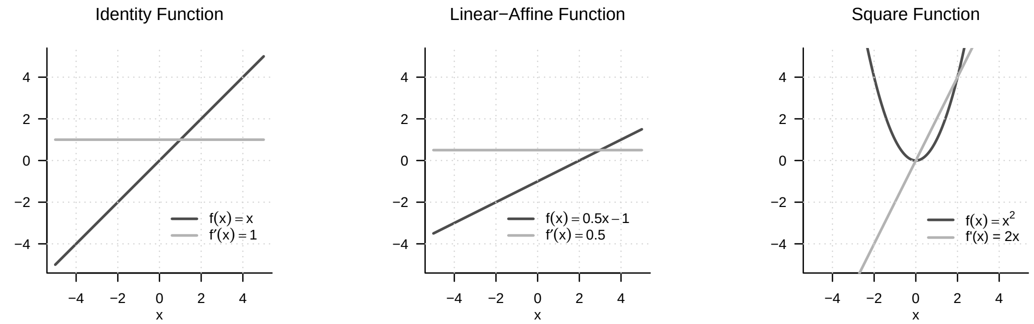

Figure 6.2 Derivatives of three elementary functions. The derivative of the identity function is the constant function with value 1, the derivative of the linear-affine function considered here is the constant function with value 0.5, and the derivative of the square function is the linear-affine function with $a = 2$ and $b = 0$.

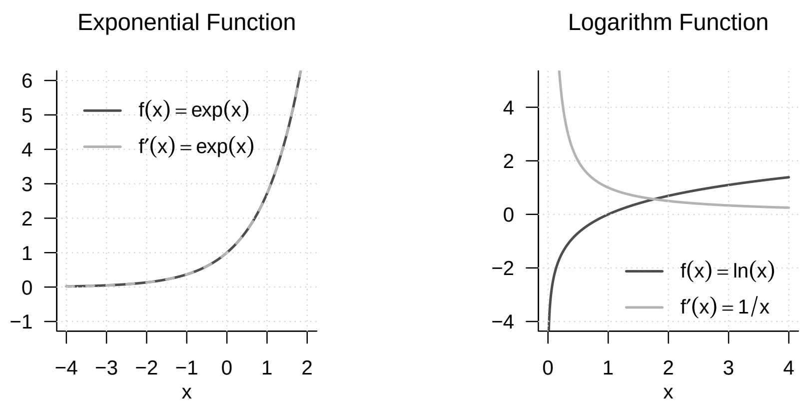

Figure 6.3 Derivatives of the exponential function and the logarithm function. The derivative of the exponential function is the exponential function itself; the derivative of the logarithm function has function values $1/x$.

Based on the definition of the derivative of a univariate real-valued function, further derivatives of such a function can be defined easily.

Definition 6.3 (Higher derivatives)**.** Let f be a univariate real-valued function and let

$$f^{(1)} := f' \tag{6.5}$$

be the derivative of f. The kth derivative of f is defined recursively by

$$f^{(k)} := \left(f^{(k-1)}\right)' \text{ for } k > 1, \tag{6.6}$$

assuming that $f^{(k-1)}$ is differentiable. In particular, the *second derivative of* f is defined as the derivative of f', that is,

$$f'' := (f')'. \tag{6.7}$$

•

In analogy to the above, in calculations we also write $\frac{d^2}{dx^2}f(x)$ for the instruction to determine the second derivative of a function f. The zeroth derivative $f^{(0)}$ of f is f itself. By tradition and for simplicity, for $k < 4$ one usually writes f', f'', and f''' according to Lagrange notation instead of $f^{(1)}, f^{(2)}$, and $f^{(3)}$.

Example

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2. \tag{6.8}$$

Then

$$f^{(1)}(x) = f'(x) = \frac{d}{dx}\left(x^2\right) = 2x. \tag{6.9}$$

Furthermore,

$$\begin{aligned}
f^{(2)}(x) &= \left(f^{(2-1)}\right)'(x) = \left(f^{(1)}\right)'(x) = \frac{d}{dx}(2x) = 2, \\
f^{(3)}(x) &= \left(f^{(3-1)}\right)'(x) = \left(f^{(2)}\right)'(x) = \frac{d}{dx}(2) = 0, \\
f^{(4)}(x) &= \left(f^{(4-1)}\right)'(x) = \left(f^{(3)}\right)'(x) = \frac{d}{dx}(0) = 0.
\end{aligned} \tag{6.10}$$

Thus, throughout this calculation, one computes only first derivatives.

A number of calculation rules are helpful for determining the derivative of a function because they allow the derivative of a function to be derived from the derivatives of its subfunctions. For proofs of the calculation rules introduced in the following theorem, we refer to the advanced literature.

Theorem 6.1 (Calculation rules for derivatives)**.** *For $i = 1, ..., n$, let g_i be real-valued univariate differentiable functions. Then the following calculation rules hold:*

(1) Sum rule

$$\textit{For } f(x) := \sum_{i=1}^{n} g_i(x),\ f'(x) = \sum_{i=1}^{n} g_i'(x). \tag{6.11}$$

(2) Product rule

$$\textit{For } f(x) := g_1(x)g_2(x),\ f'(x) = g_1'(x)g_2(x) + g_1(x)g_2'(x). \tag{6.12}$$

(3) Quotient rule

$$\text{For } f(x) := \frac{g_1(x)}{g_2(x)},\ f'(x) = \frac{g_1'(x)g_2(x) - g_1(x)g_2'(x)}{g_2^2(x)}. \tag{6.13}$$

(4) Chain rule

$$\text{For } f(x) := g_1(g_2(x)),\ f'(x) = g_1'(g_2(x))g_2'(x). \tag{6.14}$$

$\circ$

Examples

(1) Sum rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := 4x^3 + 3x^2. \tag{6.15}$$

Then f has the form

$$f(x) = \sum_{i=1}^{2} g_i(x) = g_1(x) + g_2(x) \text{ with } g_1(x) := 4x^3 \text{ and } g_2(x) := 3x^2. \tag{6.16}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(4x^3\right) = 12x^2 \text{ and } g_2'(x) = \frac{d}{dx}\left(3x^2\right) = 6x. \tag{6.17}$$

Thus, by the sum rule,

$$f'(x) = \sum_{i=1}^{2} g_i'(x) = g_1'(x) + g_2'(x) = 12x^2 + 6x. \tag{6.18}$$

(2) Product rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \sin(x). \tag{6.19}$$

Then f has the form

$$f(x) = g_1(x)g_2(x) \text{ with } g_1(x) := x^2 \text{ and } g_2(x) := \sin(x). \tag{6.20}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(x^2\right) = 2x \text{ and } g_2'(x) = \frac{d}{dx}\left(\sin x\right) = \cos(x). \tag{6.21}$$

Thus, by the product rule,

$$f'(x) = g_1'(x)g_2(x) + g_1(x)g_2'(x) = 2x\sin(x) + x^2\cos(x). \tag{6.22}$$

(3) Quotient rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \frac{x^2}{x+1}. \tag{6.23}$$

Then f has the form

$$f(x) = \frac{g_1(x)}{g_2(x)} \text{ with } g_1(x) := x^2 \text{ and } g_2(x) := x + 1. \tag{6.24}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(x^2\right) = 2x \text{ and } g_2'(x) = \frac{d}{dx}\left(1 + x\right) = 1. \tag{6.25}$$

Thus, by the quotient rule,

$$f'(x) = \frac{g_1'(x)g_2(x) - g_1(x)g_2'(x)}{g_2^2(x)} = \frac{2x(x+1) - x^2}{(x+1)^2} = \frac{2x^2 + 2x - x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}. \tag{6.26}$$

(4) Chain rule

Let

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \exp\left(-x^2\right). \tag{6.27}$$

Then f has the form

$$f(x) = g_1(g_2(x)) \text{ with } g_1(x) := \exp(x) \text{ and } g_2(x) := -x^2. \tag{6.28}$$

Moreover,

$$g_1'(x) = \frac{d}{dx}\left(\exp(x)\right) = \exp(x) \text{ and } g_2'(x) = \frac{d}{dx}\left(-x^2\right) = -2x. \tag{6.29}$$

Thus, by the chain rule,

$$f'(x) = g_1'(g_2(x))g_2'(x) = \exp\left(-x^2\right)(-2x) = -2x\exp\left(-x^2\right). \tag{6.30}$$

One can therefore remember the derivative resulting from the chain rule as: "The derivative of the outer function at the value of the inner function times the derivative of the inner function."

6.2 Analytic optimization

An important application of differential calculus is the determination of extrema of functions. At its core, this concerns the question for which values in its domain a function assumes a maximum or a minimum. For simple functions this is possible analytically. The general procedure is often also known under the keyword "curve sketching". In applications, an analytic approach to optimizing functions is usually not possible, and numerical algorithms are used to determine extrema. Understanding these algorithms, however, presupposes an understanding of the principles of analytic optimization. In this section, we give an introduction to analytic optimization of univariate real-valued functions. We begin by making precise the concepts of maxima and minima of univariate real-valued functions mentioned above.

Definition 6.4 (Extreme points and extreme values)**.** Let $U \subseteq \mathbb{R}$ and let $f : U \to \mathbb{R}$ be a univariate real-valued function. f has at the point $x_0 \in U$

- a *local minimum* if there is an interval $I :=]a,b[$ with $x_0 \in]a,b[$ and

$$f(x_0) \le f(x) \text{ for all } x \in I \cap U, \tag{6.31}$$

- a *global minimum* if

$$f(x_0) \le f(x) \text{ for all } x \in U, \tag{6.32}$$

- a *local maximum* if there is an interval $I :=]a,b[$ with $x_0 \in]a,b[$ and

$$f(x_0) \ge f(x) \text{ for all } x \in I \cap U, \tag{6.33}$$

- a *global maximum* if

$$f(x_0) \ge f(x) \text{ for all } x \in U. \tag{6.34}$$

The value $x_0 \in U$ of the domain of f is called, respectively, a *local* or *global minimizer* or *maximizer*, and the function value $f(x_0) \in \mathbb{R}$ is called, respectively, a *local* or *global minimum* or *maximum*. In general, the value $x_0 \in U$ is called an *extreme point*, and the function value $f(x_0) \in \mathbb{R}$ is called an *extreme value*.

•

Extreme points of functions are often denoted by

$$\underset{x \in I \cap U}{\operatorname{argmin}} f(x) \text{ or } \underset{x \in I \cap U}{\operatorname{argmax}} f(x), \tag{6.35}$$

and extreme values of functions are often denoted by

$$\min_{x \in I \cap U} f(x) \text{ or } \max_{x \in I \cap U} f(x). \tag{6.36}$$

Figure 6.4 visualizes the different types of extreme values according to Definition 6.4.

Analytic optimization of univariate real-valued functions is based on the so-called *necessary* and *sufficient conditions for extrema*. The former makes a statement about the behavior of the first derivative of a function at an extreme point; the latter makes a statement about the behavior of a function at a point that satisfies certain requirements on its first and second derivatives.

Theorem 6.2 (Necessary condition for extrema)**.** *Let f be a univariate real-valued function that is differentiable in a neighborhood of x_0. If x_0 is an interior extreme point of f, then*

$$x_0 \textit{ is an interior extreme point of } f \Rightarrow f'(x_0) = 0. \tag{6.37}$$

∘

If x_0 is an interior extreme point of f, then the first derivative of f at x_0 is therefore equal to zero. Instead of giving a proof, let us reason that, for example, at a local maximizer x_0 of f, f increases to the left of x_0 and decreases to the right of x_0. At x_0, however, f neither increases nor decreases, so it is plausible that $f'(x_0) = 0$.

Theorem 6.3 (Sufficient conditions for local extrema)**.** *Let f be a twice differentiable univariate real-valued function.*

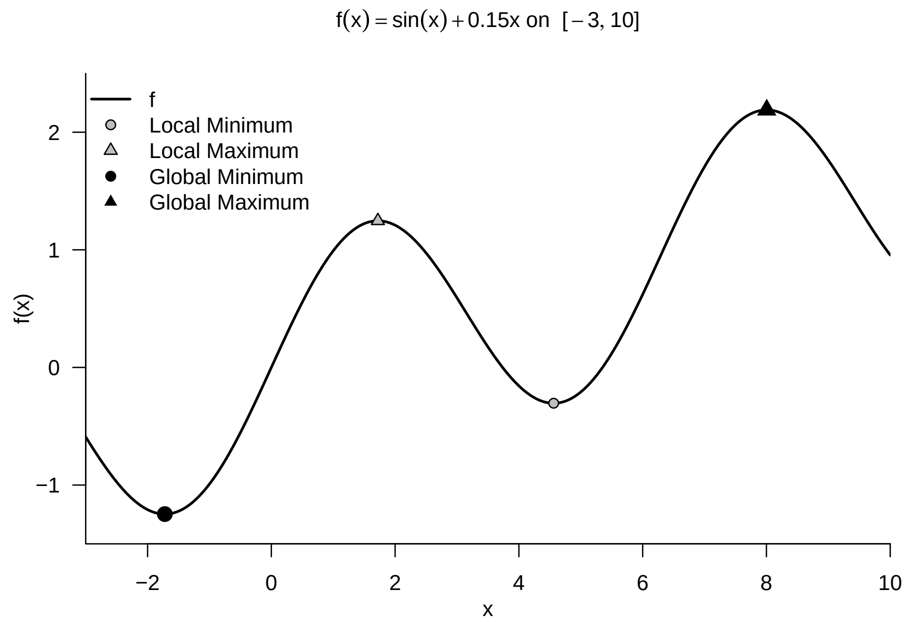

Figure 6.4 Extreme points of a function.

- *If, for $x_0 \in U \subseteq \mathbb{R}$,*
$$f'(x_0) = 0 \text{ and } f''(x_0) > 0 \tag{6.38}$$
holds, then f has a local minimum at the point x_0.
- *If, for $x_0 \in U \subseteq \mathbb{R}$,*
$$f'(x_0) = 0 \text{ and } f''(x_0) < 0 \tag{6.39}$$
holds, then f has a local maximum at the point x_0.

∘

Again we omit a proof and illustrate the condition with the example shown in Figure 6.5. Here, clearly, $x_0 = 1$ is a local minimizer of $f(x) = (x-1)^2$. We can see that to the left of x_0, f decreases, and to the right of x_0, f increases. At x_0, f neither increases nor decreases, so $f'(x_0) = 0$. Moreover, to the left and right of x_0 and at x_0, the change f'' of f' is positive: to the left of x_0, the negativity of f' weakens to 0, and to the right of x_0, the positivity of f' increases.

In particular, the sufficient conditions for local extrema suggest the following *standard procedure* for determining local extreme points.

Theorem 6.4 (Standard procedure of analytic optimization)**.** *Let f be a univariate real-valued function. Local extreme points of f can be identified by the following* standard procedure of analytic optimization*:*

1. *Compute the first and second derivatives of f.*
2. *Determine zeros x^* of f' by solving $f'(x^*) = 0$ for x^*. The zeros of f' are then candidates for extreme points of f.*

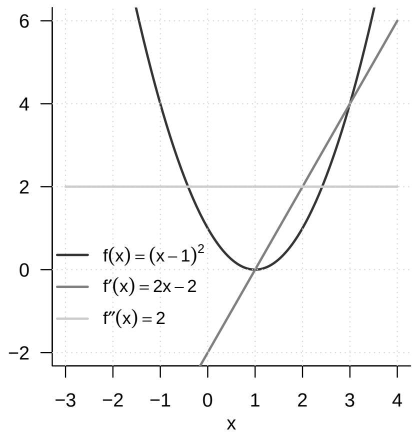

Figure 6.5 Analytic optimization of $f(x) := (x-1)^2$.

3. *Evaluate $f''(x^*)$: If $f''(x^*) > 0$, then x^* is a local minimizer of f; if $f''(x^*) < 0$, then x^* is a local maximizer of f; if $f''(x^*) = 0$, then this criterion does not allow a decision.*

∘

Example

Instead of giving a proof, we consider the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := (x-1)^2 \tag{6.40}$$

as an example. The first derivative of f from Figure 6.5 is obtained by the chain rule as

$$f'(x) = \frac{d}{dx}\left((x-1)^2\right) = 2(x-1) \cdot \frac{d}{dx}(x-1) = 2x - 2. \tag{6.41}$$

The second derivative of f is

$$f''(x) = \frac{d}{dx}f'(x) = \frac{d}{dx}(2x-2) = 2 > 0 \text{ for all } x \in \mathbb{R}. \tag{6.42}$$

Solving $f'(x^*) = 0$ for x^* yields

$$f'(x^*) = 0 \Leftrightarrow 2x^* - 2 = 0 \Leftrightarrow 2x^* = 2 \Leftrightarrow x^* = 1. \tag{6.43}$$

Consequently, $x^* = 1$ is a minimizer of f with associated minimum value $f(1) = 0$.

6.3 Differential calculus of multivariate real-valued functions

We first recall the concept of a multivariate real-valued function.

Definition 6.5 (Multivariate real-valued function)**.** A function of the form

$$f : \mathbb{R}^n \to \mathbb{R}, x \mapsto f(x) = f(x_1, ..., x_n) \tag{6.44}$$

is called a *multivariate real-valued* function.

•

The arguments of multivariate real-valued functions are thus real n-tuples of the form $x := (x_1, ..., x_n)$, whereas their function values are real numbers. An example of a multivariate real-valued function with $n := 2$ is

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2. \tag{6.45}$$

We visualize this function in Figure 6.6. The right-hand panel shows a representation by means of so-called *isocontours*, that is, lines in the domain of the function for which the function assumes identical values. The corresponding values are marked for selected isocontours in the figure.

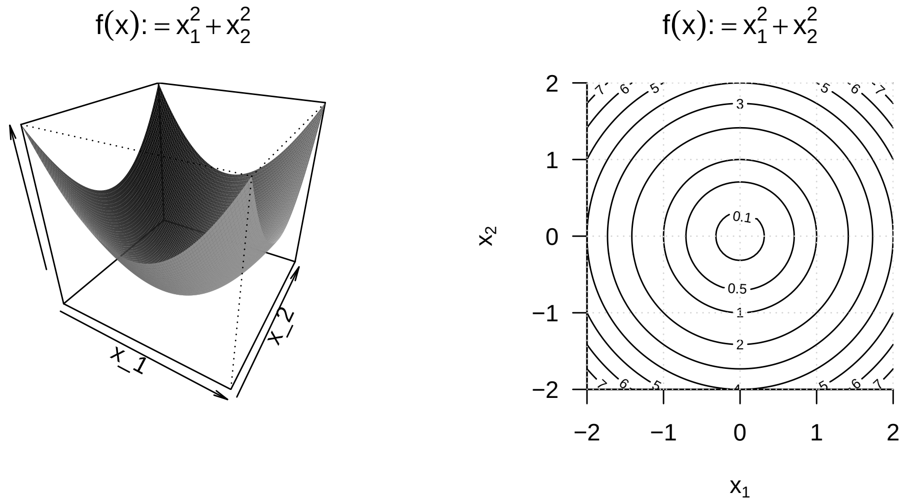

Figure 6.6 Visualizations of a bivariate function. The left-hand panel visualizes the function values as a surface above the two-dimensional domain of the function. Although this representation has a certain three-dimensional character, it is a bivariate function with a correspondingly two-dimensional domain. The right-hand panel visualizes lines with equal function values in the two-dimensional domain of the function.

We now begin to extend the concepts of differentiability and the derivative of univariate real-valued functions to the case of multivariate real-valued functions. To this end, we first introduce the concepts of *partial differentiability* and the *partial derivative*.

Definition 6.6 (Partial differentiability and partial derivative)**.** Let $D \subseteq \mathbb{R}^n$ be a set and let

$$f : D \to \mathbb{R}, x \mapsto f(x) \tag{6.46}$$

be a multivariate real-valued function. f is called *partially differentiable with respect to* x_i at $a \in D$ if the limit

$$\frac{\partial}{\partial x_i} f(a) := \lim_{h \to 0} \frac{f(a + he_i) - f(a)}{h} \tag{6.47}$$

exists. $\frac{\partial}{\partial x_i} f(a)$ is then called the *partial derivative of f with respect to x_i at the point a*. If f is partially differentiable with respect to x_i for all $x \in D$, then f is called *partially differentiable with respect to x_i*, and the function

$$\frac{\partial}{\partial x_i} f : D \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_i} f(x) \tag{6.48}$$

is called the *partial derivative of f with respect to x_i*. f is called *partially differentiable at* $x \in D$ if f is partially differentiable with respect to x_i at $x \in D$ for all $i = 1, ..., n$, and f is called *partially differentiable* if f is partially differentiable with respect to x_i at all $x \in D$ for all $i = 1, ..., n$.

•

In Definition 6.6, $e_i \in \mathbb{R}^n$ denotes the ith canonical unit vector, for which $(e_i)_j = 1$ for $i = j$ and $(e_i)_j = 0$ for $i \neq j$ with $j = 1, ..., n$. In analogy and generalization to the Newton difference quotient, the difference quotient occurring here,

$$\frac{f(x + he_i) - f(x)}{h}, \tag{6.49}$$

therefore measures the change $f(x + he_i) - f(x)$ of f per distance h in the direction e_i. We visualize the components of this quotient for the case of a bivariate function in Figure 6.7.

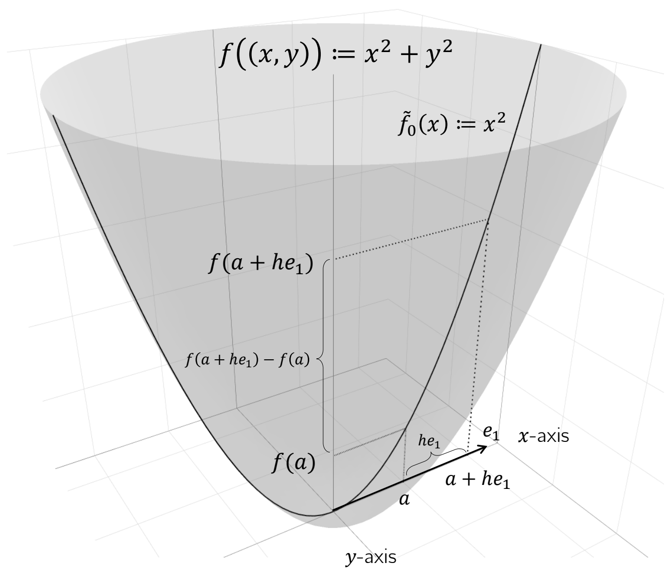

Figure 6.7 Components of the partial Newton difference quotient.

For $h \to 0$, the difference quotient correspondingly measures the *rate of change* of f at x in the direction e_i. As in the discussion of derivatives, $\frac{\partial}{\partial x_i} f(x)$ is a number, whereas $\frac{\partial}{\partial x_i} f$ is a function. In practice, one computes $\frac{\partial}{\partial x_i} f$ as the (ordinary) derivative

$$\frac{d}{dx_i} \tilde{f}_{x_1,...x_{i-1},x_{i+1},...,x_n}(x_i) \tag{6.50}$$

of the univariate real-valued function

$$\tilde{f} : \mathbb{R} \to \mathbb{R}, x_i \mapsto \tilde{f}_{x_1,\ldots x_{i-1},x_{i+1},\ldots,x_n}(x_i) := f(x_1, ..., x_i, ..., x_n). \tag{6.51}$$

Thus, for the ith partial derivative, all x_j with $j \neq i$ are regarded as constants, and one is led back to the familiar computation of derivatives of univariate real-valued functions. We illustrate the procedure for computing partial derivatives with a first example.

Example (1)

We consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2. \tag{6.52}$$

Because the domain of this function is two-dimensional, two partial derivatives can be computed:

$$\frac{\partial}{\partial x_1} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_1} f(x) \text{ and } \frac{\partial}{\partial x_2} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_2} f(x). \tag{6.53}$$

To compute the first of these partial derivatives, one considers the function

$$f_{x_2} : \mathbb{R} \to \mathbb{R}, x_1 \mapsto f_{x_2}(x_1) := x_1^2 + x_2^2, \tag{6.54}$$

where x_2 takes the role of a constant. To make explicit that x_2 is not an argument of the function, while the function still depends on x_2, we have used the subscript notation $f_{x_2}(x_1)$. To compute the partial derivative, we now compute the (ordinary) derivative of f_{x_2},

$$f'_{x_2}(x_1) = 2x_1. \tag{6.55}$$

Thus,

$$\frac{\partial}{\partial x_1} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_1} f(x) = \frac{\partial}{\partial x_1}(x_1^2 + x_2^2) = f'_{x_2}(x_1) = 2x_1. \tag{6.56}$$

Analogously, with the corresponding formulation of f_{x_1},

$$\frac{\partial}{\partial x_2} f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto \frac{\partial}{\partial x_2} f(x) = \frac{\partial}{\partial x_2}(x_1^2 + x_2^2) = f'_{x_1}(x_2) = 2x_2. \tag{6.57}$$

As for the derivative of a univariate real-valued function, it is also possible for a multivariate real-valued function to define a higher derivative recursively.

Definition 6.7 (Second partial derivatives)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function, and let $\frac{\partial}{\partial x_i} f$ be the partial derivative of f with respect to x_i. Then the second partial derivative of f with respect to x_i and x_j is defined as

$$\frac{\partial^2}{\partial x_j \partial x_i} f(x) := \frac{\partial}{\partial x_j}\left(\frac{\partial}{\partial x_i} f\right). \tag{6.58}$$

•

Note that for each partial derivative $\frac{\partial}{\partial x_i} f$ for $i = 1, ..., n$, there are in total n second partial derivatives $\frac{\partial^2}{\partial x_j \partial x_i} f$ for $j = 1, ..., n$. The resulting n^2 second partial derivatives, however, are not all distinct. This is an essential statement of *Schwarz's theorem*.

Theorem 6.5 (Schwarz's theorem)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a partially differentiable multivariate real-valued function. Then*

$$\frac{\partial^2}{\partial x_j \partial x_i} f(x) = \frac{\partial^2}{\partial x_i \partial x_j} f(x) \text{ for all } 1 \le i, j \le n. \tag{6.59}$$

∘

For a proof, we refer to the advanced literature. Schwarz's theorem states in particular that, when forming second partial derivatives, the order of partial differentiation is irrelevant. In this way, the theorem facilitates the calculation of second partial derivatives and also helps detect analytic errors when computing them. We illustrate this by continuing the example above.

Example (1)

We want to compute the second-order partial derivatives of the function

$$f : \mathbb{R}^2 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_2^2 \tag{6.60}$$

With the results for the first-order partial derivatives of this function, we obtain

$$\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_1}(2x_1) = 2 \\
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_1}(2x_2) = 0 \\
\frac{\partial^2}{\partial x_2 \partial x_1} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_2}(2x_1) = 0 \\
\frac{\partial^2}{\partial x_2 \partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_2}(2x_2) = 2.
\end{aligned} \tag{6.61}$$

Obviously,

$$\frac{\partial^2}{\partial x_1 x_2} f(x) = \frac{\partial^2}{\partial x_2 x_1} f(x). \tag{6.62}$$

Example (2)

As a further example, we want to compute the first- and second-order partial derivatives of the function

$$f : \mathbb{R}^3 \to \mathbb{R}, x \mapsto f(x) := x_1^2 + x_1 x_2 + x_2 \sqrt{x_3} \tag{6.63}$$

With the calculation rules for derivatives, the first-order partial derivatives are

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(x_1^2 + x_1 x_2 + x_2\sqrt{x_3}\right) = 2x_1 + x_2, \\
\frac{\partial}{\partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(x_1^2 + x_1 x_2 + x_2\sqrt{x_3}\right) = x_1 + \sqrt{x_3}, \\
\frac{\partial}{\partial x_3} f(x) &= \frac{\partial}{\partial x_3}\left(x_1^2 + x_1 x_2 + x_2\sqrt{x_3}\right) = \frac{x_2}{2\sqrt{x_3}}.
\end{aligned} \tag{6.64}$$

For the second partial derivatives with respect to x_1, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_1} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_1}(2x_1 + x_2) = 2, \\
\frac{\partial^2}{\partial x_2 \partial x_1} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_2}(2x_1 + x_2) = 1, \\
\frac{\partial^2}{\partial x_3 \partial x_1} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_1} f(x)\right) = \frac{\partial}{\partial x_3}(2x_1 + x_2) = 0.
\end{aligned} \tag{6.65}
$$

For the second partial derivatives with respect to x_2, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_1}(x_1 + \sqrt{x_3}) = 1, \\
\frac{\partial^2}{\partial x_2 \partial x_2} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_2}(x_1 + \sqrt{x_3}) = 0, \\
\frac{\partial^2}{\partial x_3 \partial x_2} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_2} f(x)\right) = \frac{\partial}{\partial x_3}(x_1 + \sqrt{x_3}) = \frac{1}{2\sqrt{x_3}}.
\end{aligned} \tag{6.66}
$$

For the second partial derivatives with respect to x_3, we obtain

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_3} f(x) &= \frac{\partial}{\partial x_1}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_1}\left(\frac{x_2}{2\sqrt{x_3}}\right) = 0, \\
\frac{\partial^2}{\partial x_2 \partial x_3} f(x) &= \frac{\partial}{\partial x_2}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_2}\left(\frac{x_2}{2\sqrt{x_3}}\right) = \frac{1}{2\sqrt{x_3}}, \\
\frac{\partial^2}{\partial x_3 \partial x_3} f(x) &= \frac{\partial}{\partial x_3}\left(\frac{\partial}{\partial x_3} f(x)\right) = \frac{\partial}{\partial x_3}\left(x_2 \frac{1}{2} x_3^{-\frac{1}{2}}\right) = -\frac{1}{4} x_2 x_3^{-\frac{3}{2}}.
\end{aligned} \tag{6.67}
$$

Furthermore, one sees that the order of partial differentiation is irrelevant, because

$$
\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_2} f(x) &= \frac{\partial^2}{\partial x_2 \partial x_1} f(x) = 1, \\
\frac{\partial^2}{\partial x_1 \partial x_3} f(x) &= \frac{\partial^2}{\partial x_3 \partial x_1} f(x) = 0, \\
\frac{\partial^2}{\partial x_2 \partial x_3} f(x) &= \frac{\partial^2}{\partial x_3 \partial x_2} f(x) = \frac{1}{2\sqrt{x_3}}.
\end{aligned} \tag{6.68}
$$

As seen above, for a multivariate real-valued function $f : \mathbb{R}^n \to \mathbb{R}$ there are in total n first partial derivatives and n^2 second partial derivatives. These are collected in the *gradient* and the *Hessian matrix* of a multivariate real-valued function.

Definition 6.8 (Gradient). Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. The *gradient* $\nabla f(x)$ of f at the point $x \in \mathbb{R}^n$ is defined as

$$
\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{pmatrix} \in \mathbb{R}^n. \tag{6.69}
$$

•

Note that gradients are multivariate vector-valued functions of the form

$$\nabla f : \mathbb{R}^n \to \mathbb{R}^n, x \mapsto \nabla f(x). \tag{6.70}$$

For $n = 1$, $\nabla f(x) = f'(x)$. An important property of the gradient is that $-\nabla f(x)$ indicates the direction of steepest descent of f in $\mathbb{R}^n$. This insight is not trivial, however, and will be deepened at a later point. As examples, we consider the gradients of the functions analyzed above.

Example (1)

For the function $f : \mathbb{R}^2 \to \mathbb{R}$ considered in Equation 6.60,

$$\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix} \in \mathbb{R}^2. \tag{6.71}$$

In Figure 6.8 we visualize color-coded selected values of this gradient for $(0.7, 0.7)^T$, $(-0.3, 0.1)^T$, $(-0.5, -0.4)^T$, and $(0.1, -1.0)^T$.

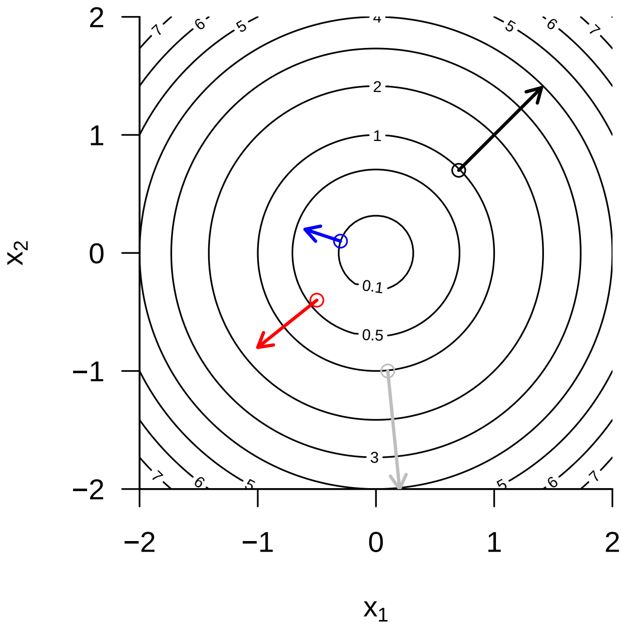

Figure 6.8 Example gradient values of the bivariate function $f(x) = x_1^2 + x_2^2$.

Example (2)

For the function $f : \mathbb{R}^3 \to \mathbb{R}$ considered in Equation 6.63,

$$\nabla f(x) := \begin{pmatrix} \frac{\partial}{\partial x_1} f(x) \\ \frac{\partial}{\partial x_2} f(x) \\ \frac{\partial}{\partial x_3} f(x) \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ x_1 + \sqrt{x_3} \\ \frac{x_2}{2\sqrt{x_3}} \end{pmatrix} \in \mathbb{R}^3. \tag{6.72}$$

Finally, we turn to collecting the second partial derivatives of a multivariate real-valued function in the *Hessian matrix*.

Definition 6.9 (Hessian matrix)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. The *Hessian matrix* $\nabla^2 f(x)$ of f at the point $x \in \mathbb{R}^n$ is defined as

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 \partial x_1} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_1 \partial x_n} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_2 \partial x_n} f(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_n \partial x_1} f(x) & \frac{\partial^2}{\partial x_n \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_n \partial x_n} f(x) \end{pmatrix} \in \mathbb{R}^{n \times n}. \tag{6.73}$$

•

Note that Hessian matrices are multivariate matrix-valued mappings of the form

$$\nabla^2 f : \mathbb{R}^n \to \mathbb{R}^{n \times n}, x \mapsto \nabla^2 f(x). \tag{6.74}$$

For $n = 1$, $\nabla^2 f(x) = f''(x)$. Furthermore, from

$$\frac{\partial^2}{\partial x_i \partial x_j} f(x) = \frac{\partial^2}{\partial x_j \partial x_i} f(x) \text{ for } 1 \le i, j \le n \tag{6.75}$$

it follows that the Hessian matrix is symmetric, that is,

$$\left(\nabla^2 f(x)\right)^T = \nabla^2 f(x). \tag{6.76}$$

Example (1)

For the function $f : \mathbb{R}^2 \to \mathbb{R}$ considered in Equation 6.60,

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 x_1} f(x) & \frac{\partial^2}{\partial x_1 x_2} f(x) \\ \frac{\partial^2}{\partial x_2 x_1} f(x) & \frac{\partial^2}{\partial x_2 x_2} f(x) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}. \tag{6.77}$$

The Hessian matrix of this function is therefore a constant function that does not depend on x.

Example (2)

For the function $f : \mathbb{R}^3 \to \mathbb{R}$ considered in Equation 6.63,

$$\nabla^2 f(x) := \begin{pmatrix} \frac{\partial^2}{\partial x_1 x_1} f(x) & \frac{\partial^2}{\partial x_1 x_2} f(x) & \frac{\partial^2}{\partial x_1 x_3} f(x) \\ \frac{\partial^2}{\partial x_2 x_1} f(x) & \frac{\partial^2}{\partial x_2 x_2} f(x) & \frac{\partial^2}{\partial x_2 x_3} f(x) \\ \frac{\partial^2}{\partial x_3 x_1} f(x) & \frac{\partial^2}{\partial x_3 x_2} f(x) & \frac{\partial^2}{\partial x_3 x_3} f(x) \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & \frac{1}{2\sqrt{x_3}} \\ 0 & \frac{1}{2\sqrt{x_3}} & -\frac{1}{4} x_2 x_3^{-3/2} \end{pmatrix}. \tag{6.78}$$

In contrast to Example (1), the Hessian matrix of the function considered here is not a constant function, and its value depends on the value of the function argument $x \in \mathbb{R}^3$.

7 Integral calculus

This chapter gives an overview of central concepts of integral calculus. The main focus throughout is on clarifying terminology, mathematical symbolism, and the intuition conveyed by it, rather than on the concrete computation of integrals.

7.1 Indefinite integrals

We begin with the definition of the indefinite integral and the concept of an antiderivative.

Definition 7.1 (Indefinite integral and antiderivative)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a univariate real-valued function. Then a differentiable function $F : I \to \mathbb{R}$ with the property

$$F' = f \tag{7.1}$$

is called an *antiderivative of* f. If F is an antiderivative of f, then

$$\int f(x)\,dx := F + c \text{ with } c \in \mathbb{R} \tag{7.2}$$

is called the *indefinite integral of the function* f. The indefinite integral of a function thus denotes the set of all antiderivatives of a function.

•

The above definition states that the derivative of an antiderivative of a function f is precisely f. In addition, the indefinite integral of a function f is the set of *all* antiderivatives of f obtained by adding different constants $c \in \mathbb{R}$. Such a constant $c \in \mathbb{R}$ is also called an *integration constant*; of course, $\frac{d}{dx}c = 0$. The symbol $\int f(x)\,dx$ is defined as $F + c$. In this expression, $f(x)$ is called the *integrand*. $\int$ and dx have no meaning of their own; they are merely symbols.

For the elementary functions introduced in the previous sections, the antiderivatives listed in Table 7.1 result. This can be verified by differentiating the respective antiderivative with the calculation rules of differential calculus. The indefinite integrals of these elementary functions then follow directly from these antiderivatives by adding an integration constant.

Table 7.1 Antiderivatives of elementary functions

Name	Definition	Antiderivative
Polynomial function	$f(x) := \sum_{i=0}^{k} a_i x^i$	$F(x) = \sum_{i=0}^{k} \frac{a_i}{i+1} x^{i+1}$
Constant function	$f(x) := a$	$F(x) = ax$
Identity function	$f(x) := x$	$F(x) = \frac{1}{2}x^2$

Table 7.1 Antiderivatives of elementary functions

Name	Definition	Antiderivative
Linear-affine function	$f(x) := ax + b$	$F(x) = \frac{1}{2}ax^2 + bx$
Square function	$f(x) := x^2$	$F(x) = \frac{1}{3}x^3$
Exponential function	$f(x) := \exp(x)$	$F(x) = \exp(x)$
Logarithm function	$f(x) := \ln(x)$	$F(x) = x\ln x - x$

The calculation rules collected in the following theorem are often helpful for determining antiderivatives of functions that are composed of functions with known antiderivatives.

Theorem 7.1 (Calculation rules for antiderivatives)**.** *Let f and g be univariate real-valued functions that possess antiderivatives, and let g be invertible. Then the following calculation rules hold for determining antiderivatives.*

(1) Sum rule

$$\int af(x) + bg(x)\,dx = a\int f(x)\,dx + b\int g(x)\,dx \text{ for } a, b \in \mathbb{R} \tag{7.3}$$

(2) Integration by parts

$$\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx \tag{7.4}$$

(3) Substitution rule

$$\int f(g(x))g'(x)\,dx = \int f(t)\,dt \text{ with } t = g(x) \tag{7.5}$$

∘

Proof. For a proof of the sum rule, we refer to the advanced literature. The calculation rule for integration by parts follows by integrating the product rule of differentiation. Recall that

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x). \tag{7.6}$$

Integrating both sides of the equation and taking into account the sum rule for antiderivatives then yields

$$\begin{aligned} \textstyle\int (f(x)g(x))'\,dx &= \textstyle\int f'(x)g(x) + f(x)g'(x)\,dx \\ \Leftrightarrow f(x)g(x) &= \textstyle\int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx \\ \Leftrightarrow \textstyle\int f'(x)g(x)\,dx &= f(x)g(x) - \textstyle\int f(x)g'(x)\,dx. \end{aligned} \tag{7.7}$$

For $F' = f$, the substitution rule follows by applying the chain rule of differentiation to the composite function $F(g)$. Specifically, first

$$(F(g(x)))' = F'(g(x))g'(x) = f(g(x))g'(x). \tag{7.8}$$

Integrating both sides of the equation

$$(F(g(x)))' = f(g(x))g'(x) \tag{7.9}$$

then yields

$$\begin{aligned} \textstyle\int (F(g(x)))'\,dx &= \textstyle\int f(g(x))g'(x)\,dx \\ \Leftrightarrow F(g(x)) + c &= \textstyle\int f(g(x))g'(x)\,dx \\ \Leftrightarrow \textstyle\int f(g(x))g'(x)\,dx &= \textstyle\int f(t)\,dt \text{ with } t := g(x). \end{aligned} \tag{7.10}$$

Here, the right-hand side of the last equation above is to be understood as $F(g(x)) + c$, that is, as the antiderivative of f evaluated at the point $t := g(x)$. The dt is not to be replaced by $dg(x)$; it is purely notational in nature. □

Indefinite integrals occupy a central place in the solution of differential equations. More immediate, however, is the use of indefinite integrals in the context of evaluating *definite integrals*, as introduced in the next section.

7.2 Definite integrals

Intuitively, a definite integral corresponds to the signed area between the graph of a function f and the x-axis, restricted to an interval $[a, b]$ (cf. Figure 7.1). *Signed* means that areas between the x-axis and positive values of f contribute positively to the area, whereas areas between the x-axis and negative values of f contribute negatively. For example, the value of the definite integral shown in Figure 7.1 A is 0.68, the value of the definite integral shown in Figure 7.1 B is 0.95 (the shaded area is clearly larger than in Figure 7.1 A), and the value of the definite integral shown in Figure 7.1 C is 0 (the shaded positive and negative areas cancel exactly). The latter example also suggests interpreting the integral as the average value of a function f over an interval $[a, b]$.

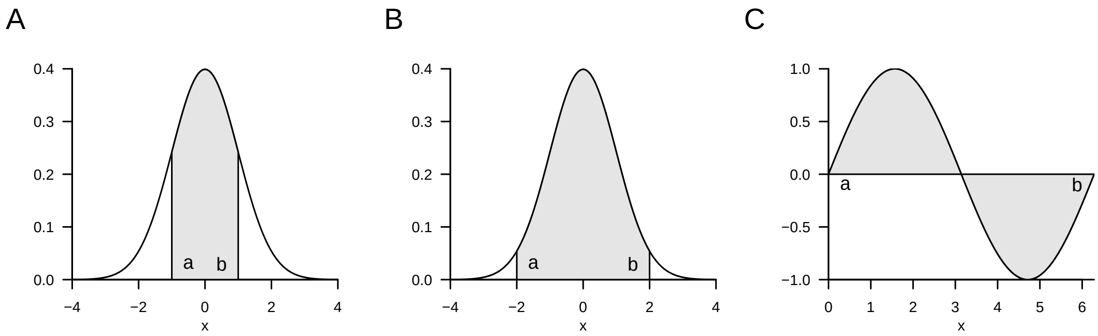

Figure 7.1 Examples of definite integrals

To introduce the concept of the *definite integral* in the sense of the *Riemann integral*, we first need to do some preliminary work. We begin by introducing a term for the subdivision of an interval into smaller sections.

Definition 7.2 (Partition of an interval and mesh)**.** Let $[a, b] \subset \mathbb{R}$ be an interval and let $x_0, x_1, x_2, ..., x_n \in [a, b]$ be a set of points with

$$a =: x_0 < x_1 < x_2 < \cdots < x_n := b \tag{7.11}$$

and

$$\Delta x_i := x_i - x_{i-1} \text{ for } i = 1, ..., n. \tag{7.12}$$

Then the set

$$Z := \{[x_0, x_1], [x_1, x_2], ..., [x_{n-1}, x_n]\} \tag{7.13}$$

of subintervals of $[a, b]$ defined by $x_0, x_1, x_2, ..., x_n$ is called a *partition of* $[a, b]$. Furthermore,

$$Z_{\max} := \max_{i \in \mathbb{N}_n} \Delta x_i, \tag{7.14}$$

that is, the largest of the subinterval lengths Δx_i, is called the *mesh of* Z.

•

Intuitively, Δx_i is the width of the rectangles in Figure 7.2, as we will see in what follows. With the concepts of the partition of an interval, we can now introduce the concept of *Riemann sums*.

Definition 7.3 (Riemann sums)**.** Let $f : [a,b] \to \mathbb{R}$ be a bounded function on $[a,b]$, that is, $|f(x)| < c$ for $0 < c < \infty$ and all $x \in [a,b]$. Let Z be a partition of $[a,b]$ with subinterval lengths Δx_i for $i = 1, ..., n$. Furthermore, let ξ_i for $i = 1, ..., n$ be an arbitrary point in the subinterval $[x_{i-1}, x_i]$ of the partition Z. Then

$$R(Z) := \sum_{i=1}^{n} f(\xi_i)\Delta x_i \tag{7.15}$$

is called the *Riemann sum of f on $[a,b]$ with respect to the partition Z.*

•

If, for example, one chooses the maximum of f in each subinterval in the Riemann sum, the so-called *upper Riemann sum* results,

$$R_o(Z) := \sum_{i=1}^{n} \left(\max_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.16}$$

If, by contrast, one chooses the minimum of f in each subinterval, the so-called *lower Riemann sum* results,

$$R_u(Z) := \sum_{i=1}^{n} \left(\min_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.17}$$

Figure 7.2 illustrates the definition of these Riemann sums: the dark gray rectangles each have area $\Delta x_i \cdot \min_{[x_{i-1},x_i]} f(\xi)$ and thus form the summands in the lower Riemann sum

$$R_u(Z) := \sum_{i=1}^{4} \left(\min_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.18}$$

The vertical combination of dark gray and light gray rectangles each has area $\Delta x_i \cdot \max_{[x_{i-1},x_i]} f(\xi)$ and thus forms the summands in the upper Riemann sum

$$R_o(Z) := \sum_{i=1}^{4} \left(\max_{[x_{i-1},x_i]} f(\xi_i) \right) \cdot \Delta x_i. \tag{7.19}$$

Now imagine letting Δx_i tend to zero for all $i = 1, ..., n$, thereby making the mesh of the partition Z smaller and smaller. The values of $\min_{[x_{i-1},x_i]} f(\xi_i)$ and $\max_{[x_{i-1},x_i]} f(\xi_i)$, and hence also the values of $R_u(Z)$ and $R_o(Z)$, will then approach each other more and more. This limiting process is used in the definition of the Riemann integral.

Definition 7.4 (Definite riemann integral)**.** Let $f : [a,b] \to \mathbb{R}$ be a bounded real-valued function on $[a,b]$. Furthermore, for Z_k with $k = 1, 2, 3...$, let there be a sequence of partitions of $[a,b]$ with corresponding mesh $Z_{\max,k}$. If, for every sequence of partitions $Z_1, Z_2, ...$ with $|Z_{\max,k}| \to 0$ as $k \to \infty$ and for arbitrarily chosen points ξ_{ki} with $i = 1, ..., n$ in the subinterval $[x_{k,i-1}, x_{k,i}]$ of the partition Z_k, the sequence of corresponding Riemann sums $R(Z_1), R(Z_2), ...$ tends to the same limit, then f is called *integrable* on $[a,b]$. The

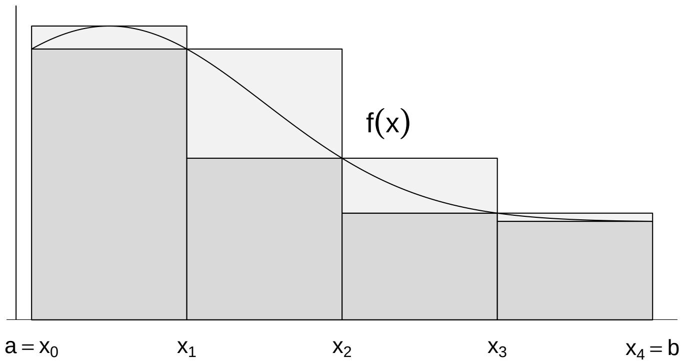

Figure 7.2 Riemann sums

corresponding limit of the sequence of Riemann sums is called the *definite Riemann integral* and denoted by

$$\int_a^b f(x)\,dx := \lim_{k\to\infty} R(Z_k) \text{ for } |Z_{\max,k}| \to 0. \tag{7.20}$$

In this context, the values a and b are called the *lower* and *upper* limits of integration, respectively, $f(x)$ is called the *integrand*, and x is called the *variable of integration*.

•

The Riemann integrability of a function and the value of a definite Riemann integral are thus defined in terms of a limiting process. However, the theory of Riemann integrals can be extended by the fundamental theorems of differential and integral calculus, so that the concrete computation of a definite integral rarely requires forming partitions and determining a limit. For simplicity, in what follows we omit the designation *Riemann* and simply speak of *definite integrals*.

A first step toward simplifying the computation of definite integrals is to record the following calculation rules, for whose proof we refer to the advanced literature.

Theorem 7.2 (Calculation rules for definite integrals)**.** *Let f and g be integrable functions on $[a,b]$. Then the following calculation rules hold.*

(1) Linearity. For $c_1, c_2 \in \mathbb{R}$,

$$\int_a^b (c_1 f(x) + c_2 g(x))\,dx = c_1 \int_a^b f(x)\,dx + c_2 \int_a^b g(x)\,dx. \tag{7.21}$$

(2) Additivity. For $a < c < b$,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx. \tag{7.22}$$

(3) Change of sign when reversing the limits of integration

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx. \tag{7.23}$$

(4) Independence from the choice of the variable of integration

$$\int_a^b f(x)\,dx = \int_a^b f(y)\,dy. \tag{7.24}$$

(5) Independence of the integral from the type of interval

$$\int_a^b f(x)\,dx = \int_{]a,b[} f(x)\,dx = \int_{[a,b[} f(x)\,dx = \int_{]a,b]} f(x)\,dx = \int_{[a,b]} f(x)\,dx, \tag{7.25}$$

where $\int_I$ denotes the definite integral of f on the interval $I \subseteq \mathbb{R}$.

∘

Note that the linearity of the definite integral is analogous to associativity for sums of equal length and distributivity when multiplying by a constant (cf. Theorem 3.1),

$$\sum_{i=1}^n (c_1 x_i + c_2 y_i) = c_1 \sum_{i=1}^n x_i + c_2 \sum_{i=1}^n y_i,$$

that the additivity of definite integrals is analogous to splitting sums, and that the independence of the value of an integral from the choice of the variable of integration has its counterpart in the independence of the value of a sum from its summation index. A graphical representation of the additivity calculation rule for integrals is shown in Figure 7.3. The sum of the areas given by the definite integrals $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ with $a < c < b$ is the area of $\int_a^b f(x)\,dx$.

With the help of additivity and the zero-interval integral

$$\int_a^a f(x)\,dx = 0 \tag{7.26}$$

the change of sign when reversing the limits of integration, which will be important below, can be justified. To this end, as in Figure 7.3, let

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx. \text{ and } \int_b^a f(x)\,dx = \int_b^c f(x)\,dx + \int_c^a f(x)\,dx. \tag{7.27}$$

Adding both equations gives

$$\begin{aligned} \int_a^b f(x)\,dx + \int_b^a f(x)\,dx &= \int_a^c f(x)\,dx + \int_c^b f(x)\,dx + \int_b^c f(x)\,dx + \int_c^a f(x)\,dx \\ \Leftrightarrow \int_b^a f(x)\,dx &= \int_a^b f(x)\,dx + \int_b^a f(x)\,dx \\ \Leftrightarrow 0 &= \int_a^b f(x)\,dx + \int_b^a f(x)\,dx \\ \Leftrightarrow \int_a^b f(x)\,dx &= -\int_b^a f(x)\,dx. \end{aligned} \tag{7.28}$$

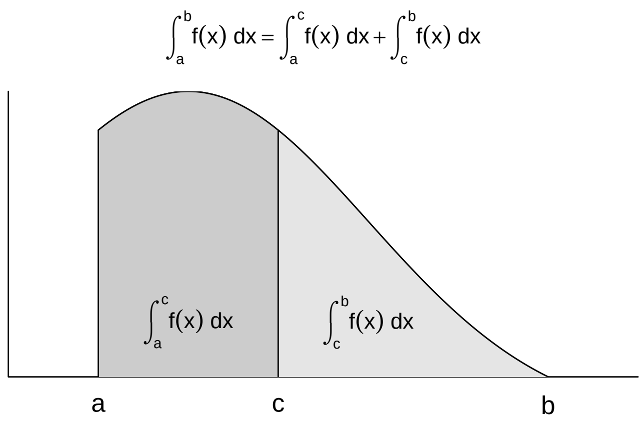

Figure 7.3 Additivity of definite integrals

The *fundamental theorems of differential and integral calculus* finally make it possible to compute definite integrals of a function f directly with the help of the antiderivative F of f. For the proof of the first fundamental theorem of differential and integral calculus, we need the *mean value theorem of integral calculus*, which we state here without proof and illustrate in Figure 7.4.

Theorem 7.3 (Mean value theorem of integral calculus). *For a continuous function $f : [a,b] \to \mathbb{R}$, there exists a $\xi \in]a,b[$ such that*

$$\int_a^b f(x)\,dx = f(\xi)(b-a) \tag{7.29}$$

∘

The mean value theorem of integral calculus guarantees the existence of a $\xi \in [a,b]$ such that the definite integral $\int_a^b f(x)\,dx$ equals the product of the "rectangle height" $f(\xi)$ and the "rectangle width" $(b-a)$. In Figure 7.4, this ξ lies exactly halfway between a and b. That the resulting gray rectangle area equals $\int_a^b f(x)\,dx$ follows from the visually at least plausible fact that the areas between $f(x)$ and $f(\xi)$ on the interval $[a,\xi]$ and between $f(\xi)$ and $f(x)$ on the interval $[\xi,b]$ have the same magnitude, but the former has a negative sign. In general, however, the mean value theorem of integral calculus only guarantees the existence of a $\xi \in [a,b]$ with the discussed property; it does not provide a formula for determining ξ.

With this preliminary work, we can now formulate and prove the first fundamental theorem of differential and integral calculus.

Theorem 7.4 (First fundamental theorem of calculus). *If $f : I \to \mathbb{R}$ is a continuous function on the interval $I \subset \mathbb{R}$, then the function*

$$F : I \to \mathbb{R}, x \mapsto F(x) := \int_a^x f(t)\,dt \text{ with } x,a \in I \tag{7.30}$$

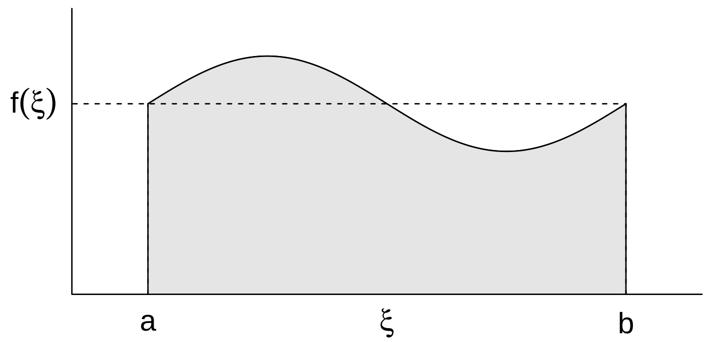

Figure 7.4 On the mean value theorem of integral calculus

is an antiderivative of f.

∘

Proof. We consider the difference quotient

$$\frac{1}{h}(F(x+h)-F(x)) \tag{7.31}$$

With the definition $F(x) := \int_a^x f(t)\,dt$ and the additivity of the definite integral, it follows that

$$\frac{1}{h}(F(x+h)-F(x)) = \frac{1}{h}\left(\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right) = \frac{1}{h}\int_x^{x+h} f(t)\,dt \tag{7.32}$$

By the mean value theorem of integral calculus, there is therefore a $\xi \in]x, x+h[$ such that

$$\frac{1}{h}(F(x+h)-F(x)) = f(\xi) \tag{7.33}$$

Taking the limit then yields

$$\lim_{h\to 0}\frac{1}{h}(F(x+h)-F(x)) = \lim_{h\to 0} f(\xi) \text{ for } \xi \in]x, x+h[\Leftrightarrow F'(x) = f(x). \tag{7.34}$$

□

The second fundamental theorem of differential and integral calculus states how to compute a definite integral with the help of the antiderivative.

Theorem 7.5 (Second fundamental theorem of calculus). *If F is an antiderivative of a continuous function $f : I \to \mathbb{R}$ on an interval I, then, for $a, b \in I$ with $a \le b$,*

$$\int_a^b f(x)\,dx = F(b) - F(a) =: F(x)|_a^b. \tag{7.35}$$

∘

Proof. Using the calculation rules for definite integrals and the first fundamental theorem of differential and integral calculus, we obtain

$$F(b) - F(a) = \int_{\alpha}^{b} f(t)\,dt - \int_{\alpha}^{a} f(t)\,dt = \int_{a}^{\alpha} f(t)\,dt + \int_{\alpha}^{b} f(t)\,dt = \int_{a}^{b} f(x)\,dx. \tag{7.36}$$

□

We want to apply the second fundamental theorem of differential and integral calculus in three examples (cf. Figure 7.5).

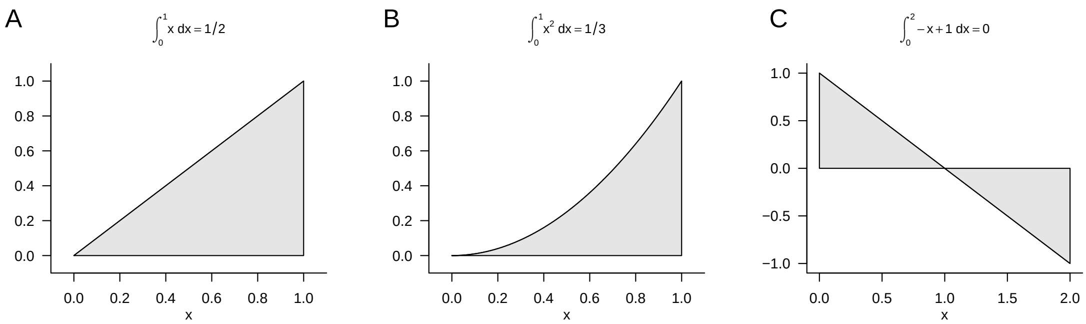

Figure 7.5 Examples of the second fundamental theorem of differential and integral calculus

Example (1)

We consider the identity function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x \tag{7.37}$$

and want to compute the definite integral of this function on the interval $[0, 1]$, that is,

$$\int_0^1 f(x)\,dx = \int_0^1 x\,dx. \tag{7.38}$$

To do so, recall that an antiderivative of f is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := \frac{1}{2}x^2 \tag{7.39}$$

because

$$F'(x) = \frac{d}{dx}\left(\frac{1}{2}x^2\right) = 2 \cdot \frac{1}{2}x^{2-1} = x. \tag{7.40}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^1 x\,dx = \frac{1}{2}1^2 - \frac{1}{2}0^2 = \frac{1}{2}. \tag{7.41}$$

This result agrees with the intuition suggested by the gray area in Figure 7.5 A.

Example (2)

Next, we consider the square function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := x^2 \tag{7.42}$$

and want to compute the definite integral of this function on the interval $[0, 1]$, that is,

$$\int_0^1 f(x)\,dx = \int_0^1 x^2\,dx. \tag{7.43}$$

To do so, recall that an antiderivative of f is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := \frac{1}{3}x^3 \tag{7.44}$$

because

$$F'(x) = \frac{d}{dx}\left(\frac{1}{3}x^3\right) = 3 \cdot \frac{1}{3}x^{3-1} = x^2. \tag{7.45}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^1 x^2\,dx = \frac{1}{3}1^3 - \frac{1}{3}0^3 = \frac{1}{3}. \tag{7.46}$$

This result agrees with the intuition that follows from comparing the gray areas in Figure 7.5 A and Figure 7.5 B.

Example (3)

Finally, we consider the linear-affine function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := -x + 1 \tag{7.47}$$

and want to compute the definite integral of this function on the interval $[0, 2]$, that is,

$$\int_0^2 f(x)\,dx = \int_0^2 -x + 1\,dx. \tag{7.48}$$

To do so, recall that an antiderivative of the linear function with $a = -1$ and $b = 1$ (cf. Table 7.1) is given by

$$F : \mathbb{R} \to \mathbb{R}, x \mapsto F(x) := -\frac{1}{2}x^2 + x \tag{7.49}$$

because

$$F'(x) = \frac{d}{dx}\left(-\frac{1}{2}x^2 + x\right) = -2 \cdot \frac{1}{2}x^{2-1} + 1 \cdot x^{1-1} = -x + 1. \tag{7.50}$$

Substitution into the second fundamental theorem of differential and integral calculus then gives

$$\int_0^2 -x + 1\,dx = \left(-\frac{1}{2}2^2 + 2\right) - \left(-\frac{1}{2}0^2 + 0\right) = -2 + 2 - 0 = 0. \tag{7.51}$$

This result agrees with the intuition that the positive and negative gray areas in Figure 7.5 C cancel each other.

7.3 Improper integrals

Improper integrals are definite integrals in which at least one limit of integration is not a real number, but rather $-\infty$ or ∞. We illuminate the nature of improper integrals with the following definition and an example.

Definition 7.5 (Improper integrals)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a univariate real-valued function. With the definitions

$$\int_{-\infty}^{b} f(x)\,dx := \lim_{a \to -\infty} \int_{a}^{b} f(x)\,dx \text{ and } \int_{a}^{\infty} f(x)\,dx := \lim_{b \to \infty} \int_{a}^{b} f(x)\,dx \tag{7.52}$$

and the additivity of integrals

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{b} f(x)\,dx + \int_{b}^{\infty} f(x)\,dx \tag{7.53}$$

the concept of the definite integral is extended to the unbounded intervals of integration $]-\infty, b]$, $[a, \infty[$, and $]-\infty, \infty[$. Integrals with unbounded intervals of integration are called *improper integrals*. If the corresponding limits exist, one says that the improper integrals *converge*.

•

As an example, we consider the improper integral of the function

$$f : \mathbb{R} \to \mathbb{R}, x \mapsto f(x) := \frac{1}{x^2} \tag{7.54}$$

on the interval $[1, \infty[$, that is,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx. \tag{7.55}$$

According to the conventions in the definition of improper integrals,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx = \lim_{b \to \infty} \int_{1}^{b} \frac{1}{x^2}\,dx. \tag{7.56}$$

With the antiderivative $F(x) = -x^{-1}$ of $f(x) = x^{-2}$, the definite integral in the above equation becomes

$$\int_{1}^{b} \frac{1}{x^2}\,dx = F(b) - F(1) = -\frac{1}{b} - \left(-\frac{1}{1}\right) = -\frac{1}{b} + 1. \tag{7.57}$$

Thus,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx = \lim_{b \to \infty} \int_{1}^{b} \frac{1}{x^2}\,dx = \lim_{b \to \infty} \left(-\frac{1}{b} + 1\right) = -\lim_{b \to \infty} \frac{1}{b} + \lim_{b \to \infty} 1 = 0 + 1 = 1. \tag{7.58}$$

7.4 Multidimensional integrals

So far, we have only considered integrals of univariate real-valued functions. The concept of the integral can also be extended to multivariate real-valued functions. In that case, however, the integration domain of the function is not necessarily as easy to describe as an interval. In particular, arbitrarily shaped integration domains are already possible for bivariate functions, for example. Here, we now want to consider the simplest case of a *hyperrectangle*. In this case, we can define multidimensional definite integrals as follows.

Definition 7.6 (Multidimensional definite integrals on hyperrectangles)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a multivariate real-valued function. Then integrals of the form

$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} f(x)\,dx = \int_{a_1}^{b_1}\cdots\int_{a_n}^{b_n} f(x_1,...,x_n)\,dx_1...\,dx_n \tag{7.59}$$

are called *multidimensional definite integrals on hyperrectangles.* Furthermore, integrals of the form

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1,...,x_n)\,dx_1...\,dx_n \tag{7.60}$$

are called *multidimensional improper integrals.*

•

As an example of Definition 7.6, we consider the two-dimensional definite integral of the function

$$f : \mathbb{R}^2 \to \mathbb{R}, (x_1,x_2) \mapsto f(x_1,x_2) := x_1^2 + 4x_2 \tag{7.61}$$

on the rectangle $[0,1]\times[0,1]$. Fubini's theorem from the theory of multidimensional integrals states that multidimensional integrals can be evaluated in any coordinate order. Thus, for example,

$$\int_{a_1}^{b_1}\left(\int_{a_2}^{b_2} f(x_1,x_2)\,dx_2\right)dx_1 = \int_{a_2}^{b_2}\left(\int_{a_1}^{b_1} f(x_1,x_2)\,dx_1\right)dx_2. \tag{7.62}$$

In this sense, for the example we consider

$$\int_0^1\int_0^1 x_1^2 + 4x_2\,dx_1\,dx_2 = \int_0^1\left(\int_0^1 x_1^2 + 4x_2\,dx_1\right)dx_2 \tag{7.63}$$

and therefore first the inner integral. Here, x_2 takes on the role of a constant. An antiderivative of $g(x_1) := x_1^2 + 4x_2$ is $G(x_1) = \frac{1}{3}x_1^3 + 4x_2x_1$, which can be verified by differentiating G. Thus, for the inner integral,

$$\begin{aligned}\int_0^1 x_1^2 + 4x_2\,dx_1 &= G(1) - G(0)\\ &= \frac{1}{3}\cdot 1^3 + 4x_2\cdot 1 - \frac{1}{3}\cdot 0^3 - 4x_2\cdot 0\\ &= \frac{1}{3} + 4x_2.\end{aligned} \tag{7.64}$$

Considering the outer integral

$$\int_0^1 4x_2 + \frac{1}{3}\,dx_2$$

then gives, with the antiderivative

$$H(x_2) = 2x_2^2 + \frac{1}{3}x_2 \tag{7.65}$$

of

$$h(x_2) := 4x_2 + \frac{1}{3}, \tag{7.66}$$

that

$$
\begin{aligned}
\int_0^1 \int_0^1 x_1^2 + 4x_2 \, dx_1 \, dx_2 &= \int_0^1 4x_2 + \frac{1}{3} \, dx_2 \\
&= H(1) - H(0) \\
&= 2 \cdot 1^2 + \frac{1}{3} \cdot 1 - 2 \cdot 0^2 - \frac{1}{3} \cdot 0 \\
&= \frac{7}{3}.
\end{aligned}
\tag{7.67}
$$

8 Vectors

In scientific modeling, one often considers phenomena characterized by several quantitative features. For example, the position of an object in three-dimensional space is determined by three coordinates with respect to the three axes of a Cartesian coordinate system. Similarly, a person's health state may be characterized by three measured values, for example a self-report score, a biomarker, and an expert rating. For modeling and analyzing such multidimensional quantitative phenomena, mathematics provides a versatile tool in the form of the real vector space. In this chapter, we first introduce the concept of a real vector space and the basic arithmetic of vectors (Section 8.1). A vector space structure that is strongly guided by three-dimensional spatial intuition is then provided by the Euclidean vector space (Section 8.2). With vector arithmetic, all vectors of a vector space can be formed from a small collection of distinguished vectors. We discuss the concepts underlying this principle in Section 8.3 and Section 8.4.

8.1 Real vector space

We begin with the general definition of a vector space, which specifies basic rules for computing with vectors.

Definition 8.1 (Vector space)**.** Let V be a nonempty set and let S be a set of scalars. Furthermore, let a mapping

$$+ : V \times V \to V, (v_1, v_2) \mapsto +(v_1, v_2) =: v_1 + v_2, \tag{8.1}$$

called *vector addition*, be defined. Finally, let a mapping

$$\cdot : S \times V \to V, (s, v) \mapsto \cdot(s, v) =: sv, \tag{8.2}$$

called *scalar multiplication*, be defined. Then the tuple $(V, S, +, \cdot)$ is called a *vector space* if and only if, for arbitrary elements $v, w, u \in V$ and $a, b \in S$, the following conditions hold:

(1) *Commutativity of vector addition.*

$$v + w = w + v.$$

(2) *Associativity of vector addition.*

$$(v + w) + u = v + (w + u)$$

(3) *Existence of an identity element for vector addition.*

$$\text{There exists a vector } 0 \in V \text{ with } v + 0 = 0 + v = v.$$

(4) *Existence of inverse elements for vector addition.*

$$\text{For all vectors } v \in V \text{ there exists a vector } -v \in V \text{ with } v + (-v) = 0.$$

(5) *Existence of an identity element for scalar multiplication.*

$$\text{There exists a scalar } 1 \in S \text{ with } 1 \cdot v = v.$$

(6) *Associativity of scalar multiplication.*

$$a \cdot (b \cdot v) = (a \cdot b) \cdot v.$$

(7) *Distributivity with respect to vector addition.*

$$a \cdot (v + w) = a \cdot v + a \cdot w.$$

(8) *Distributivity with respect to scalar addition.*

$$(a + b) \cdot v = a \cdot v + b \cdot v.$$

•

It is striking that Definition 8.1 specifies how one should compute with vectors, but does not state what a vector actually is beyond being an element of a set. This is due to the fact that there are many different mathematical objects for which vector space structures can be defined. Examples include the set of real m-tuples, the set of matrices, the set of polynomials, the set of solutions of a linear system of equations, the set of real sequences, the set of continuous functions, and many others.

Here we are initially interested only in the vector space of the set of real m-tuples. We recall that we denote the real m-tuples by

$$\mathbb{R}^m := \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} | x_i \in \mathbb{R} \text{ for all } 1 \le i \le m \right\} \tag{8.3}$$

and pronounce $\mathbb{R}^m$ as "$\mathbb{R}$ to the power of m". The elements $x \in \mathbb{R}^m$ are called *real vectors* or simply *vectors*. We now want to use the set $\mathbb{R}^m$ as the basis for the definition of a vector space. To this end, we first define vector addition for elements of $\mathbb{R}^m$ and scalar multiplication for elements of $\mathbb{R}$ and $\mathbb{R}^m$.

Definition 8.2 (Vector addition and scalar multiplication in $\mathbb{R}^m$)**.** For all $x, y \in \mathbb{R}^m$ and $a \in \mathbb{R}$, let *vector addition* be defined by

$$+ : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}^m, (x, y) \mapsto x + y = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_m + y_m \end{pmatrix} \tag{8.4}$$

and let *scalar multiplication* be defined by

$$\cdot : \mathbb{R} \times \mathbb{R}^m \to \mathbb{R}^m, (a, x) \mapsto ax = a \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} := \begin{pmatrix} ax_1 \\ \vdots \\ ax_m \end{pmatrix} \tag{8.5}$$

•

This yields the following result.

Theorem 8.1 (Real vector space)**.** $(\mathbb{R}^m, \mathbb{R}, +, \cdot)$ *with the arithmetic rules of addition and multiplication in* $\mathbb{R}$ *is a vector space.*

∘

For a proof, which we omit here, one has to verify conditions (1) to (8) from Definition 8.1 for the set considered here and the forms of vector addition and scalar multiplication specified here. These follow easily from the arithmetic rules of addition and multiplication in $\mathbb{R}$ and from the fact that vector addition and scalar multiplication for elements of $\mathbb{R}^m$ were defined componentwise in Definition 8.2. We thus define the concept of the *real vector space*.

Definition 8.3 (Real vector space)**.** For $\mathbb{R}^m$, let $+$ and $\cdot$ be the vector addition and scalar multiplication defined in Definition 8.2. Then, based on Theorem 8.1, we call the vector space $(\mathbb{R}^m, \mathbb{R}, +, \cdot)$ the *real vector space*.

•

On the basis of Definition 8.3, we now illustrate computing with real vectors by means of a few examples.

Examples

(1) For

$$x := \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \in \mathbb{R}^4 \text{ and } y := \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

we have

$$x + y = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 2+1 \\ 3+0 \\ 4+1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \\ 5 \end{pmatrix} \in \mathbb{R}^4.$$

In **R**, one implements this example as follows.

```
x = matrix(c(1,2,3,4), nrow = 4)       # vector definition
y = matrix(c(2,1,0,1), nrow = 4)       # vector definition
x + y                                  # vector addition
```

```
     [,1]
[1,]    3
[2,]    3
[3,]    3
[4,]    5
```

(2) For

$$x := \begin{pmatrix} 2 \\ 3 \end{pmatrix} \in \mathbb{R}^2 \text{ and } y := \begin{pmatrix} 1 \\ 3 \end{pmatrix} \in \mathbb{R}^2$$

we have

$$x - y = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2-1 \\ 3-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^2.$$

In **R**, one implements this example as follows.

```
x = matrix(c(2,3), nrow = 2)          # vector definition
y = matrix(c(1,3), nrow = 2)          # vector definition
x - y                                 # vector subtraction
```

```
     [,1]
[1,]    1
[2,]    0
```

(3) For

$$x := \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \in \mathbb{R}^3 \text{ and } a := 3 \in \mathbb{R}$$

we have

$$ax = 3 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \cdot 2 \\ 3 \cdot 1 \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

In **R**, one implements this example as follows.

```
x = matrix(c(2,1,3), nrow = 3)        # vector definition
a = 3                                 # scalar definition
a*x                                   # scalar multiplication
```

```
     [,1]
[1,]    6
[2,]    3
[3,]    9
```

For $m \in \{1, 2, 3\}$, real vectors and computations with them can be visualized. For $m > 3$, for example when more than three quantitative features of a person's health state are available, as is regularly the case in applications, this is not possible. Nevertheless, visual intuition for $m \leq 3$ may facilitate an initial understanding of vector spaces. We focus here on the case $m := 2$. In this case, the real vectors under consideration lie in the two-dimensional plane and are usually visualized as points or arrows (Figure 8.1).

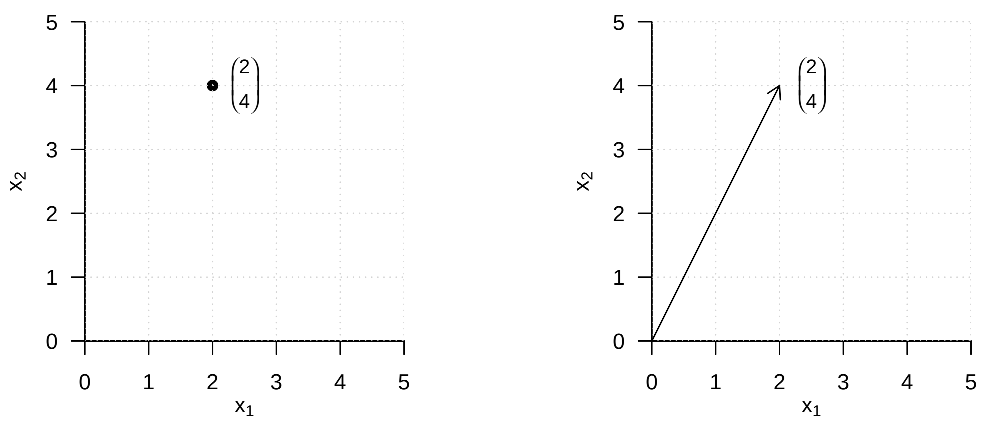

Figure 8.1 Visualization of vectors in $\mathbb{R}^2$

Figure 8.2 visualizes the vector addition

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}. \tag{8.6}$$

The sum vector corresponds to the diagonal of the parallelogram spanned by the two summands.

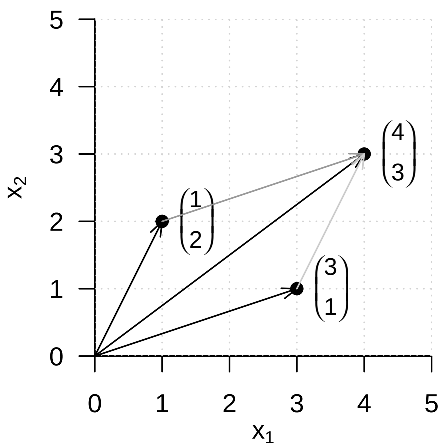

Figure 8.2 Vector addition in $\mathbb{R}^2$

Figure 8.3 visualizes the vector subtraction

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \tag{8.7}$$

The resulting vector corresponds to the diagonal of the parallelogram spanned by the first vector and the opposite vector of the second vector.

Finally, Figure 8.4 visualizes the scalar multiplication

$$3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \tag{8.8}$$

Multiplication of a vector by a positive scalar changes only its length, but not its direction.

8.2 Euclidean vector space

By defining the *inner product*, the real vector space can be endowed with spatial-geometric intuition in the sense of a *Euclidean vector space*. In particular, this makes it possible to define and compute concepts such as the *length of a vector*, the *distance between two vectors*, and, not least, the *angle between two vectors*. We first introduce the *inner product*.

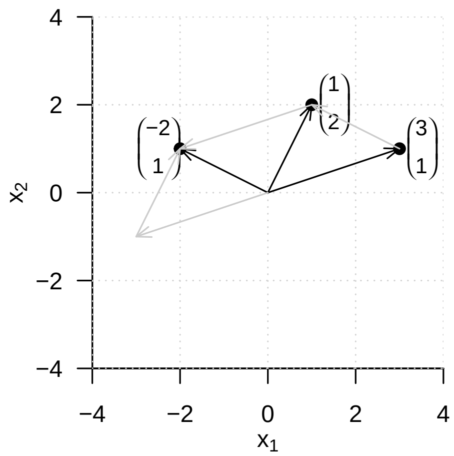

Figure 8.3 Vector subtraction in $\mathbb{R}^2$

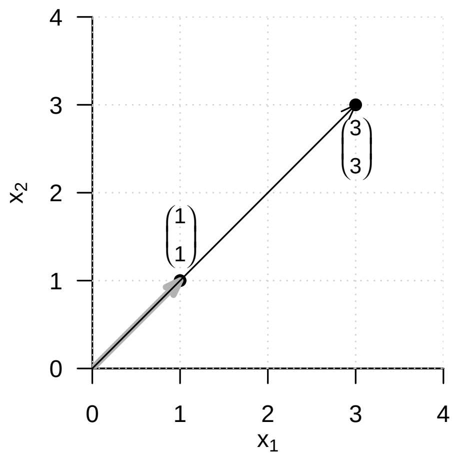

Figure 8.4 Scalar multiplication in $\mathbb{R}^2$

Definition 8.4 (Inner product on $\mathbb{R}^m$)**.** The *inner product on* $\mathbb{R}^m$ is defined as the mapping

$$\langle\rangle : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}, (x, y) \mapsto \langle(x, y)\rangle := \langle x, y\rangle := \sum_{i=1}^{m} x_i y_i. \tag{8.9}$$

•

The inner product is called an inner product because it yields a scalar, not because it involves multiplication by scalars. The inner product is closely related to the matrix product, as we will see later. We first consider an example and its implementation in **R**.

Example

Let

$$x := \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \text{ and } y := \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \tag{8.10}$$

Then

$$\langle x, y\rangle = x_1y_1 + x_2y_2 + x_3y_3 = 1 \cdot 2 + 2 \cdot 0 + 3 \cdot 1 = 2 + 0 + 3 = 5. \tag{8.11}$$

In **R**, there are several ways to evaluate an inner product. We list two of them for the given example below.

```
# vector definitions
x = matrix(c(1,2,3), nrow = 3)
y = matrix(c(2,0,1), nrow = 3)

# inner product using R's componentwise multiplication and sum() function
sum(x*y)
```

```
[1] 5
```

```
# inner product using R's matrix transposition and multiplication
t(x) %*% y
```

```
     [,1]
[1,]    5
```

With the help of the inner product, the concept of the real vector space can be extended to the concept of the *real canonical Euclidean vector space.*

Definition 8.5 (Euclidean vector space)**.** The tuple $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ consisting of the real vector space $(\mathbb{R}^m, +, \cdot)$ and the inner product $\langle\rangle$ on $\mathbb{R}^m$ is called the *real canonical Euclidean vector space.*

•

In general, every tuple consisting of a vector space and an inner product is called a “Euclidean vector space”. Informally, however, we often simply speak of $\mathbb{R}^m$ as a “Euclidean vector space” and, in particular, of $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ as the “Euclidean vector space”. A Euclidean vector space is a vector space with geometric structure induced by the inner product. In particular, the geometric concepts of *length*, *distance*, and *angle* now acquire meaning in the Euclidean vector space. We define them as follows.

Definition 8.6. $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ is the Euclidean vector space.

(1) The *length* of a vector $x \in \mathbb{R}^m$ is defined as

$$\|x\| := \sqrt{\langle x, x\rangle}. \tag{8.12}$$

(2) The *distance* between two vectors $x, y \in \mathbb{R}^m$ is defined as

$$d(x, y) := \|x - y\|. \tag{8.13}$$

(3) The *angle* α between two vectors $x, y \in \mathbb{R}^m$ with $x, y \neq 0$ is defined by

$$0 \le \alpha \le \pi \text{ and } \cos\alpha := \frac{\langle x, y\rangle}{\|x\|\|y\|} \tag{8.14}$$

•

The length $\|x\|$ of a vector $x \in \mathbb{R}^m$ is also called the *Euclidean norm of* x, the ℓ_2*-norm of* x, or simply the *norm of* x. It is often denoted by $\|x\|_2$. We consider three examples for determining the length of a vector and their corresponding **R** implementation. We illustrate these examples in Figure 8.5.

Example (1)

$$\left\|\begin{pmatrix}2\\0\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}2\\0\end{pmatrix}\right\rangle} = \sqrt{2^2 + 0^2} = \sqrt{4} = 2.00 \tag{8.15}$$

```
norm(matrix(c(2,0),nrow = 2), type = "2")          # vector length = l_2 norm
```

```
[1] 2
```

Example (2)

$$\left\|\begin{pmatrix}2\\2\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\2\end{pmatrix}, \begin{pmatrix}2\\2\end{pmatrix}\right\rangle} = \sqrt{2^2 + 2^2} = \sqrt{8} \approx 2.83 \tag{8.16}$$

```
norm(matrix(c(2,2),nrow = 2), type = "2")          # vector length = l_2 norm
```

```
[1] 2.828427
```

Example (3)

$$\left\|\begin{pmatrix}2\\4\end{pmatrix}\right\| = \sqrt{\left\langle\begin{pmatrix}2\\4\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\rangle} = \sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.47 \tag{8.17}$$

```
norm(matrix(c(2,4),nrow = 2), type = "2")          # vector length = l_2 norm
```

```
[1] 4.472136
```

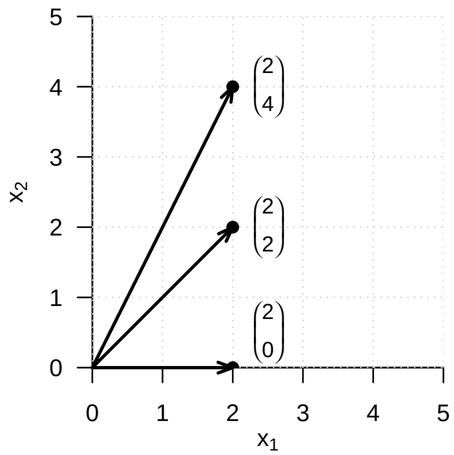

Figure 8.5 Vector length in $\mathbb{R}^2$

For the distance $d(x, y)$ between two vectors $x, y \in \mathbb{R}^m$, we note without proof that it is nonnegative and symmetric, that is,

$$d(x,y) \geq 0, d(x,x) = 0 \text{ and } d(x,y) = d(y,x) \tag{8.18}$$

hold. Moreover, $d(x, y)$ satisfies the so-called *triangle inequality*, which states that the direct path between two points in space is always shorter than an indirect path through a third point,

$$d(x,y) \leq d(x,z) + d(z,y). \tag{8.19}$$

Thus, $d(x, y)$ satisfies important aspects of spatial intuition. We give two examples for determining distances between vectors in $\mathbb{R}^2$, which we visualize in Figure 8.6.

Example (1)

$$d\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}2\\2\end{pmatrix}\right) = \left\|\begin{pmatrix}1\\1\end{pmatrix} - \begin{pmatrix}2\\2\end{pmatrix}\right\| = \left\|\begin{pmatrix}-1\\-1\end{pmatrix}\right\| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2} \approx 1.41 \tag{8.20}$$

```
norm(matrix(c(1,1),nrow = 2) - matrix(c(2,2),nrow = 2), type = "2")
```

```
[1] 1.414214
```

Example (2)

$$d\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}4\\1\end{pmatrix}\right) = \left\|\begin{pmatrix}1\\1\end{pmatrix} - \begin{pmatrix}4\\1\end{pmatrix}\right\| = \left\|\begin{pmatrix}-3\\0\end{pmatrix}\right\| = \sqrt{(-3)^2 + 0^2} = \sqrt{9} = 3 \tag{8.21}$$

```
norm(matrix(c(1,1),nrow = 2) - matrix(c(1,4),nrow = 2), type = "2")
```

```
[1] 3
```

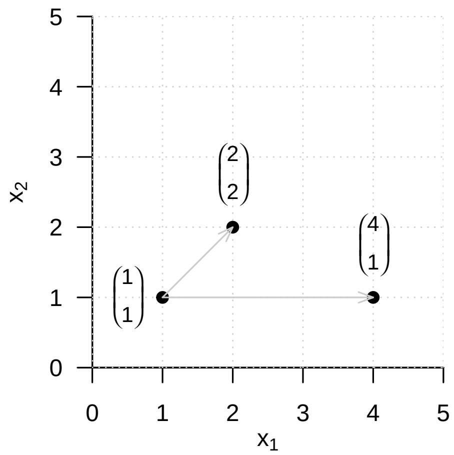

Figure 8.6 Vector distances in $\mathbb{R}^2$

Finally, we note that for the calculation of the angle between two vectors based on the above definition, the cosine function cos is bijective on $[0, \pi]$ and hence invertible with inverse function *acos*, the arccosine function. We also consider two examples for the concept of angle. Note in particular that Definition 8.6 gives the angle in radians. For a specification in degrees, a corresponding conversion is required.

Example (1)

$$\text{acos}\left(\frac{\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \end{pmatrix} \right\rangle}{\left\| \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\| \left\| \begin{pmatrix} 3 \\ 3 \end{pmatrix} \right\|}\right) = \text{acos}\left(\frac{3 \cdot 3 + 3 \cdot 0}{\sqrt{3^2 + 0^2} \cdot \sqrt{3^2 + 3^2}}\right) = \text{acos}\left(\frac{9}{3 \cdot \sqrt{18}}\right) = \frac{\pi}{4} \approx 0.785 \tag{8.22}$$

The conversion to degrees then gives

$$0.785 \cdot \frac{180^\circ}{\pi} = 45^\circ \tag{8.23}$$

In **R**, one implements this as follows.

```
x = matrix(c(3,0), nrow = 2)                                  # vector 1
y = matrix(c(3,3), nrow = 2)                                  # vector 2
w = acos(sum(x*y)/(sqrt(sum(x*x))*sqrt(sum(y*y)))) * 180/pi  # angle in degrees
print(w)
```

```
[1] 45
```

Example (2)

$$\alpha = \text{acos}\left(\frac{\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right\rangle}{\left\| \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\| \left\| \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right\|}\right) = \text{acos}\left(\frac{3 \cdot 0 + 0 \cdot 3}{\sqrt{3^2 + 0^2} \cdot \sqrt{0^2 + 3^2}}\right) = \text{acos}\left(\frac{0}{3 \cdot 3}\right) = \frac{\pi}{2} \approx 1.57 \tag{8.24}$$

The conversion to degrees then gives

$$\frac{\pi}{2} \cdot \frac{180^\circ}{\pi} = 90^\circ \tag{8.25}$$

The corresponding **R** implementation is as follows.

```
x = matrix(c(3,0), nrow = 2)                                  # vector 1
y = matrix(c(0,3), nrow = 2)                                  # vector 2
w = acos(sum(x*y)/(sqrt(sum(x*x))*sqrt(sum(y*y)))) * 180/pi   # angle in degrees
print(w)
```

```
[1] 90
```

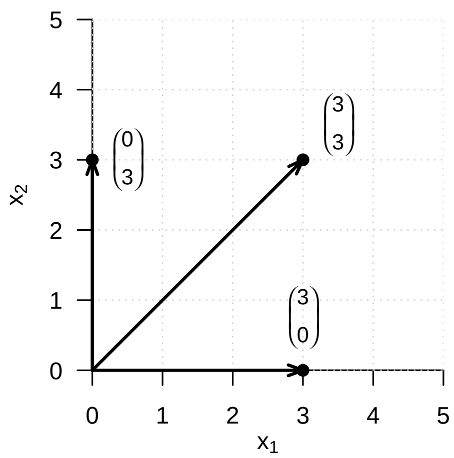

Figure 8.7 Angles in $\mathbb{R}^2$

The fact that two vectors can form a right angle, that is, can in a sense be maximally non-parallel, is an important geometric principle and is therefore singled out by the following definition.

Definition 8.7 (Orthogonality and orthonormality of vectors)**.** Let $((\mathbb{R}^m, +, \cdot), \langle\rangle)$ be the Euclidean vector space.

(1) Two vectors $x, y \in \mathbb{R}^m$ are called *orthogonal* if

$$\langle x, y\rangle = 0 \tag{8.26}$$

(2) Two vectors $x, y \in \mathbb{R}^m$ are called *orthonormal* if

$$\langle x, y\rangle = 0 \text{ and } \|x\| = \|y\| = 1. \tag{8.27}$$

•

For orthogonal and orthonormal vectors, in particular, we therefore also have

$$\cos\alpha = \frac{\langle x, y\rangle}{\|x\|\|y\|} = \frac{0}{\|x\|\|y\|} = 0, \tag{8.28}$$

and hence

$$\alpha = \frac{\pi}{2} = 90^\circ. \tag{8.29}$$

8.3 Linear independence

In this section, we introduce the concept of *linear independence* of vectors. To this end, we first define the concept of a *linear combination* of vectors.

Definition 8.8 (Linear combination)**.** Let $\{v_1, v_2, ..., v_k\}$ be a set of k vectors of a vector space V, and let $a_1, a_2, ..., a_k$ be scalars. Then the *linear combination* of the vectors in $\{v_1, v_2, ..., v_k\}$ with the *coefficients* $a_1, a_2, ..., a_k$ is defined as the vector

$$w := \sum_{i=1}^{k} a_i v_i \in V. \tag{8.30}$$

•

Example

Let

$$v_1 := \begin{pmatrix} 2 \\ 1 \end{pmatrix}, v_2 := \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_3 := \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } a_1 := 2, a_2 := 3, a_3 := 0. \tag{8.31}$$

Then the linear combination of v_1, v_2, v_3 with coefficients a_1, a_2, a_3 is

$$\begin{aligned} w &= a_1 v_1 + a_2 v_2 + a_3 v_3 \\ &= 2 \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ 5 \end{pmatrix}. \end{aligned} \tag{8.32}$$

Based on the concept of a linear combination, one can now define the concept of *linear independence* of vectors.

Definition 8.9 (Linear independence)**.** Let V be a vector space. A set $W := \{w_1, w_2, ..., w_k\}$ of vectors in V is called *linearly independent* if the only representation of the zero element $0 \in V$ by a linear combination of the $w \in W$ is the so-called *trivial representation*

$$0 = a_1 w_1 + a_2 w_2 + \cdots + a_k w_k \text{ with } a_1 = a_2 = \cdots = a_k = 0 \tag{8.33}$$

If the set W is not linearly independent, it is called *linearly dependent.*

•

To check whether a given set of vectors is linearly dependent or independent, in principle one has to check whether a possible linear combination of the given vectors is zero. Theorem 8.2 and Theorem 8.3 show how this can be achieved with less effort for two and for finitely many vectors, respectively.

Theorem 8.2 (Linear dependence of two vectors)**.** *Let V be a vector space. Two vectors $v_1, v_2 \in V$ are linearly dependent if one of the vectors is a scalar multiple of the other vector.*

$\circ$

Proof. Let v_1 be a scalar multiple of v_2, that is,

$$v_1 = \lambda v_2 \text{ with } \lambda \neq 0. \tag{8.34}$$

Then

$$v_1 - \lambda v_2 = 0. \tag{8.35}$$

But this corresponds to the linear combination

$$a_1 v_1 + a_2 v_2 = 0 \tag{8.36}$$

with $a_1 = 1 \neq 0$ and $a_2 = -\lambda \neq 0$. Thus, there exists a linear combination of the zero element that is not the trivial representation, and hence v_1 and v_2 are not linearly independent. $\square$

Theorem 8.3 (Linear dependence of a set of vectors)**.** *Let V be a vector space, and let $w_1, ..., w_k \in V$ be a set of vectors in V. If one of the vectors w_i with $i = 1, ..., k$ is a linear combination of the other vectors, then the set of vectors is linearly dependent.*

$\circ$

Proof. The vectors $w_1, ..., w_k$ are linearly dependent if and only if $\sum_{i=1}^k a_i w_i = 0$ with at least one $a_i \neq 0$. Thus, for example, let $a_j \neq 0$. Then

$$0 = \sum_{i=1}^k a_i w_i = \sum_{i=1, i\neq j}^k a_i w_i + a_j w_j \tag{8.37}$$

Therefore,

$$a_j w_j = -\sum_{i=1, i\neq j}^k a_i w_i \tag{8.38}$$

and thus

$$w_j = -a_j^{-1} \sum_{i=1, i\neq j}^k a_i w_i = -\sum_{i=1, i\neq j}^k (a_j^{-1} a_i) w_i \tag{8.39}$$

Thus, w_j is a linear combination of the w_i for $i = 1, ..., k$ with $i \neq j$. $\square$

8.4 Vector space bases

In this section, we introduce the concept of a *vector space basis*. A basis of a vector space is a subset of vectors of the vector space that can be used to represent all vectors of the vector space. In the sense of linear combinations of vectors, a vector space basis therefore contains all information needed to construct the corresponding vector space. However, a vector space basis is usually not unique, and many vector spaces indeed have infinitely many bases. The following definition first states how infinitely many vectors can be formed from a limited number of vectors by means of linear combinations.

Definition 8.10 (Span and spanning)**.** Let V be a vector space, and let $W := \{w_1, ..., w_k\} \subset V$. Then the *span* of W is defined as the set of all linear combinations of the elements of W,

$$\operatorname{Span}(W) := \left\{ \sum_{i=1}^{k} a_i w_i | a_1, ..., a_k \text{ are scalar coefficients} \right\} \tag{8.40}$$

A set of vectors $W \subseteq V$ is said to *span a vector space* V if every $v \in V$ can be written as a linear combination of vectors in W.

•

We now define the concept of a *basis* of a vector space.

Definition 8.11 (Basis)**.** Let V be a vector space, and let $B \subseteq V$. B is called a *basis of* V if

(1) the vectors in B are linearly independent, and
(2) the vectors in B span the vector space V.

•

Bases of vector spaces have the following important properties.

Theorem 8.4 (Properties of bases)**.**

(1) All bases of a vector space contain the same number of vectors.
(2) Every set of m linearly independent vectors is a basis of an m-dimensional vector space.

∘

For a proof of this very deep theorem, we refer to the advanced literature. The unique number of vectors in a basis of a vector space named by the above theorem is called the *dimension of the vector space*. Since there are usually infinitely many sets of m linearly independent vectors in a vector space, vector spaces usually have infinitely many bases.

If one considers a single vector in a vector space, one may ask how this vector can be represented with the help of a vector space basis. This leads to the following concepts.

Definition 8.12 (Basis representation and coordinates)**.** Let $B := \{b_1, ..., b_m\}$ be a basis of an m-dimensional vector space V, and let $v \in V$. Then the linear combination

$$v = \sum_{i=1}^{m} c_i b_i \tag{8.41}$$

is called the *representation of v with respect to the basis B*, and the coefficients $c_1, ..., c_m$ are called the *coordinates of v with respect to the basis B*.

•

For a fixed basis, the coordinates of a vector with respect to this basis are also fixed and unique. This is the statement of the following theorem.

Theorem 8.5 (Uniqueness of basis representation). *The basis representation of a $v \in V$ with respect to a basis B is unique.*

$\circ$

Proof. Without loss of generality, assume that the vector space has dimension m. Assume that two representations of v with respect to the basis B exist, that is,

$$\begin{aligned} v &= a_1 b_1 + \cdots + a_m b_m \\ v &= c_1 b_1 + \cdots + c_m b_m \end{aligned} \tag{8.42}$$

Subtracting the lower equation from the upper equation gives

$$0 = (a_1 - c_1)b_1 + \cdots + (a_m - c_m)b_m \tag{8.43}$$

Because $b_1, ..., b_m$ are linearly independent, however, $(a_i - c_i) = 0$ for all $i = 1, ..., m$, and thus the two representations of v with respect to the basis B are identical. $\square$

At the end of this section, we consider a special basis of the real vector space.

Definition 8.13 (Orthonormal basis of $\mathbb{R}^m$). A set of m vectors $v_1, ..., v_m \in \mathbb{R}^m$ is called an *orthonormal basis* of $\mathbb{R}^m$ if $v_1, ..., v_m$ each have length 1 and are mutually orthogonal, that is, if

$$\langle v_i, v_j \rangle = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases}. \tag{8.44}$$

$\bullet$

We first consider an example of an orthonormal basis.

Example (1)

$$B_1 := \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \tag{8.45}$$

is an orthonormal basis of $\mathbb{R}^2$, because B_1 consists of two vectors and

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle = 1 \cdot 1 + 0 \cdot 0 = 1 + 0 = 1 \tag{8.46}$$

as well as

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1 \tag{8.47}$$

and

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle = 1 \cdot 0 + 0 \cdot 1 = 0 + 0 = 0 \tag{8.48}$$

hold.

For general real vector spaces, bases of the form of B_1 are singled out by the concept of the *canonical basis*.

Definition 8.14 (Canonical basis and standard unit vectors)**.** The orthonormal basis

$$B := \left\{e_1, ..., e_m | (e_i)_j = 1 \text{ for } i = j \text{ and } (e_i)_j = 0 \text{ for } i \neq j, \, i, j = 1, ..., m\right\} \subset \mathbb{R}^m \quad (8.49)$$

is called the *canonical basis* of $\mathbb{R}^m$, and the e_i are called *standard unit vectors.*

•

B_1 from Example (1) is therefore the canonical basis of $\mathbb{R}^2$.

The canonical basis of $\mathbb{R}^3$ is

$$B := \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}. \quad (8.50)$$

However, there are also non-canonical orthonormal bases. To see this, we consider another example.

Example (2)

$$B_2 := \left\{ \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\} \quad (8.51)$$

is also an orthonormal basis of $\mathbb{R}^2$, because B_2 consists of two vectors and

$$\left\langle \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{1}{2} = 1, \quad (8.52)$$

as well as

$$\left\langle \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\rangle = \left(-\frac{1}{\sqrt{2}}\right) \cdot \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{1}{2} = 1 \quad (8.53)$$

and

$$\left\langle \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\rangle = -\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = -\frac{1}{2} + \frac{1}{2} = 0 \quad (8.54)$$

hold.

We visualize the two orthonormal bases B_1 and B_2 of $\mathbb{R}^2$ in Figure 8.8.

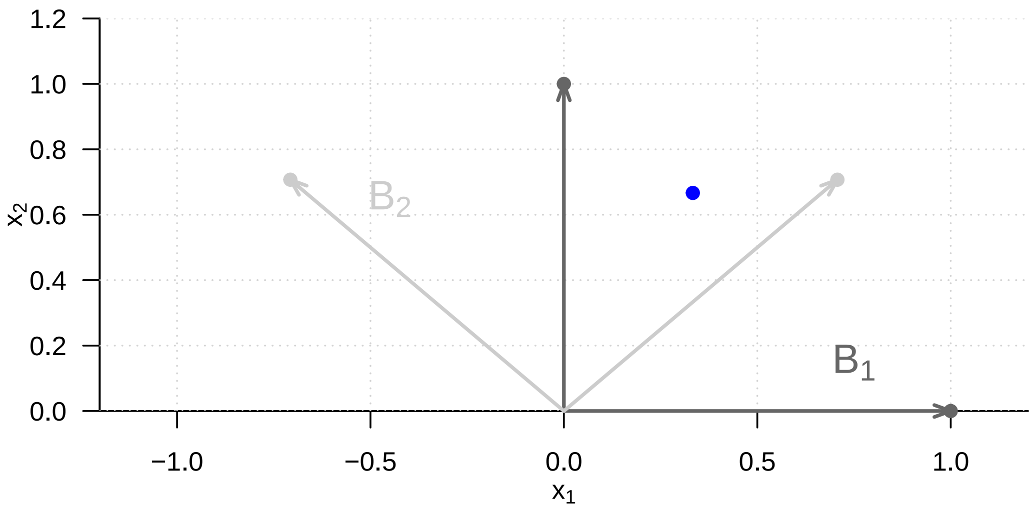

Figure 8.8 Two bases of $\mathbb{R}^2$

9 Matrices

Matrices are the words of the language of modern data analysis. An understanding of modern data-analytic methods and their implementation is not possible without a basic understanding of the concept of a matrix and knowledge of the fundamental matrix operations. Matrices can play very different roles. For example, matrices can represent data, experimental designs, and model parameters. In the context of linear algebra, matrices are used to represent linear mappings and vector spaces; here vectors are then understood as special matrices.

In this chapter, we introduce working with matrices while largely avoiding abstract terminology from linear algebra. We first introduce the concept of a matrix and then discuss the first fundamental matrix operations, namely matrix addition, matrix subtraction, scalar multiplication, and matrix transposition (Section 9.1 and Section 9.2). We then introduce the central concepts of matrix multiplication and matrix inversion (Section 9.3 and Section 9.4). With the matrix determinant, we discuss a first measure for describing matrices in Section 9.5. We close in Section 9.6 with an overview of particularly common matrices.

9.1 Definition

We begin with the definition of a matrix.

Definition 9.1. A matrix is a rectangular arrangement of numbers denoted as follows:

$$A := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} := (a_{ij})_{1 \le i \le n,\, 1 \le j \le m}. \tag{9.1}$$

•

Matrices consist of *rows* and *columns*. The matrix entries a_{ij} are indexed with a *row index* i and a *column index* j. For example, for

$$A := \begin{pmatrix} 2 & 7 & 5 & 2 \\ 8 & 2 & 5 & 6 \\ 6 & 4 & 0 & 9 \\ 9 & 2 & 1 & 2 \end{pmatrix}, \tag{9.2}$$

we have $a_{32} = 4$. The *size* or *dimension* of a matrix is determined by the number of its rows $n \in \mathbb{N}$ and columns $m \in \mathbb{N}$. Matrices with $n = m$ are called *square matrices*.

In what follows, we only need matrices with real entries, that is, $a_{ij} \in \mathbb{R}$ for all $i = 1, ..., n$ and $j = 1, ..., m$. We call matrices with real entries *real matrices* and denote the set of real matrices with n rows and m columns by $\mathbb{R}^{n\times m}$. From the expression

$$A \in \mathbb{R}^{n\times m} \tag{9.3}$$

we can therefore see that A is a real matrix with n rows and m columns. We identify the set $\mathbb{R}^{1\times 1}$ with the set $\mathbb{R}$ and the set $\mathbb{R}^{n\times 1}$ with the set $\mathbb{R}^n$. Thus, real matrices with one column and n rows correspond to n-dimensional real vectors, and real matrices with one column and one row correspond to real numbers.

Defining matrices in R

In **R**, matrices are defined by transforming **R** vectors into the representation of a mathematical matrix with the `matrix()` function. The entries of an **R** vector are distributed over the matrix according to their total number and the specified number of rows `nrow`. For example, if we want to define the matrix

$$A := \begin{pmatrix} 2 & 3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \tag{9.4}$$

in **R**, we obtain

```
# columnwise definition of A (R default)
A = matrix(c(2,1,3,6,0,5), nrow = 2)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

By default, **R** follows a so-called *column-major order*, that is, the elements of the **R** vector `c(2,1,3,6,0,5)` are transferred one after another from top to bottom into the columns of the matrix from left to right. A somewhat clearer connection between the visual layout of the **R** code and the resulting matrix is obtained by formatting the **R** vector with line breaks according to the intended matrix layout and then changing the column-major order to *row-major order* with the argument `byrow = TRUE`. Then the first row of the matrix is filled from left to right with the elements of the **R** vector, then the second row, and so on until all elements of the vector are distributed over the matrix.

```
# rowwise definition of A (byrow = TRUE)
A = matrix(c(2,3,0,
             1,6,5),
             nrow = 2,
             byrow = TRUE)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

```
# rowwise definition of B
B = matrix(c(4,1,0,
            -4,2,0),
            nrow = 2,
            byrow = TRUE)
print(B)
```

```
     [,1] [,2] [,3]
[1,]    4    1    0
[2,]   -4    2    0
```

9.2 Basic matrix operations

One can compute with matrices. The following matrix operations are fundamental:

- the addition of matrices of the same size, called *matrix addition*,
- the subtraction of matrices of the same size, called *matrix subtraction*,
- the multiplication of a matrix by a scalar, called *scalar multiplication*,
- the interchange of the rows and columns of a matrix, called *matrix transposition*.

In what follows, we introduce these operations in operator form, that is, as functions. This serves in particular to emphasize, for each operation, by means of its domain what kind of objects the respective operation takes, and by means of its codomain what kind of object the result of the respective operation is.

9.2.1 Matrix addition

Definition 9.2. Let $A, B \in \mathbb{R}^{n\times m}$. Then the *addition* of A and B is defined as the mapping

$$+ : \mathbb{R}^{n\times m} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m},\ (A,B) \mapsto +(A,B) := A + B \tag{9.5}$$

with

$$\begin{aligned}
A + B &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{pmatrix} \\
&:= \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1m}+b_{1m} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2m}+b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}+b_{n1} & a_{n2}+b_{n2} & \cdots & a_{nm}+b_{nm} \end{pmatrix}.
\end{aligned} \tag{9.6}$$

•

In particular, the definition of matrix addition specifies that only matrices of the same size can be added and that the operation of matrix addition is defined elementwise.

Example

Let $A, B \in \mathbb{R}^{2\times 3}$ be defined as

$$A := \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \text{ and } B := \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix}. \tag{9.7}$$

Because A and B have the same size, we can add them:

$$\begin{aligned}
C = A + B &= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} + \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 2+4 & -3+1 & 0+0 \\ 1-4 & 6+2 & 5+0 \end{pmatrix} \\
&= \begin{pmatrix} 6 & -2 & 0 \\ -3 & 8 & 5 \end{pmatrix}.
\end{aligned} \tag{9.8}$$

In **R**, one carries out the above calculation as follows.

```
# definition
A = matrix(c(2, -3, 0,
             1,  6, 5),
             nrow  = 2,
             byrow = TRUE)
B = matrix(c( 4, 1, 0,
             -4, 2, 0),
              nrow = 2,
             byrow = TRUE)

# addition
C = A + B
print(C)
```

```
     [,1] [,2] [,3]
[1,]    6   -2    0
[2,]   -3    8    5
```

9.2.2 Matrix subtraction

The subtraction of matrices of the same size is defined analogously to addition.

Definition 9.3 (Matrix subtraction)**.** Let $A, B \in \mathbb{R}^{n \times m}$. Then the *subtraction* of A and B is defined as the mapping

$$-: \mathbb{R}^{n\times m} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m},\ (A,B) \mapsto -(A,B) := A - B \tag{9.9}$$

with

$$\begin{aligned} A - B &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{pmatrix} \\ &:= \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} & \cdots & a_{1m} - b_{1m} \\ a_{21} - b_{21} & a_{22} - b_{22} & \cdots & a_{2m} - b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} - b_{n1} & a_{n2} - b_{n2} & \cdots & a_{nm} - b_{nm} \end{pmatrix}. \end{aligned} \tag{9.10}$$

•

As with matrix addition, the definition of matrix subtraction specifies that only matrices of the same size can be subtracted from one another and that the subtraction of two equally sized matrices is defined elementwise.

Example

We can also subtract the matrices A and B defined in the example on matrix addition from one another:

$$\begin{aligned} D = A - B &= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} - \begin{pmatrix} 4 & 1 & 0 \\ -4 & 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2-4 & -3-1 & 0-0 \\ 1+4 & 6-2 & 5-0 \end{pmatrix} \\ &= \begin{pmatrix} -2 & -4 & 0 \\ 5 & 4 & 5 \end{pmatrix}. \end{aligned} \tag{9.11}$$

In **R**, one carries out this calculation as follows.

```
# subtraction
D = A - B
print(D)
```

```
     [,1] [,2] [,3]
[1,]   -2   -4    0
[2,]    5    4    5
```

9.2.3 Scalar multiplication

The *scalar multiplication* of a matrix denotes the multiplication of a scalar by a matrix.

Definition 9.4 (Scalar multiplication)**.** Let $c \in \mathbb{R}$ be a scalar and let $A \in \mathbb{R}^{n\times m}$. Then the *scalar multiplication* of c and A is defined as the mapping

$$\cdot : \mathbb{R} \times \mathbb{R}^{n\times m} \to \mathbb{R}^{n\times m}, \ (c, A) \mapsto \cdot(c, A) := cA \tag{9.12}$$

with

$$cA = c\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} := \begin{pmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{pmatrix}. \tag{9.13}$$

•

With this definition, scalar multiplication is therefore defined elementwise.

Example

Let $c := -3$ and let $A \in \mathbb{R}^{4\times 3}$ be defined as

$$A := \begin{pmatrix} 3 & 1 & 1 \\ 5 & 2 & 5 \\ 2 & 7 & 1 \\ 3 & 4 & 2 \end{pmatrix}. \tag{9.14}$$

Then

$$B := cA = -3\begin{pmatrix} 3 & 1 & 1 \\ 5 & 2 & 5 \\ 2 & 7 & 1 \\ 3 & 4 & 2 \end{pmatrix} = \begin{pmatrix} -3\cdot 3 & -3\cdot 1 & -3\cdot 1 \\ -3\cdot 5 & -3\cdot 2 & -3\cdot 5 \\ -3\cdot 2 & -3\cdot 7 & -3\cdot 1 \\ -3\cdot 3 & -3\cdot 4 & -3\cdot 2 \end{pmatrix} = \begin{pmatrix} -9 & -3 & -3 \\ -15 & -6 & -15 \\ -6 & -21 & -3 \\ -9 & -12 & -6 \end{pmatrix}. \tag{9.15}$$

In **R**, one carries out this scalar multiplication as follows.

```
# definitions
A = matrix(c(3,1,1,
             5,2,5,
             2,7,1,
             3,4,2),
           nrow = 4,
           byrow = TRUE)
c = -3

# scalar multiplication
B = c*A
print(B)
```

```
     [,1] [,2] [,3]
[1,]   -9   -3   -3
[2,]  -15   -6  -15
[3,]   -6  -21   -3
[4,]   -9  -12   -6
```

With the help of the definition of matrix addition and scalar multiplication, it is possible to define a vector space whose elements are the real matrices. In particular, this definition also specifies the arithmetic rules for working with matrix addition and scalar multiplication.

Theorem 9.1 (Vector space of real-valued matrices)**.** *The triple* $(\mathbb{R}^{n\times m}, +, \cdot)$ *with the matrix addition and scalar multiplication defined above is a vector space. In particular, for* $A, B, C \in \mathbb{R}^{n\times m}$ *and* $r, s, t \in \mathbb{R}$*, the following arithmetic rules therefore hold:*

(1) Commutativity of addition: $A + B = B + A$.
(2) Associativity of addition: $(A + B) + C = A + (B + C)$.
(3) Existence of an identity element for addition: $\exists\, 0 \in \mathbb{R}^{n\times m}$ *with* $A + 0 = 0 + A = A$.
(4) Existence of inverse elements for addition: $\forall A\, \exists -A \in \mathbb{R}^{n\times m}$ *with* $A + (-A) = 0$.
(5) Existence of an identity element for scalar multiplication: $\exists\, 1 \in \mathbb{R}$ *with* $1 \cdot A = A$.
(6) Associativity of scalar multiplication: $r \cdot (s \cdot A) = (r \cdot s) \cdot A$.
(7) Distributivity with respect to matrix addition: $r \cdot (A + B) = r \cdot A + r \cdot B$.
(8) Distributivity with respect to scalar addition: $(r + s) \cdot A = r \cdot A + s \cdot A$.

∘

We omit a proof, which follows directly, with some notational effort, from the elementwise character of matrix addition and scalar multiplication as well as the arithmetic rules known from working with real numbers. The identity element of addition mentioned in the theorem is called the *zero matrix*; we will introduce a general notation for it later. The inverse elements of addition are given by

$$-A := (-a_{ij})_{1\le i\le n, 1\le j\le m} \tag{9.16}$$

and make it possible to view matrix subtraction as a special case of matrix addition.

9.2.4 Matrix transposition

Another frequently occurring basic matrix operation is the interchange of the row and column arrangement of a matrix, called *matrix transposition*.

Definition 9.5 (Matrix transposition)**.** Let $A \in \mathbb{R}^{n\times m}$. Then the *transposition* of A is defined as the mapping

$$\cdot^T : \mathbb{R}^{n\times m} \to \mathbb{R}^{m\times n},\ A \mapsto \cdot^T(A) := A^T \tag{9.17}$$

with

$$A^T = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{pmatrix}. \tag{9.18}$$

•

Thus, for $A \in \mathbb{R}^{n\times m}$, we always have $A^T \in \mathbb{R}^{m\times n}$. Furthermore, the following arithmetic rules of matrix transposition hold, as one can see from examples:

(1) For $A \in \mathbb{R}^{1\times 1}$,
$$A^T = A. \tag{9.19}$$
(2) We have
$$\left(A^T\right)^T = A. \tag{9.20}$$
(3) We have
$$(a_{ii})_{1\le i\le \min(n,m)} = (a_{ii})^T_{1\le i\le \min(n,m)}\,. \tag{9.21}$$

The latter property of transposition states that the elements on the main diagonal of a matrix remain unchanged under transposition.

Example

Let $A \in \mathbb{R}^{2\times 3}$ be defined by
$$A := \begin{pmatrix} 2 & 3 & 0 \\ 1 & 6 & 5 \end{pmatrix}, \tag{9.22}$$
Then $A^T \in \mathbb{R}^{3\times 2}$ and, specifically,
$$A^T := \begin{pmatrix} 2 & 1 \\ 3 & 6 \\ 0 & 5 \end{pmatrix}. \tag{9.23}$$
Furthermore, apparently $\min(m,n) = 2$, and consequently
$$(a_{11}) = (a_{11})^T \text{ and } (a_{22}) = (a_{22})^T\,. \tag{9.24}$$
In **R**, one transposes a matrix as follows.

```
# definition
A = matrix(c(2,3,0,
             1,6,5),
           nrow = 2,
           byrow = TRUE)
print(A)
```

```
     [,1] [,2] [,3]
[1,]    2    3    0
[2,]    1    6    5
```

```
# transposition
AT = t(A)
print(AT)
```

```
     [,1] [,2]
[1,]    2    1
[2,]    3    6
[3,]    0    5
```

Finally, in connection with matrix addition, matrix subtraction, and scalar multiplication, the following arithmetic rules hold, as one can see from examples:

(1) For $A, B \in \mathbb{R}^{n\times m}$,
$$(A+B)^T = A^T + B^T. \tag{9.25}$$
(2) For $A, B \in \mathbb{R}^{n\times m}$,
$$(A-B)^T = A^T - B^T. \tag{9.26}$$
(3) For $c \in \mathbb{R}$ and $A \in \mathbb{R}^{n\times m}$,
$$(cA)^T = cA^T. \tag{9.27}$$

9.3 Matrix multiplication

Matrix multiplication is the central operation when computing with matrices. It is defined as follows.

Definition 9.6 (Matrix multiplication)**.** Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$. Then the *matrix multiplication* of A and B is defined as the mapping

$$\cdot : \mathbb{R}^{n\times m} \times \mathbb{R}^{m\times k} \to \mathbb{R}^{n\times k},\ (A,B) \mapsto \cdot(A,B) := AB \tag{9.28}$$

with

$$\begin{aligned}
AB &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mk} \end{pmatrix} \\
&:= \begin{pmatrix} \sum_{i=1}^m a_{1i}b_{i1} & \sum_{i=1}^m a_{1i}b_{i2} & \cdots & \sum_{i=1}^m a_{1i}b_{ik} \\ \sum_{i=1}^m a_{2i}b_{i1} & \sum_{i=1}^m a_{2i}b_{i2} & \cdots & \sum_{i=1}^m a_{2i}b_{ik} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^m a_{ni}b_{i1} & \sum_{i=1}^m a_{ni}b_{i2} & \cdots & \sum_{i=1}^m a_{ni}b_{ik} \end{pmatrix} \\
&= \left(\sum_{i=1}^m a_{ji}b_{il}\right)_{1\le j\le n, 1\le l\le k}
\end{aligned} \tag{9.29}$$

•

The matrix product AB is therefore defined only if A has exactly as many columns as B has rows. Informally, the following mnemonic applies to the involved matrix sizes:

$$(n\times m)(m\times k) = (n\times k). \tag{9.30}$$

The entry $(AB)_{ij}$ in AB corresponds to the sum of the multiplied ith row of A and jth column of B. To compute $(AB)_{ij}$ for $i = 1, ..., n$ and $j = 1, ..., k$, one can therefore proceed mentally as follows:

(1) Place the transpose of the ith row of A over the jth column of B.
(2) Because A has exactly m columns and B has exactly m rows, each element of the row from A has a corresponding element in the column of B.
(3) Multiply the corresponding elements with one another.
(4) The sum of these products is then the entry with index ij in AB.

Example

Let $A \in \mathbb{R}^{2\times 3}$ and $B \in \mathbb{R}^{3\times 2}$ be defined as

$$A := \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \text{ and } B := \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix}. \tag{9.31}$$

We want to compute $C := AB$ and $D := BA$. With $n = 2, m = 3$, and $k = 2$, we already know that $C \in \mathbb{R}^{2\times 2}$ and $D \in \mathbb{R}^{3\times 3}$, because

$$(2\times 3)(3\times 2) = (2\times 2) \tag{9.32}$$

and

$$(3 \times 2)(2 \times 3) = (3 \times 3). \tag{9.33}$$

Thus here surely $AB \neq BA$. For C, we obtain

$$\begin{aligned}
C &= AB \\
&= \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix} \\
&= \begin{pmatrix} 2 \cdot 4 + (-3) \cdot (-1) + 0 \cdot 1 & 2 \cdot 2 + (-3) \cdot 0 + 0 \cdot 3 \\ 1 \cdot 4 + 6 \cdot (-1) + 5 \cdot 1 & 1 \cdot 2 + 6 \cdot 0 + 5 \cdot 3 \end{pmatrix} \\
&= \begin{pmatrix} 8 + 3 + 0 & 4 + 0 + 0 \\ 4 - 6 + 5 & 2 + 0 + 15 \end{pmatrix} \\
&= \begin{pmatrix} 11 & 4 \\ 3 & 17 \end{pmatrix}.
\end{aligned} \tag{9.34}$$

In **R**, the %*% operator is used for matrix multiplication.

```
# definitions
A = matrix(c(2,-3,0,
             1, 6,5),
           nrow  = 2,
           byrow = TRUE)
B = matrix(c( 4,2,
             -1,0,
              1,3),
           nrow  = 3,
           byrow = TRUE)

# matrix multiplication
C = A %*% B
print(C)
```

```
     [,1] [,2]
[1,]   11    4
[2,]    3   17
```

For D, we further obtain

$$\begin{aligned}
D &= BA \\
&= \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & -3 & 0 \\ 1 & 6 & 5 \end{pmatrix} \\
&= \begin{pmatrix} 4 \cdot 2 + 2 \cdot 1 & 4 \cdot (-3) + 2 \cdot 6 & 4 \cdot 0 + 2 \cdot 5 \\ (-1) \cdot 2 + 0 \cdot 1 & (-1) \cdot (-3) + 0 \cdot 6 & (-1) \cdot 0 + 0 \cdot 5 \\ 1 \cdot 2 + 3 \cdot 1 & 1 \cdot (-3) + 3 \cdot 6 & 1 \cdot 0 + 3 \cdot 5 \end{pmatrix} \\
&= \begin{pmatrix} 8 + 2 & -12 + 12 & 0 + 5 \\ -2 + 0 & 3 + 0 & 0 + 0 \\ 2 + 3 & -3 + 18 & 0 + 15 \end{pmatrix} \\
&= \begin{pmatrix} 10 & 0 & 10 \\ -2 & 3 & 0 \\ 5 & 15 & 15 \end{pmatrix}
\end{aligned} \tag{9.35}$$

In **R**, one checks this calculation as follows.

```r
# definitions
A = matrix(c(2,-3,0,
             1, 6,5),
           nrow  = 2,
           byrow = TRUE)
B = matrix(c( 4,2,
             -1,0,
              1,3),
           nrow  = 3,
           byrow = TRUE)

# matrix multiplication
D = B %*% A
print(D)
```

```
     [,1] [,2] [,3]
[1,]   10    0   10
[2,]   -2    3    0
[3,]    5   15   15
```

However, if a matrix multiplication is not defined because of incompatible matrix sizes, then it also cannot be evaluated numerically.

```r
# example of an undefined matrix multiplication
E = t(A) %*% B      # (3 x 2)(3 x 2)
```

```
Error in `t(A) %*% B`:
! nicht passende Argumente
```

The following theorem, which we do not prove, establishes the relation between the inner product of two vectors and the multiplication of two matrices. This essentially follows from the identification of $\mathbb{R}^n$ and $\mathbb{R}^{n\times 1}$ and from the fact that, by definition, the entry $(AB)_{ij}$ in the product of $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$ corresponds to the vector inner product of the ith column of A^T and the jth column of B.

Theorem 9.2 (Matrix multiplication and vector inner product)**.** *Let $x, y \in \mathbb{R}^n$. Then*

$$\langle x, y\rangle = x^T y. \tag{9.36}$$

Furthermore, for $A \in \mathbb{R}^{n\times m}$ and $i = 1, ..., n$, let

$$\bar{a}_i := (a_{ij})_{1\le j\le m} \in \mathbb{R}^m \tag{9.37}$$

be the columns of A^T, and for $B \in \mathbb{R}^{m\times k}$ and $j = 1, ..., k$, let

$$\bar{b}_j := (b_{ij})_{1\le i\le m} \in \mathbb{R}^m \tag{9.38}$$

be the columns of B, that is,

$$A^T = \begin{pmatrix} \bar{a}_1 & \bar{a}_2 & \cdots & \bar{a}_n \end{pmatrix} \in \mathbb{R}^{m\times n} \text{ and } B = \begin{pmatrix} \bar{b}_1 & \bar{b}_2 & \cdots & \bar{b}_k \end{pmatrix} \in \mathbb{R}^{m\times k}. \tag{9.39}$$

Then

$$AB = \left(\langle \bar{a}_i, \bar{b}_j\rangle\right)_{1\le i\le n, 1\le j\le k}. \tag{9.40}$$

○

9.3.1 Arithmetic rules for matrix multiplication

In the following, we collect a few basic arithmetic rules for matrix multiplication, especially also in combination with other matrix operations.

For proofs of the following two theorems on associativity and distributivity, which essentially follow from the corresponding arithmetic rules for sums and products of real numbers, we refer to the advanced literature.

Theorem 9.3 (Associativity). *Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{m\times k}$, $C \in \mathbb{R}^{k\times p}$, and $c \in \mathbb{R}$. Then:*

(1) Matrix multiplication is associative, that is,

$$A(BC) = (AB)C. \tag{9.41}$$

(2) The combination of matrix multiplication and scalar multiplication is associative,

$$c(AB) = (cA)B = A(cB). \tag{9.42}$$

∘

The associativity of matrix multiplication and scalar multiplication can be seen easily by considering the j,lth element of $c(AB)$, $(cA)B$, and $A(cB)$:

$$c\left(\sum_{i=1}^{m} a_{ji}b_{il}\right) = \sum_{i=1}^{m}\left(ca_{ji}\right)b_{il} = \sum_{i=1}^{m} a_{ji}\left(cb_{il}\right). \tag{9.43}$$

Theorem 9.4 (Distributivity). *Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{n\times m}$, and $C \in \mathbb{R}^{m\times p}$. Then*

$$(A + B)C = AC + BC \tag{9.44}$$

and

$$C^T(A + B) = C^T A + C^T B \tag{9.45}$$

∘

In contrast to the commutativity of multiplication of real numbers, matrix multiplication is in general not commutative.

Theorem 9.5 (Noncommutativity). *Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times p}$. In general,*

$$AB \neq BA. \tag{9.46}$$

∘

Proof. If $p \neq n$, then BA is not defined, so we consider only the case $p = n$. We show by giving a counterexample with $A, B \in \mathbb{R}^{2\times 2}$ that in general $AB = BA$ does *not* hold. Let

$$A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ and } B := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{9.47}$$

Then

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = BA. \tag{9.48}$$

□

Theorem 9.6 (Combination of matrix multiplication and transposition)**.** *Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$. Then*

$$(AB)^T = B^T A^T. \tag{9.49}$$

∘

Proof. A proof is obtained as follows:

$$\begin{aligned}
(AB)^T &= \left(\left(\sum_{i=1}^n a_{ji} b_{il}\right)_{1 \le j \le m, 1 \le l \le k}\right)^T \\
&= \left(\sum_{i=1}^n a_{li} b_{ij}\right)_{1 \le j \le k, 1 \le l \le m} \\
&= \left(\sum_{i=1}^n b_{ij} a_{li}\right)_{1 \le j \le k, 1 \le l \le m} \\
&= B^T A^T.
\end{aligned} \tag{9.50}$$

□

9.4 Matrix inversion

To motivate the concept of the inverse matrix, we first consider the problem of *solving a system of linear equations.* To this end, let $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^n$, and suppose that

$$Ax = b. \tag{9.51}$$

Let A and b be known, and let x be unknown. Specifically, for example, let

$$A := \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } b := \begin{pmatrix} 5 \\ 11 \end{pmatrix}. \tag{9.52}$$

Then the following system of linear equations with two equations and two unknowns is given:

$$Ax = b \Leftrightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} \Leftrightarrow \begin{matrix} 1x_1 + 2x_2 & = 5 \\ 3x_1 + 4x_2 & = 11 \end{matrix}. \tag{9.53}$$

The goal of solving systems of linear equations is, as is well known, to find those x for which the system is satisfied. To introduce the concept of the inverse matrix of A in this context, we simplify the situation further. We assume that $A = a$ is a 1×1 matrix, that is, a scalar, and likewise that x and b are scalars, so that for $a, x, b \in \mathbb{R}$ we have the equation

$$ax = b \tag{9.54}$$

To solve this equation for x, one would naturally multiply both sides of the equation by the *multiplicative inverse* of a, where the *multiplicative inverse* of a denotes the value which, when multiplied by a, yields 1. As is well known, this is given by

$$a^{-1} = \frac{1}{a} \tag{9.55}$$

Then

$$ax = b \Leftrightarrow a^{-1}ax = a^{-1}b \Leftrightarrow 1 \cdot x = a^{-1}b \Leftrightarrow x = \frac{b}{a}. \tag{9.56}$$

For example, quite concretely,

$$2x = 6 \Leftrightarrow 2^{-1}2x = 2^{-1}6 \Leftrightarrow \frac{1}{2}2x = \frac{1}{2}6 \Leftrightarrow x = 3. \tag{9.57}$$

Analogously to the case in which all matrices in $Ax = b$ are scalars, in the case of a system of linear equations one would like to be able to multiply both sides of the equation by the *multiplicative inverse* A^{-1} of A, so that an equation of the form

$$A^{-1}A = \text{"1"}. \tag{9.58}$$

results. Then one would have

$$Ax = b \Leftrightarrow A^{-1}Ax = A^{-1}b \Leftrightarrow x = A^{-1}b. \tag{9.59}$$

This intuitive idea of the multiplicative inverse of a matrix A is formalized below under the concept of the *inverse matrix.* To this end, we first need the concept of the *identity matrix.*

Definition 9.7 (Identity matrix)**.** The matrix

$$I_n := (a_{ij})_{1\le i\le n, 1\le j\le n} \in \mathbb{R}^{n\times n} := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \tag{9.60}$$

with $a_{ij} = 1$ for $i = j$ and $a_{ij} = 0$ for $i \neq j$ is called the *n-dimensional identity matrix.*

•

In **R**, I_n is generated with the command `diag(n)`. For matrix multiplication, the identity matrix is the analogue of the 1 in the multiplication of real numbers. This is the statement of the following theorem.

Theorem 9.7 (Identity element of matrix multiplication)**.** *I_n is the identity element of matrix multiplication, that is, for $A \in \mathbb{R}^{n\times m}$ we have*

$$I_nA = A \text{ and } AI_m = A. \tag{9.61}$$

∘

Proof. Let $B = (b_{ij}) = I_nA \in \mathbb{R}^{n\times m}$. Then, for all $1 \le i \le n$ and all $1 \le j \le m$,

$$b_{ij} = 0 \cdot a_{1j} + 0 \cdot a_{2j} + \cdots + 0 \cdot a_{i-1,j} + 1 \cdot a_{ij} + \cdots + 0 \cdot a_{i+1,j} + 0 \cdot a_{nj} = a_{ij}. \tag{9.62}$$

The proof for AI_m is analogous. □

With the concept of the identity matrix, we can now define the concepts of the inverse matrix and the invertible matrix:

Definition 9.8 (Invertible matrix and inverse matrix)**.** A square matrix $A \in \mathbb{R}^{n\times n}$ is called *invertible* if there exists a square matrix $A^{-1} \in \mathbb{R}^{n\times n}$ such that

$$A^{-1}A = AA^{-1} = I_n \tag{9.63}$$

The matrix A^{-1} is called the *inverse matrix of* A.

•

Note that the concepts of the inverse matrix and invertibility refer *only* to square matrices. In particular, square matrices can be invertible, but need not be (systems of linear equations can therefore have solutions, but need not). Non-invertible matrices are also called *singular* matrices, and invertible matrices are sometimes called *nonsingular* matrices. Finally, note that Definition 9.8 merely states what an inverse matrix is, but not how to compute it.

Example of an invertible matrix

The matrix

$$A := \begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix} \tag{9.64}$$

is invertible, and its inverse matrix is given by

$$A^{-1} = \begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix}, \tag{9.65}$$

because

$$\begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix}\begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{pmatrix}\begin{pmatrix} 2.0 & 1.0 \\ 3.0 & 4.0 \end{pmatrix}, \tag{9.66}$$

as can be verified by direct calculation.

Example of a non-invertible matrix

The matrix

$$B := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{9.67}$$

is not invertible, because if B were invertible, then there would exist

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{9.68}$$

with

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{9.69}$$

But this would mean that $0 = 1$ in $\mathbb{R}$, which is a contradiction. Therefore, B cannot be invertible.

On computing inverse matrices

2×2 up to about 5×5 matrices can in principle be inverted by hand; linear algebra provides various methods for this. We omit an introduction to matrix inversion by hand here, since matrices are inverted numerically by default in applications. Numerical matrix inversion is also a large field of research in numerical mathematics, which provides a variety of algorithms for this purpose. In **R**, matrices are inverted by default with the function `solve()`, in analogy to solving systems of linear equations. For the above example of an invertible matrix, this yields the following **R** code.

```
# definition
A = matrix(c(2,1,
             3,4),
           nrow  = 2,
           byrow = TRUE)

# computing A^{-1}
print(solve(A))
```

```
     [,1] [,2]
[1,]  0.8 -0.2
[2,] -0.6  0.4
```

```
# checking the properties of an inverse matrix
print(solve(A) %*% A)
```

```
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

```
# the reverse calculation yields small rounding errors
print(A %*% solve(A))
```

```
     [,1]          [,2]
[1,]    1 -5.551115e-17
[2,]    0  1.000000e+00
```

Non-invertible matrices are, of course, also numerically non-invertible, as the following error message in **R** demonstrates for the above example of a non-invertible matrix.

```
B = matrix(c(1,0,
             0,0),
           nrow  = 2,
           byrow = TRUE)
solve(B)
```

```
Error in `solve.default()`:
! Lapackroutine dgesv: System ist genau singulär: U[2,2] = 0
```

9.5 Determinants

The determinant is a versatile measure of a square matrix. For understanding *eigenanalysis* and *matrix decomposition*, the concept of the determinant is fundamental in the context of the *characteristic polynomial*.

In general, a determinant is a nonlinear mapping of the form

$$|\cdot| : \mathbb{R}^{n\times n} \to \mathbb{R}, A \mapsto |A|, \tag{9.70}$$

that is, a determinant assigns the real number $|A|$ to a square matrix A. The number $|A|$ is determined recursively by the following definition.

Definition 9.9 (Determinant)**.** For $A = (a_{ij})_{1\le i,j\le n} \in \mathbb{R}^{n\times n}$ with $n > 1$, let $A_{ij} \in \mathbb{R}^{n-1\times n-1}$ be the matrix obtained from A by removing the ith row and the jth column. Then the number

$$|A| := a_{11} \quad \text{for } n = 1 \tag{9.71}$$

$$|A| := \sum_{j=1}^{n} a_{1j}(-1)^{1+j} \det\left(A_{1j}\right) \quad \text{for } n > 1 \tag{9.72}$$

is called the *determinant of A*.

•

The definition thus successively reduces the determination of the determinant of a square matrix, by deleting rows and columns, to the determinant of a 1×1 matrix, which is given by its only element. For

$$A := \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \in \mathbb{R}^{3\times 3} \tag{9.73}$$

we obtain, for example, the following matrices of the form $A_{ij} \in \mathbb{R}^{3-1\times 3-1}$:

$$A_{11} = \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix}, A_{12} = \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix}, A_{21} = \begin{pmatrix} 2 & 3 \\ 8 & 9 \end{pmatrix}, A_{22} = \begin{pmatrix} 1 & 3 \\ 7 & 9 \end{pmatrix}. \tag{9.74}$$

For computing determinants of two- and three-dimensional square matrices, there are direct, non-recursive arithmetic rules, which are recorded in the following theorem.

Theorem 9.8 (Determinants of two- and three-dimensional matrices)**.**
Let $A = (a_{ij})_{1\le i,j\le 2} \in \mathbb{R}^{2\times 2}$. Then

$$|A| = a_{11}a_{22} - a_{12}a_{21}. \tag{9.75}$$

Let $A = (a_{ij})_{1\le i,j\le 3} \in \mathbb{R}^{3\times 3}$. Then

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}. \tag{9.76}$$

∘

Proof. For $A \in \mathbb{R}^{2\times 2}$, by definition

$$\begin{aligned} |A| &= \sum_{j=1}^{n} a_{1j}(-1)^{1+j}|A_{1j}| \\ &= a_{11}(-1)^{1+1}|A_{11}| + a_{12}(-1)^{1+2}|A_{12}| \\ &= a_{11}|(a_{22})| - a_{12}|(a_{21})| \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned} \tag{9.77}$$

For $A \in \mathbb{R}^{3\times 3}$, by definition and with the formula for determinants of 2×2 matrices,

$$
\begin{aligned}
|A| &= \sum_{j=1}^{n} a_{1j}(-1)^{1+j}|A_{1j}| \\
&= a_{11}(-1)^{1+1}|A_{11}| + a_{12}(-1)^{1+2}|A_{12}| + a_{13}(-1)^{1+3}|A_{13}| \\
&= a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}| \\
&= a_{11}\left|\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}\right| - a_{12}\left|\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix}\right| + a_{13}\left|\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}\right| \\
&= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
&= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \\
&= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}.
\end{aligned}
\tag{9.78}
$$

□

For determining the determinants of 2×2 and 3×3 matrices, the so-called *rule of Sarrus* therefore applies:

> "sum of the products on the diagonals minus sum of the products on the counterdiagonals."

For 3×3 matrices, the mnemonic refers to the scheme

$$
\left(\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right). \tag{9.79}
$$

Examples for determinants of 2×2 and 3×3 matrices

Let

$$
A := \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, B := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } C := \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \tag{9.80}
$$

Then

$$
|A| = 2 \cdot 4 - 1 \cdot 3 = 8 - 3 = 5 \tag{9.81}
$$

and

$$
|B| = 1 \cdot 0 - 0 \cdot 0 = 0 - 0 = 0 \tag{9.82}
$$

and

$$
|C| = 2 \cdot 1 \cdot 3 + 0 \cdot 0 \cdot 0 + 0 \cdot 0 \cdot 0 - 0 \cdot 0 \cdot 3 - 0 \cdot 0 \cdot 0 - 0 \cdot 1 \cdot 0 = 2 \cdot 1 \cdot 3 = 6. \tag{9.83}
$$

In **R**, one checks this with the `det()` function as follows.

```r
# matrix definition and determinant calculation
A = matrix(c(2,1,
             3,4),
           nrow = 2,
           byrow = TRUE)
det(A)
```

```
[1] 5
```

```r
# matrix definition and determinant calculation
B = matrix(c(1,0,
             0,0),
           nrow = 2,
           byrow = TRUE)
det(B)
```

```
[1] 0
```

```
# matrix definition and determinant calculation
C = matrix(c(2,0,0,
             0,1,0,
             0,0,3),
           nrow = 3,
           byrow = TRUE)
det(C)
```

```
[1] 6
```

There are numerous arithmetic rules for determinants in conjunction with matrix multiplication and matrix inversion. Without proof, we collect these in the following theorem.

Theorem 9.9 (Arithmetic rules for determinants)**.**

(Multiplication theorem for determinants.) For $A, B \in \mathbb{R}^{n \times n}$,

$$|AB| = |A||B|. \tag{9.84}$$

(Transposition.) For $A \in \mathbb{R}^{n \times n}$,

$$|A| = \left|A^T\right|. \tag{9.85}$$

(Inversion.) For an invertible matrix $A \in \mathbb{R}^{n \times n}$,

$$\left|A^{-1}\right| = \frac{1}{|A|}. \tag{9.86}$$

(Triangular matrices.) For matrices $A = (a_{ij})_{1 \le i,j \le n} \in \mathbb{R}^{n \times n}$ *with* $a_{ij} = 0$ *for* $i > j$ *or* $a_{ij} = 0$ *for* $j > i$,

$$|A| = \prod_{i=1}^{n} a_{ii}. \tag{9.87}$$

∘

The following very deep theorem, which we do not want to prove completely, provides a way to determine from the determinant of a square matrix whether it is invertible.

Theorem 9.10. $A \in \mathbb{R}^{n \times n}$ *is invertible if and only if* $|A| \neq 0$. *Thus,*

$$A \text{ is invertible} \Leftrightarrow |A| \neq 0 \text{ and } A \text{ is not invertible} \Leftrightarrow |A| = 0. \tag{9.88}$$

∘

Proof. We only indicate a proof and show that invertibility of A implies that $|A|$ cannot be zero. Suppose, then, that A is invertible. Then there exists a matrix B with $AB = I_n$, and by the multiplication theorem for determinants it follows that

$$|AB| = |A||B| = |I_n| = 1. \tag{9.89}$$

Thus, $|A| = 0$ cannot hold, because otherwise $0 = 1$. □

Visual Intuition

The abstract concept of the determinant of a square matrix can be illustrated somewhat with the concept of a vector space. To this end, let $a_1, ..., a_n \in \mathbb{R}^n$ be the columns of $A \in \mathbb{R}^{n\times n}$. Then (as we do not prove) $|A|$ corresponds to the signed volume of the parallelotope spanned by $a_1, ..., a_n \in \mathbb{R}^n$. To illustrate this visually, we consider the matrices

$$A_1 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, A_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, A_3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \tag{9.90}$$

with the respective determinants

$$|A_1| = 3 \cdot 2 - 1 \cdot 1 = 5, \quad |A_2| = 2 \cdot 2 - 0 \cdot 0 = 4, \quad |A_3| = 2 \cdot 2 - 2 \cdot 2 = 0. \tag{9.91}$$

Figure 9.1 visualizes the corresponding intuition.

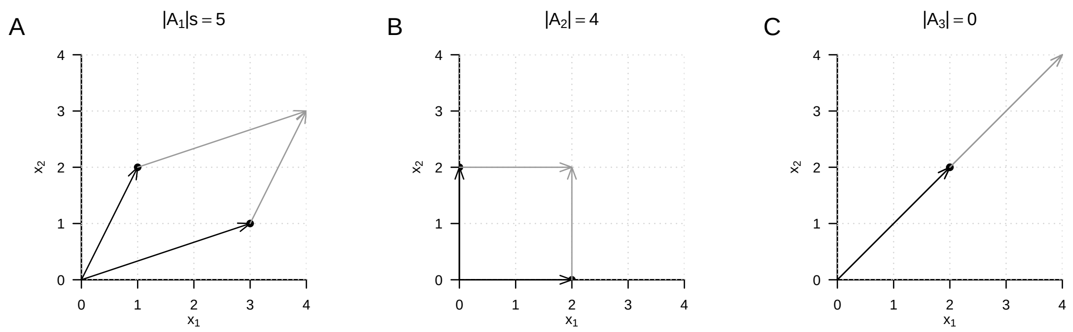

Figure 9.1 Determinants as parallelotope volumes.

9.6 Special matrices

In this section, we collect several frequently occurring types of matrices and their properties. For proofs of the vast majority of properties, we refer to the advanced literature.

9.6.1 Identity matrices

We have already encountered the identity matrix and the unit vectors. We collect them here once more in a joint definition.

Definition 9.10 (Identity matrix and unit vectors)**.** We denote the *identity matrix* by

$$I_n := (i_{jk})_{1\le j\le n, 1\le k\le n} \in \mathbb{R}^{n\times n} \text{ with } i_{jk} = 1 \text{ for } j = k \text{ and } i_{jk} = 0 \text{ for } j \neq k. \tag{9.92}$$

We denote the *unit vectors* $e_i, i = 1, ..., n$ by

$$e_i := (e_{i_j})_{1\le j\le n} \in \mathbb{R}^n \text{ with } e_{i_j} = 1 \text{ for } i = j \text{ and } e_{i_j} = 0 \text{ for } i \neq j. \tag{9.93}$$

•

The identity matrix I_n consists only of zeros and diagonal elements equal to one; the unit vectors consist only of zeros and one single 1 in the respectively indexed component. We have

$$I_n = \begin{pmatrix} e_1 & \cdots & e_n \end{pmatrix} \in \mathbb{R}^{n \times n} \tag{9.94}$$

For $n = 3$, for example,

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{9.95}$$

Furthermore, as is well known, for the unit vectors and $1 \le i, j \le n$,

$$e_i^T e_j = 0 \text{ for } i \neq j, e_i^T e_i = 1 \text{ and } e_i^T v = v^T e_i = v_i \text{ for } v \in \mathbb{R}^n. \tag{9.96}$$

9.6.2 One matrices and zero matrices

Definition 9.11 (Zero matrices, zero vectors, one matrices, one vectors)**.** We denote *zero matrices* and *zero vectors* by

$$0_{nm} := (0)_{1 \le i \le n, 1 \le j \le m} \in \mathbb{R}^{n \times m} \text{ and } 0_n := (0)_{1 \le i \le n} \in \mathbb{R}^n. \tag{9.97}$$

We denote *one matrices* and *one vectors* by

$$1_{nm} := (1)_{1 \le i \le n, 1 \le j \le m} \in \mathbb{R}^{n \times m} \text{ and } 1_n := (1)_{1 \le i \le n} \in \mathbb{R}^n. \tag{9.98}$$

•

0_{nm} and 0_n therefore consist only of zeros, and 1_{nm} and 1_n consist only of ones. For example,

$$0_{32} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, 0_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, 1_{32} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \text{ and } 1_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \tag{9.99}$$

Furthermore, for example,

$$0_n 0_n^T = 0_{nn} \text{ and } 1_n 1_n^T = 1_{nn}, \tag{9.100}$$

as can be verified by direct calculation.

9.6.3 Diagonal matrices

Definition 9.12 (Diagonal matrix)**.** A matrix $D \in \mathbb{R}^{n \times m}$ is called a *diagonal matrix* if $d_{ij} = 0$ for $1 \le i \le n, 1 \le j \le m$ with $i \neq j$.

•

A square diagonal matrix $D \in \mathbb{R}^{n \times n}$ with diagonal elements $d_1, ..., d_n \in \mathbb{R}$ is also written as

$$D = \text{diag}(d_1, ..., d_n). \tag{9.101}$$

For example,

$$D := \text{diag}(1, 2, 3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \tag{9.102}$$

and for $\sigma^2 \in \mathbb{R}$,

$$\Sigma = \text{diag}(\sigma^2, \sigma^2, \sigma^2) = \begin{pmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{pmatrix} = \sigma^2 I_3. \tag{9.103}$$

In the following theorem, we collect a few important properties of square diagonal matrices.

Theorem 9.11 (Properties of square diagonal matrices)**.**

(Determinant.) Let $D := \text{diag}(d_1, ..., d_n) \in \mathbb{R}^{n \times n}$ *be a square diagonal matrix. Then*

$$|D| = \prod_{i=1}^{n} d_i. \tag{9.104}$$

∘

9.6.4 Symmetric matrices

Symmetric matrices are square matrices that remain unchanged under transposition:

Definition 9.13. A matrix $S \in \mathbb{R}^{n \times n}$ is called *symmetric* if $S^T = S$.

•

An example of a symmetric matrix is

$$S := \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}. \tag{9.105}$$

In the following theorem, we collect a few important properties of symmetric matrices.

Theorem 9.12 (Properties of symmetric matrices)**.**

(Summation.) Let $S_1 \in \mathbb{R}^{n \times n}$ *and* $S_2 \in \mathbb{R}^{n \times n}$ *be symmetric matrices. Then*

$$S_1 + S_2 = (S_1 + S_2)^T. \tag{9.106}$$

(Inverse.) Let S *be an invertible symmetric matrix and let* S^{-1} *be its inverse. Then* S^{-1} *is also a symmetric matrix, that is,*

$$\left(S^{-1}\right)^T = S^{-1}. \tag{9.107}$$

∘

9.6.5 Orthogonal matrices

Definition 9.14. A matrix $Q \in \mathbb{R}^{n\times n}$ is called *orthogonal* if $Q^TQ = I_n$.

•

The columns of an orthogonal matrix are therefore pairwise orthogonal. For

$$Q = \begin{pmatrix} q_1 & \cdots & q_n \end{pmatrix} \text{ with } q_i \in \mathbb{R}^n \text{ for } 1 \le i \le n, \tag{9.108}$$

we have

$$q_i^T q_j = 0 \text{ for } i \ne j \text{ and } q_i^T q_j = 1 \text{ for } i = j \text{ with } 1 \le i, j \le n. \tag{9.109}$$

Theorem 9.13 (Properties of orthogonal matrices)**.** *Let $Q \in \mathbb{R}^{n\times n}$ be an orthogonal matrix. Then the following properties of Q hold.*

(Inverse.) The inverse of Q is Q^T, that is,

$$Q^{-1} = Q^T. \tag{9.110}$$

(Transposition.) The rows of Q are orthonormal, that is,

$$QQ^T = I_n \tag{9.111}$$

∘

Proof. (Inverse.) Under the assumption that Q^{-1} exists, we have

$$Q^TQ = I_n \Leftrightarrow Q^TQQ^{-1} = I_nQ^{-1} \Leftrightarrow Q^{-1} = Q^T. \tag{9.112}$$

(Transposition.) We have

$$Q^TQ = I_n \Leftrightarrow QQ^TQ = QI_n \Leftrightarrow QQ^TQQ^T = QQ^T \Leftrightarrow QQ^T = I_n. \tag{9.113}$$

□

9.6.6 Positive-definite matrices

Positive-definite matrices are central to probabilistic modeling using multivariate normal distributions.

Definition 9.15. A square matrix $C \in \mathbb{R}^{n\times n}$ is called positive-definite (p.d.) if

- C is a symmetric matrix and
- for all $x \in \mathbb{R}^n, x \ne 0_n$, $x^TCx > 0$.

•

In the following theorem, we collect a few important properties of positive-definite matrices.

Theorem 9.14 (Properties of positive-definite matrices)**.**

(Inverse.) Let $C \in \mathbb{R}^{n\times n}$ be a positive-definite matrix. Then C^{-1} exists and is also positive-definite.

∘

9.7 Eigenanalysis

With the *eigenanalysis of a square matrix*, the *orthonormal decomposition of a symmetric matrix*, and the *singular value decomposition of an arbitrary matrix*, we discuss in this section three closely related concepts of matrix theory that play central roles in many areas of data-analytic applications. However, the meaning of these concepts becomes apparent above all in the respective application context, so this section may necessarily seem somewhat abstract.

Eigenvectors and eigenvalues

The *eigenanalysis* of a square matrix is understood as determining its *eigenvectors* and *eigenvalues*. These are defined for a square matrix as follows.

Definition 9.16 (Eigenvector and eigenvalue)**.** Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. Then every vector $v \in \mathbb{R}^m$ different from the zero vector 0_m for which, with a scalar $\lambda \in \mathbb{R}$,

$$Av = \lambda v \tag{9.114}$$

holds is called an *eigenvector of* A, and λ is then called an *eigenvalue of* A.

•

By definition, every eigenvector therefore has an associated eigenvalue, although the eigenvalues of different eigenvectors can certainly be identical. Intuitively, the definition of eigenvector and eigenvalue means that an eigenvector of a matrix is changed in its length, but not in its direction, by multiplication with this very matrix. The associated eigenvalue of the eigenvector corresponds to the factor of the change in length. However, the assignment of eigenvectors and eigenvalues is not unique, as the following theorem shows.

Theorem 9.15 (Multiplicativity of eigenvectors)**.** *Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. If $v \in \mathbb{R}^m$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{R}$, then for $c \in \mathbb{R} \setminus \{0\}$, $cv \in \mathbb{R}^m$ is also an eigenvector of A, again with eigenvalue $\lambda \in \mathbb{R}$.*

∘

Proof. This follows from

$$A(cv) = cAv = c\lambda v = \lambda(cv). \tag{9.115}$$

Thus, cv is an eigenvector of A with eigenvalue λ. □

To resolve the non-uniqueness in the definition of the eigenvector associated with an eigenvalue, we use the convention of considering only those vectors as eigenvectors for an eigenvalue λ that have length 1 and hence are elements of the unit sphere in $\mathbb{R}^m$:

$$\mathbb{S}^{m-1} := \{x \in \mathbb{R}^m : \|x\|_2 = 1\}. \tag{9.116}$$

Thus, if we should have found an eigenvector v for an eigenvalue λ of a matrix A with length other than 1, then

$$\frac{v}{\|v\|_2} \tag{9.117}$$

is of length 1 and, by Theorem 9.15, is likewise an eigenvector of A with eigenvalue λ. Before turning to the determination of eigenvalues and eigenvectors, we illustrate the concepts of eigenvalue and eigenvector for the case of a 2×2 matrix using an example.

Example

Consider the matrix

$$A := \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.118}$$

and the vector

$$v := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{9.119}$$

Then v is an eigenvector of A with eigenvalue $\lambda = 3$, because

$$\begin{aligned} Av &= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda v. \end{aligned} \tag{9.120}$$

Inspection of Figure 9.2 accordingly shows that for the matrix A defined here, the vectors v and Av point in the same direction, but that Av is three times as long as v. Now consider the vector

$$w := \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{9.121}$$

This vector also has length 1, but in contrast to v it is not an eigenvector of A. We have

$$\begin{aligned} Aw &= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \end{aligned} \tag{9.122}$$

Because the second entry of w is 0, there can be no scalar λ that would transform the second entry of w into the second entry of Aw by multiplication. Figure 9.2 therefore shows accordingly that the vector resulting from multiplication of w by A no longer points in the same direction as w.

Determining eigenvalues and eigenvectors

The following theorem states how the eigenvalues and eigenvectors of a square matrix can be computed.

Theorem 9.16 (Determining eigenvalues and eigenvectors)**.** *Let $A \in \mathbb{R}^{m \times m}$ be a square matrix. Then the eigenvalues of A are obtained as the roots of the characteristic polynomial*

$$\chi_A(\lambda) := |A - \lambda I_m| \tag{9.123}$$

of A. Furthermore, let $\lambda_i^*, i = 1, 2, ...$ be the eigenvalues of A determined in this way. The corresponding eigenvectors $v_i, i = 1, 2, ...$ of A can then be determined by solving the systems of linear equations

$$(A - \lambda_i^* I_m) v_i = 0_m \text{ for } i = 1, 2, ... \tag{9.124}$$

∘

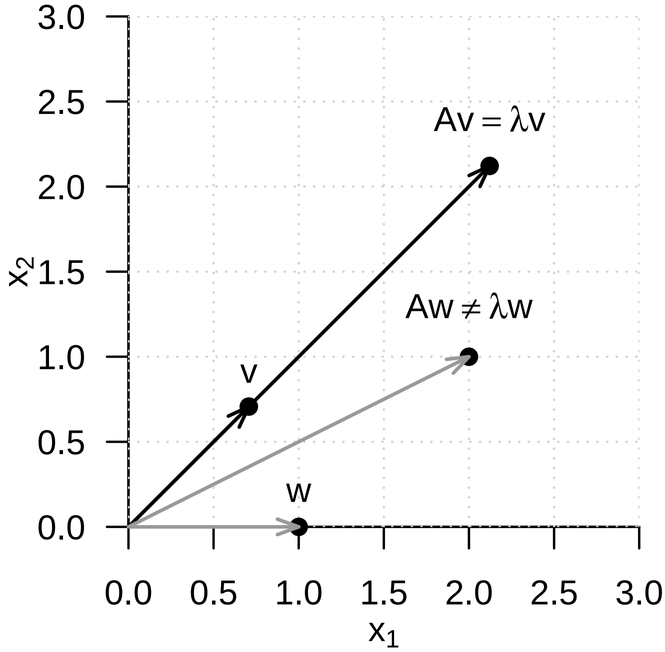

Figure 9.2 Eigenvector of a 2×2 matrix. For the matrix A defined in the text, v is an eigenvector, but w is not.

Proof. (1) Determining eigenvalues

We first note that, by the definition of eigenvectors and eigenvalues,

$$Av = \lambda v \Leftrightarrow Av - \lambda v = 0_m \Leftrightarrow (A - \lambda I_m)v = 0_m. \tag{9.125}$$

For the eigenvalue λ, the eigenvector v is therefore mapped to the zero vector 0_m by multiplication with $(A - \lambda I_m)$. But because by definition $v \neq 0_m$, the matrix $(A - \lambda I_m)$ is not invertible: both the zero vector and v are mapped to 0_m by $(A - \lambda I_m)$, so the mapping

$$f : \mathbb{R}^m \to \mathbb{R}^m, x \mapsto (A - \lambda I_m)x \tag{9.126}$$

is not bijective, and $(A - \lambda I_m)^{-1}$ cannot exist. The fact that $(A - \lambda I_m)$ is not invertible, however, is equivalent to the determinant of $(A - \lambda I_m)$ being equal to zero. Thus,

$$\chi_A(\lambda) = |A - \lambda I_m| = 0 \tag{9.127}$$

is a necessary and sufficient condition for λ to be an eigenvalue of A.

(2) Determining eigenvectors

Let λ_i^* be an eigenvalue of A. Then the above considerations imply that solving

$$(A - \lambda_i^* I_m)v_i^* = 0_m \tag{9.128}$$

for v_i^* yields an eigenvector for the eigenvalue λ_i^*. □

In general, determining eigenvalues and eigenvectors therefore requires determining polynomial roots and solving systems of linear equations. For small matrices with $m \leq 4$, this can indeed be done manually. The matrices that occur in applications, however, are usually much larger, so numerical methods for root finding and for solving systems of linear equations are used for eigenanalysis, for example in functions such as **R**'s `eigen()`, **SciPy's** `linalg.eig()`, or **Julia**'s `eigvals()` and `eigvecs()`. For details on these methods, we refer to the advanced literature, for example Burden et al. (2016) and Richter & Wick (2017). Here we merely illustrate Theorem 9.16 by means of an example.

Example

To this end, let again

$$A := \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.129}$$

We first want to compute the eigenvalues of A. By Theorem 9.16, these are the roots of the characteristic polynomial of A. We therefore first compute the characteristic polynomial of A by

$$\chi_A(\lambda) = \left| \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} \right| = (2-\lambda)^2 - 1. \tag{9.130}$$

Using the quadratic formula to solve quadratic equations, one then finds

$$(2 - \lambda_{1/2}^*)^2 - 1 = 0 \Leftrightarrow \lambda_1^* = 3 \text{ or } \lambda_2^* = 1. \tag{9.131}$$

The eigenvalues of A are therefore $\lambda_1 = 3$ and $\lambda_2 = 1$. The associated eigenvectors are then obtained for $i = 1, 2$ by solving the system of linear equations

$$(A - \lambda_i I_2)v_i = 0_2. \tag{9.132}$$

Specifically, for $\lambda_1 = 3$,

$$(A - 3I_2)v_1 = 0_2 \Leftrightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_{1_1} \\ v_{1_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{9.133}$$

implies that

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{9.134}$$

is an eigenvector for the eigenvalue λ_1, and for $\lambda_2 = 1$,

$$(A - I_2)v_2 = 0_2 \Leftrightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_{2_1} \\ v_{2_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{9.135}$$

implies that

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{9.136}$$

is an eigenvector for the eigenvalue $\lambda_2 = 1$. Furthermore, here obviously

$$v_1^T v_2 = 0 \text{ and } \|v_1\| = \|v_2\| = 1. \tag{9.137}$$

The following **R** code demonstrates the determination of the eigenvalues and eigenvectors of the matrix considered here with the help of the `eigen()` function.

```
# matrix definition
A = matrix(c(2,1,
             1,2),
           nrow  = 2,
           byrow = TRUE)

# eigenanalysis
eigen(A)
```

```
eigen() decomposition
$values
[1] 3 1

$vectors
          [,1]       [,2]
[1,] 0.7071068 -0.7071068
[2,] 0.7071068  0.7071068
```

At the end of this section, we consider two technical theorems that make statements about the relation between special matrix products and their eigenvalues and eigenvectors. We need these theorems in the context of canonical correlation analysis.

Theorem 9.17 (Eigenvalues and eigenvectors of matrix products)**.** *For $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times n}$, the eigenvalues of $AB \in \mathbb{R}^{n\times n}$ and $BA \in \mathbb{R}^{m\times m}$ are equal. Furthermore, if v is an eigenvector for a nonzero eigenvalue λ of AB, then $w := Bv$ is an eigenvector of BA for the eigenvalue λ.*

$\circ$

For a proof, we refer to Mardia et al. (1979), p. 468. We demonstrate the statement of this theorem using the **R** code below.

```
A   = matrix(1:6, nrow = 2,  byrow = T)          # matrix A \in \mathbb{R}^{2 x 3}
B   = matrix(1:6, ncol = 2,  byrow = T)          # matrix B \in \mathbb{R}^{3 x 2}
EAB = eigen(A %*% B)                             # eigenanalysis of AB \in \mathbb{R}^{2 \times 2}
EBA = eigen(B %*% A)                             # eigenanalysis of BA \in \mathbb{R}^{3 \times 3}
w   = B %*% EAB$vectors[,1]                      # eigenvector of BA
cat("Eigenvalues of AB :"  , EAB$values[1:2],
    "\nEigenvalues of BA :", EBA$values[1:2],
    "\nBAw with w = Bv    :", B %*% A %*% w,
    "\nlw with w = Bv    :", EBA$values[1] * w)
```

```
Eigenvalues of AB : 85.57934 0.4206623
Eigenvalues of BA : 85.57934 0.4206623
BAw with w = Bv   : -191.1333 -416.7586 -642.3839
lw with w = Bv    : -191.1333 -416.7586 -642.3839
```

Theorem 9.18. *For $A \in \mathbb{R}^{n \times m}$, $B \in \mathbb{R}^{p \times n}$, $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^p$, the only nonzero eigenvalue of $Aab^TB \in \mathbb{R}^{n \times n}$ is equal to b^TBAa, with associated eigenvector Aa.*

∘

For a proof, we refer to Mardia et al. (1979), p. 468. We demonstrate the statement of this theorem using the **R** code below.

```
A      = matrix(1:6, nrow = 2,  byrow = T)     # matrix A \in \mathbb{R}^{2 x 3}
B      = matrix(1:8, ncol = 2,  byrow = T)     # matrix B \in \mathbb{R}^{4 x 2}
a      = matrix(1:3, nrow = 3,  byrow = T)     # vector a \in \mathbb{R}^{3 x 1}
b      = matrix(1:4, nrow = 4,  byrow = T)     # vector b \in \mathbb{R}^{4 x 1}
EAabTB = eigen(A %*% a %*% t(b) %*% B)         # eigenanalysis of Aab^TB \in \mathbb{R}^{2 x 2}
cat("Eigenvalues of AabTB :", EAabTB$values,
    "\nbTBAa                :", t(b) %*% B %*% A %*% a,
    "\nAa                   :", A %*% a,
    "\n(AabTB)Aa            :",(A %*% a %*% t(b) %*% B) %*% A %*% a,           # mv
    "\n(bTBAa)Aa            :",as.vector((t(b) %*% B %*% A %*% a)) * (A %*% a)) # = \lambda v
```

```
Eigenvalues of AabTB : 2620 0
bTBAa                : 2620
Aa                   : 14 32
(AabTB)Aa            : 36680 83840
(bTBAa)Aa            : 36680 83840
```

9.8 Orthonormal decomposition

The term *decomposition* of a matrix denotes the splitting of a given matrix into the matrix product of several matrices. A wide variety of matrix decompositions play an important role in many mathematical applications; for an overview, see for example Golub & Van Loan (2013). In this section, we introduce a special matrix decomposition, the *orthonormal decomposition of a symmetric matrix*, which builds directly on eigenanalysis. We first record the following fundamental theorem on the eigenvalues and eigenvectors of symmetric matrices.

Theorem 9.19 (Eigenvalues and eigenvectors of symmetric matrices)**.**
Let $S \in \mathbb{R}^{m \times m}$ be a symmetric matrix. Then:

(1) The eigenvalues of S are real.

(2) The eigenvectors corresponding to any two distinct eigenvalues of S are orthogonal.

∘

Proof. We take the fact that a symmetric matrix has m real eigenvalues as given and show only that the eigenvectors corresponding to any two distinct eigenvalues of a symmetric matrix are orthogonal. Without loss of generality, let $\lambda_i, \lambda_j \in \mathbb{R}$ with $1 \le i, j \le m$ and $\lambda_i \neq \lambda_j$ be two distinct eigenvalues of S with associated eigenvectors q_i and q_j, respectively. Then, as shown below,

$$\lambda_i q_i^T q_j = \lambda_j q_i^T q_j. \tag{9.138}$$

With $q_i \neq 0_m, q_j \neq 0_m$ and $\lambda_i \neq \lambda_j$, it follows that $q_i^T q_j = 0$, because there is no number c other than zero for which, with $a, b \in \mathbb{R}$ and $a \neq b$,

$$ac = bc \tag{9.139}$$

holds. To finally show

$$\lambda_i q_i^T q_j = \lambda_j q_i^T q_j, \tag{9.140}$$

we first note that

$$Sq_i = \lambda_i q_i \Leftrightarrow (Sq_i)^T = (\lambda_i q_i)^T \Leftrightarrow q_i^T S^T = \lambda_i q_i^T \Leftrightarrow q_i^T S = \lambda_i q_i^T \Leftrightarrow q_i^T S q_j = \lambda_i q_i^T q_j \tag{9.141}$$

and

$$Sq_j = \lambda_j q_j \Leftrightarrow q_j^T S = \lambda_j q_j^T \Leftrightarrow q_j^T S q_i = \lambda_j q_j^T q_i \Leftrightarrow (q_j^T S q_i)^T = (\lambda_j q_j^T q_i)^T \Leftrightarrow q_i^T S q_j = \lambda_j q_i^T q_j \tag{9.142}$$

hold. Thus, both $\lambda_i q_i^T q_j$ and $\lambda_j q_i^T q_j$ are identical with $q_i^T S q_j$ and therefore also with each other. □

Obviously, we have proven only statement (2) of Theorem 9.19. A complete proof of the theorem can be found, for example, in Strang (2009). We also note that, because by convention we consider eigenvectors of length 1, the orthogonal eigenvectors addressed in Theorem 9.19 are in particular also orthonormal. With the help of Theorem 9.19, we can now formulate the orthonormal decomposition of a symmetric matrix and prove its existence.

Theorem 9.20 (Orthonormal decomposition of a symmetric matrix)**.** *Let $S \in \mathbb{R}^{m\times m}$ be a symmetric matrix with m distinct eigenvalues. Then S can be written as*

$$S = Q\Lambda Q^T, \tag{9.143}$$

where $Q \in \mathbb{R}^{m\times m}$ is an orthogonal matrix and $\Lambda \in \mathbb{R}^{m\times m}$ is a diagonal matrix.

∘

Proof. Let $\lambda_1 > \lambda_2 > \ldots > \lambda_m$ be the eigenvalues of S ordered by size, and let $q_1, \ldots, q_m$ be the associated orthonormal eigenvectors. With

$$Q := \begin{pmatrix} q_1 & q_2 & \cdots & q_m \end{pmatrix} \in \mathbb{R}^{m\times m} \text{ and } \Lambda := \operatorname{diag}\left(\lambda_1, \lambda_2, \ldots, \lambda_m\right) \in \mathbb{R}^{m\times m}, \tag{9.144}$$

the definitions of eigenvalues and eigenvectors first imply that

$$Sq_i = \lambda_i q_i \text{ for } i = 1, \ldots, m \Leftrightarrow SQ = Q\Lambda. \tag{9.145}$$

Right multiplication by Q^T then gives, with $QQ^T = I_m$,

$$SQQ^T = Q\Lambda Q^T \Leftrightarrow SI_m = Q\Lambda Q^T \Leftrightarrow S = Q\Lambda Q^T. \tag{9.146}$$

□

Because of the diagonality of Λ, the splitting of S into the matrix product $Q\Lambda Q^T$ is also called a *diagonalization of S*. As shown in the proof, to represent S in diagonal form, the diagonal elements of Λ are chosen as the eigenvalues of S ordered by size, and the columns of Q are chosen as the corresponding eigenvectors of S. We illustrate this with an example.

Example

For the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{9.147}$$

with the eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 1$ determined above and the associated orthonormal eigenvectors

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{9.148}$$

let

$$Q := \begin{pmatrix} v_1 & v_2 \end{pmatrix} \text{ and } \Lambda = \operatorname{diag}(\lambda_1, \lambda_2). \tag{9.149}$$

Then obviously

$$\begin{aligned}
Q\Lambda Q^T &= \begin{pmatrix} v_1 & v_2 \end{pmatrix} \operatorname{diag}(\lambda_1, \lambda_2) \begin{pmatrix} v_1 & v_2 \end{pmatrix}^T \\
&= \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right) \\
&= \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} \right) \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right) \\
&= \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \\
&= A
\end{aligned}$$

and we have verified Theorem 9.20 for this example.

Symmetric square root of a matrix

The definition of the orthonormal decomposition of a symmetric matrix allows us to introduce the concept of the *symmetric square root of a matrix*.

Definition 9.17 (Symmetric square root of a matrix)**.** Let $S \in \mathbb{R}^{m \times m}$ be an invertible symmetric matrix with positive eigenvalues. Then for $r \in \mathbb{N}^0$ and $s \in \mathbb{N}$, the rational powers of S are defined, with the orthonormal matrix $Q \in \mathbb{R}^{m \times m}$ of the eigenvectors of S and the diagonal matrix $\Lambda = \operatorname{diag}(\lambda_i) \in \mathbb{R}^{m \times m}$ of the associated eigenvalues $\lambda_1, ..., \lambda_m$ of S, as

$$S^{r/s} = Q\Lambda^{r/s}Q^T \text{ with } \Lambda^{r/s} = \operatorname{diag}\left(\lambda_i^{r/s}\right). \tag{9.150}$$

The special case $r := 1, s := 2$ is called the *symmetric square root of* S and has the form

$$S^{1/2} = Q\Lambda^{1/2}Q^T \text{ with } \Lambda^{1/2} = \operatorname{diag}\left(\lambda_i^{1/2}\right). \tag{9.151}$$

•

We note that Definition 9.17 immediately implies

$$\left(S^{1/2}\right)^2 = Q\Lambda^{1/2}Q^TQ\Lambda^{1/2}Q^T = Q\Lambda^{1/2}\Lambda^{1/2}Q^T = Q\Lambda Q^T = S. \tag{9.152}$$

Furthermore,

$$\left(S^{-1/2}\right)^2 = Q\Lambda^{-1/2}Q^TQ\Lambda^{-1/2}Q^T = Q\Lambda^{-1/2}\Lambda^{-1/2}Q^T = Q\Lambda^{-1}Q^T = S^{-1}. \tag{9.153}$$

Finally,

$$\begin{aligned}
S^{-1/2}SS^{-1/2} &= Q\Lambda^{-1/2}Q^TQ\Lambda Q^TQ\Lambda^{-1/2}Q^T \\
&= Q\Lambda^{-1/2}\Lambda\Lambda^{-1/2}Q^T \\
&= Q\Lambda\Lambda^{-1}Q^T \\
&= I_m
\end{aligned} \tag{9.154}$$

9.9 Singular value decomposition

A versatile matrix decomposition of an arbitrary matrix is the *singular value decomposition*. At this point, we are interested only in the relation between singular value decomposition and eigenanalysis and refer to the advanced literature, for example Strang (2009), for a detailed discussion of the singular value decomposition. The concept of singular value decomposition is defined as follows.

Definition 9.18 (Singular value decomposition)**.** Let $Y \in \mathbb{R}^{m \times n}$ be a matrix. Then the decomposition

$$Y = USV^T, \tag{9.155}$$

where $U \in \mathbb{R}^{m \times m}$ is an orthogonal matrix, $S \in \mathbb{R}^{m \times n}$ is a diagonal matrix, and $V \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, is called the *singular value decomposition* of Y. The diagonal elements of S are called the *singular values* of Y.

•

Singular value decompositions are abbreviated as *SVD*. We omit a discussion of the computation of a singular value decomposition and merely note that singular value decompositions can be computed, for example, in **R** with the function `svd()`, in **SciPy** with `scipy.linalg.svd()`, and in **Julia** with `svd()`. The following theorem describes the relation between singular value decomposition and eigenanalysis and is used in many places.

Theorem 9.21 (Singular value decomposition and eigenanalysis)**.**

Let $Y \in \mathbb{R}^{m \times n}$ be a matrix and

$$Y = USV^T \tag{9.156}$$

be its singular value decomposition. Then:

- *the columns of U are the eigenvectors of YY^T,*
- *the columns of V are the eigenvectors of Y^TY, and*
- *the singular values are the square roots of the associated eigenvalues.*

∘

Proof. We first note that, with

$$\left(YY^T\right)^T = YY^T \text{ and } \left(Y^TY\right)^T = Y^TY, \tag{9.157}$$

YY^T and Y^TY are symmetric matrices and thus have orthonormal decompositions. We further note that, with $V^TV = I_n$ and $U^TU = I_m$,

$$YY^T = USV^T\left(USV^T\right)^T = USV^TVS^TU^T = USS^TU^T =: U\Lambda_U U^T \tag{9.158}$$

and

$$Y^TY = \left(USV^T\right)^T USV^T = VS^TU^TUSV^T = VS^TSV^T =: V\Lambda_V V^T \tag{9.159}$$

where we have defined $\Lambda_U := SS^T$ and $\Lambda_V := S^TS$. Because the product of diagonal matrices is again a diagonal matrix, Λ_U and Λ_V are diagonal matrices, and by definition U and V are orthogonal matrices. Thus, we have written YY^T and Y^TY in the form of the orthonormal decompositions

$$YY^T = U\Lambda_U U^T \text{ and } Y^TY = V\Lambda_V V^T \tag{9.160}$$

where, for the nonzero diagonal elements of Λ_U and Λ_V, these elements are the squared singular values, that is, the squared diagonal elements of S. □

10 Descriptive statistics

The aim of evaluating descriptive statistics is to make data sets accessible to human observation as efficiently as possible. As soon as the number of data points in a data set exceeds a certain number, pure numerical representations are usually difficult for human cognition to grasp. This is where descriptive statistics comes in and, in addition to graphical representations of larger data sets, also offers measures that describe certain characteristic aspects of a data set, such as its "average" value or its variability, in the sense of data reduction. In this section we want to use univariate descriptive statistics to get to know the first procedures and measures that make it possible to summarize data sets in which there is one number per study unit. We look in particular at *frequency distributions* and *histograms*, *distribution functions* and *quantiles*, as well as *measures of central tendency*, which allow statements about "average" data values, and *measures of data variability*, which allow statements about the spread of data values.

In contrast to probabilistic modeling in the context of frequentist and Bayesian inference, descriptive statistics is not based on probability theory and can therefore be viewed independently of it. However, many aspects of descriptive statistics ultimately only make sense against the background of probability theory considerations and, conversely, many concepts in probability theory are motivated by descriptive statistical concepts. If the present section does not have a background in probability theory, its full meaning will certainly only become apparent in the context of the concepts of probability theory.

In this section, we focus on descriptive statistics for describing a data set of the form

$$y := (y_1, ..., y_n) \text{ with } y_i \in \mathbb{R} \text{ for } i = 1, ..., n. \tag{10.1}$$

It is initially irrelevant whether such a data set is in the form of a column vector $y \in \mathbb{R}^{n \times 1}$ or a row vector $y \in \mathbb{R}^{1 \times n}$. Furthermore, we focus on the practical evaluation of the descriptive statistics considered, so we provide many examples for their calculation with **R**.

Application example

As an application example, we consider a data set for the evidence-based evaluation of psychotherapy for depression. To do this, we assume that $n = 100$ patients were randomly divided into a treatment and a control study condition and were asked about their depression severity using the BDI-II questionnaire at the beginning of the respective intervention. We do not want to consider the BDI-II scores collected after the interventions in this chapter. So we assume that we have a data set as shown by the following **R** code in Table 10.1.

```
D = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
knitr::kable(D[,1:2], digits = 2, align = "c")          # r markdown table output
```

Table 10.1 BDI-II scores (PRE) of n = 100 patients in a treatment (T) and a control study group (C) before the start of a psychological intervention

GROUP	PRE
T	17
T	20
T	16
T	18
T	21
T	17
T	17
T	17
T	18
T	18
T	20
T	17
T	16
T	18
T	16
T	18
T	17
T	14
T	18
T	18
T	20
T	20
T	21
T	19
T	19
T	17
T	20
T	21
T	17
T	17
T	19
T	20
T	22
T	20
T	21
T	20
T	16
T	15
T	15
T	18
T	21
T	17
T	18
T	21
T	17
T	19
T	16
T	17
T	17
T	18
C	22
C	19
C	21
C	18
C	19
C	17
C	20
C	19
C	19
C	19
C	17
C	20
C	18
C	20
C	18
C	18
C	15
C	18
C	17
C	19
C	19
C	19
C	20
C	19
C	20
C	19
C	21
C	19
C	19
C	18
C	20
C	16
C	21
C	19
C	20
C	18
C	23
C	18
C	19
C	18

Table 10.1 BDI-II scores (PRE) of n = 100 patients in a treatment (T) and a control study group (C) before the start of a psychological intervention

GROUP	PRE
C	21
C	17
C	20
C	21
C	21
C	20
C	18
C	18
C	19
C	19

10.1 Frequency distributions

We start with the concepts of *absolute* and *relative frequency distribution.* For data sets with values in which the same value occurs multiple times, frequency distributions can provide a first impression of the nature of the data set.

Definition 10.1 (Absolute and relative frequency distributions)**.** $y := (y_1, ..., y_n)$ is a data set and $W := \{w_1, ..., w_k\}$ with $k \le n$ are the values occurring in the data set. Then the function is called

$$h : W \to \mathbb{N}, w \mapsto h(w) := \text{number of } y_i \text{ in } y \text{ with } y_i = w \tag{10.2}$$

the *absolute frequency distribution* of the values of y and the function

$$r : W \to [0,1], w \mapsto r(w) := \frac{h(w)}{n} \tag{10.3}$$

is called the *relative frequency distribution* of the values of y.

•

Frequency distributions can be evaluated and displayed in **R** by combining the `table()` and `barplot()` functions. The following **R** code first generates the absolute and relative frequency distributions of the pre-intervention BDI-II data of the example data set.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                               # double vector of the PRE values
n        = length(y)                                           # number of data values (100)
H        = as.data.frame(table(y))                             # absolute frequency distribution (data frame)
names(H) = c("w", "ah")                                        # column naming
H$rh     = H$ah/n                                              # relative frequency distribution
print(H, digits = 1)                                           # output
```

```
    w ah   rh
1  14  1 0.01
2  15  3 0.03
3  16  6 0.06
4  17 17 0.17
5  18 21 0.21
6  19 20 0.20
7  20 17 0.17
8  21 12 0.12
9  22  2 0.02
10 23  1 0.01
```

From the output of the data frame `H` we can see that in the data set 10 under consideration, different values `w` occur between 14 and 23, whereby, for example, the values 18 and 22 occur 21 times and twice respectively. Both the absolute and the relative frequency distribution show that values around 18 occur more frequently in the present data set, whereas extreme values such as 13 or 23 occur rather rarely. Note that the relative frequency distribution is simply a scaled form of the absolute frequency distribution. Qualitatively, i.e. with regard to the frequent or rare occurrence of certain values, the two frequency distributions do not differ. This becomes particularly clear when both frequency distributions are visualized, as in Figure 10.1. The following **R** code uses the `barplot()` function for this purpose. The height of the bar plotted above a value w corresponds to the function values $h(w)$ or $r(w)$. The visual insight naturally corresponds to the inspection of the `H` data frame. Both frequency distributions show that values around 18 occur more frequently in the present data set, whereas lower or higher values such as 13 or 23 are rather rare.

```
pdf(                                                        # pdf copy function
file     = "./_figures/110-frequency-distributions.pdf",    # filename
width    = 7,                                               # width (inches)
height   = 4)                                               # height (inches)
par(                                                        # figure parameters
mfcol    = c(1,2),                                          # rows and columns layout
las      = 1,                                               # x-tick orientation
cex      = 0.7,                                             # annotation enlargement
cex.main = 1.3)                                             # title enlargement
h        = H$ah                                             # h(w) values
r        = H$rh                                             # r(w) values
names(h) = H$w                                              # barplot needs w values as names
names(r) = H$w                                              # barplot needs w values as names
barplot(                                                    # bar chart
h,                                                          # absolute frequencies
col      = "gray90",                                        # bar color
xlab     = "w",                                             # x axis label
ylab     = "h(w)",                                          # y axis label
ylim     = c(0,25),                                         # y axis limits
main     = "Absolute frequency distribution")               # title
barplot(                                                    # bar chart
r,                                                          # absolute frequencies
col      = "gray90",                                        # bar color
xlab     = "w",                                             # x axis label
ylab     = "r(w)",                                          # y axis label
ylim     = c(0,.25),                                        # y axis limits
main     = "Relative frequency distribution")               # title
dev.off()                                                   # end image editing
```

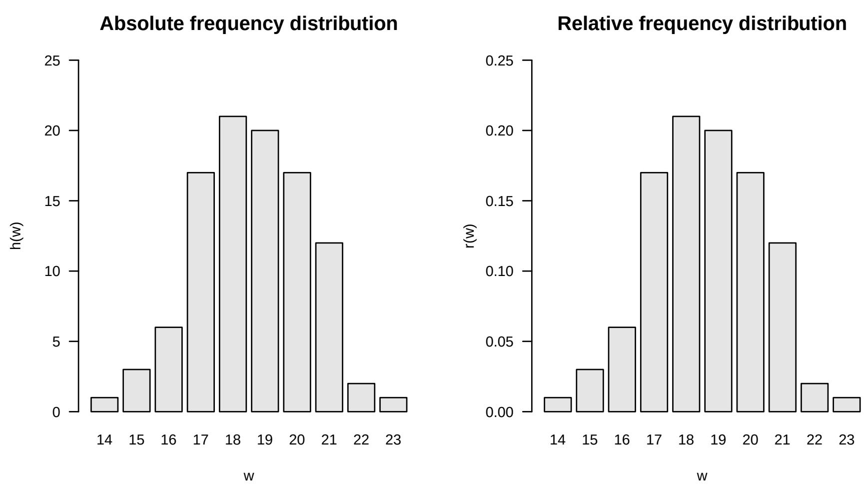

Figure 10.1 Absolute and relative frequency distributions of pre-intervention BDI-II scores.

10.2 Histograms

If no values appear multiple times in a data set, the absolute and relative frequency distributions for each of these values take the value 1 and $1/n$, respectively, and then provide no value beyond visual inspection of the data set. However, the histograms presented in this section offer a way to generate visual representations of frequency distributions in this case too. Different values that are "close to each other" in a well-defined sense are combined into classes and the absolute or relative frequencies of these classes are determined. Since in data sets in which the same values occur multiple times, these values are naturally in the same class, histograms can also be usefully used in the case considered in Section 10.1. Histograms generalize the frequency distributions considered in the previous section for data sets with multiple values.

We first give a definition of the concept of histogram. Note that in this definition the process of class formation, the determination of their absolute and relative frequencies, and the visualization of these frequencies are closely interlinked. This is due to the convention that the term histogram describes both a specific form of data analysis and its visualization.

Definition 10.2 (Histogram)**.** A *histogram* is a diagram in which, for a data set $y = (y_1, ..., y_n)$ with values $W := \{w_1, ..., w_m\}$ with $m \le n$ for $j = 1, ..., k$ over adjacent intervals $[b_{j-1}, b_j[$ with $b_0 := \min W$ and $b_k := \max W$, rectangles with the *width* $d_j := b_j - b_{j-1}$ and the *height* $h(w)$ or $r(w)$ for $w \in [b_{j-1}, b_j[$ are shown. The intervals $[b_{j-1}, b_j[$ are called *classes* (*bins*).

•

Of course, the visual appearance of a histogram depends heavily on how precisely the values of a data set are grouped into classes. According to Definition 10.2, the decisive parameter for this is the number of classes k that are selected between the minimum value b_0 and the maximum value b_k of the data set. Below we present some conventional values for the number of classes k and calculate and visualize them using the example data set as an example. We always assume that the neighboring intervals $[b_{j-1}, b_j[$ with $j = 1, ..., k$ are chosen for the number of classes k with a constant width $d = b_j - b_{j-1}$.

To calculate and visualize histograms we use the **R** function `hist()`. In this case, the classes $[b_{j-1}, b_j[$ are specified for $j = 1, ..., k$ as the argument `breaks`. Specifically, `breaks` is the **R** vector `c(b_0,b_1,...,b_k)` with length $k+1$. To illustrate the effect of different choices of the number of classes, we first read the data set and determine its minimum and maximum values b_0 and b_k.

```
# dataset
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                                # pre-intervention BDI-II data
b_0 = min(y)                                               # b_0
b_k = max(y)                                               # b_k
```

```
Minimum b_0              :  14
Maximum b_k              :  23
```

A first possibility for choosing the classes k is to achieve a desired constant width d as precisely as possible. This is what you rely on

$$k := \left\lceil \frac{b_k - b_0}{d} \right\rceil. \tag{10.4}$$

Note that the application of the rounding function implies that the resulting class width is at most the desired value, but can also be smaller. The following **R** code illustrates this effect.

```
# histogram with desired class width d = 2
d   = 2                                 # desired class width
b_0 = min(y)                            # b_0
b_k = max(y)                            # b_k
k   = ceiling((b_k - b_0)/d)            # number of classes
b   = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Desired class width d      :  2
Number of classes k        :  5
Intervals [b_j-1, b_j]    :  14 15.8 17.6 19.4 21.2 23
Achieved class width d     :  1.8
```

Choosing a desired class width of $d = 1.5$, on the other hand, leads to a resulting class width for the present data set that corresponds to the desired one.

```
# histogram with desired class width d = 1.5
d = 1.5                               # desired class width
k = ceiling((b_k - b_0)/d)            # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Desired class width d      :  1.5
Number of classes k        :  6
Intervals [b_j-1, b_j]    :  14 15.5 17 18.5 20 21.5 23
Achieved class width d     :  1.5
```

Another option is to make the number of classes dependent on the number of data points in the data set. The Excel standard choice is included

$$k := \lceil \sqrt{n} \rceil. \tag{10.5}$$

The following **R** code implements this choice for the present data set.

```
# histogram with Excel standard
n = length(y)                         # number of data points
k = ceiling(sqrt(n))                  # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Number of classes k        :  10
Intervals [b_j-1, b_j]    :  14 14.9 15.8 16.7 17.6 18.5 19.4 20.3 21.2 22.1 23
Achieved class width d     :  0.9
```

The number of classes suggested by Sturges (1926) is similar to the Excel standard, which optimizes the representation under an implicit normal distribution assumption

$$k := \lceil \log_2 n + 1 \rceil \tag{10.6}$$

is given. In this case the result is:

```
# class number according to Sturges (1926)
n = length(y)                           # number of data points
k = ceiling(log2(n)+1)                  # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Number of classes k        :  8
Intervals [b_j-1, b_j]    :  14 15.125 16.25 17.375 18.5 19.625 20.75 21.875 23
Achieved class width d     :  1.125
```

Another possibility is to incorporate descriptive statistics of the data set into the choice of the number of classes. As a measure of the dispersion of the data in the data set, the number of classes according to Scott (1979) uses the empirical standard deviation (cf. Section 10.6) in order to use the histogram to achieve a density estimate for a normal distribution. We do not want to delve into the non-parametric background of this approach here, but simply note that the choice of the number of classes according to Scott (1979) by determining the class width

$$d := 3.49S/\sqrt[3]{n} \tag{10.7}$$

and the subsequent choice of the number of classes

$$k := \left\lceil \frac{b_k - b_0}{d} \right\rceil \tag{10.8}$$

is given, where S denotes the standard deviation of the data set. In this case the result is:

```
# number of classes according to Scott (1979)
n = length(y)                           # number of data values
S = sd(y)                               # empirical standard deviation
d = ceiling(3.49*S/(n^(1/3)))           # class width
k = ceiling((b_k - b_0)/d)              # number of classes
b = seq(b_0, b_k, length.out = k + 1)  # classes [b_{j-1}, b_j[ (breaks)
```

```
Number of data points n    :  100
Standard deviation         :  1.740167
Class width by Scott d     :  2
Number of classes k        :  5
Intervals [b_j-1, b_j]    :  14 15.8 17.6 19.4 21.2 23
Achieved class width d     :  1.8
```

By default, the `hist()` function implemented in **R** uses a modification of the number of classes suggested by Sturges (1926) and offers a number of additional options for selecting classes. `?hist` provides an overview of this. The following **R** code visualizes the absolute frequencies of the values of the example data set using the classes specified above.

```
# figure parameters
pdf(                                                                            # pdf copy function
file      = "./_figures/110-histograms.pdf",                                    # filename
width     = 6,                                                                  # width (inches)
height    = 9)                                                                  # height (inches)
par(                                                                            # figure parameters
mfcol     = c(3,2),                                                             # rows and columns layout
las       = 1,                                                                  # x-tick orientation
cex       = .7,                                                                 # annotation enlargement
cex.main = .9)                                                                  # title enlargement
x_min     = 12                                                                  # x-axis limit
x_max     = 25                                                                  # x-axis limit
y_min     = 0                                                                   # y-axis limit
y_max     = 45                                                                  # y-axis limit
label     = c("", "", "Excel", "Sturges", "Scott")                              # title

# histograms with predefined number of classes and class width
for(i in 1:5){
    hist(                                                                       # histogram
    y,                                                                          # dataset
    breaks   = bs[[i]],                                                         # breaks argument
    xlim     = c(x_min, x_max),                                                 # x-axis limits
```

```
    ylim     = c(y_min, y_max),                                                # y-axis limits
    ylab     = "frequency",                                                    # y-axis label
    xlab     = "BDI",                                                          # x-axis label
    main     = sprintf(paste(label[i], "k = %.0f, d = %.2f"), ks[i],ds[i]))  # title
}

# r default histogram
hist(                                                                          # histogram
y,                                                                             # dataset
xlim     = c(x_min, x_max),                                                    # x-axis limits
ylim     = c(y_min, y_max),                                                    # y-axis limits
ylab     = "frequency",                                                        # y-axis label
xlab     = "",                                                                 # x-axis label
main     = "R Default")                                                        # title
mtext(LETTERS[6], adj=0, line=2, cex = 1.2, at = 8)                           # subplot letter
dev.off()                                                                      # end image editing
```

10.3 Empirical distribution functions

In addition to the question of how frequently certain values or classes of values occur in a data set, it can be interesting to examine how frequently values that are smaller than a certain value occur. Although the answer to this question is of course implicit in the absolute and relative frequencies or the corresponding histograms, it is sometimes advantageous to have this information directly available. The concepts of *cumulative absolute and relative frequency distribution* and *empirical distribution function* introduced in this section denote the corresponding analogues to the concepts discussed in Section 10.1 and Section 10.2. We begin by defining the *cumulative absolute* and *cumulative relative frequency distributions*.

Definition 10.3 (Cumulative absolute and relative frequency distributions)**.** Let $y = (y_1, ..., y_n)$ be a data set, $W := \{w_1, ..., w_k\}$ with $k \le n$ the numerical values occurring in the data set and h and r the absolute and relative frequency distributions of the values of y, respectively. Then the function is called

$$H : W \to \mathbb{N}, w \mapsto H(w) := \sum_{w' \le w} h(w') \tag{10.9}$$

the *cumulative absolute frequency distribution* of y and the function

$$R : W \to [0, 1], w \mapsto R(w) := \sum_{w' \le w} r(w') \tag{10.10}$$

the *cumulative relative frequency distribution* of the numerical values of y.

•

In the sense of the definition Definition 10.1 applies to the cumulative frequency distributions

$$H(w) = \text{number of } y_i \text{ in } y \text{ with } y_i \le w \tag{10.11}$$

and

$$R(w) = \text{number of } y_i \text{ in } y \text{ with } y_i \le w \text{ divided by } n. \tag{10.12}$$

In **R**, the function `cumsum()` allows the calculation of cumulative sums and thus enables the evaluation of cumulative frequency distributions, as the following **R** code demonstrates using the example data set.

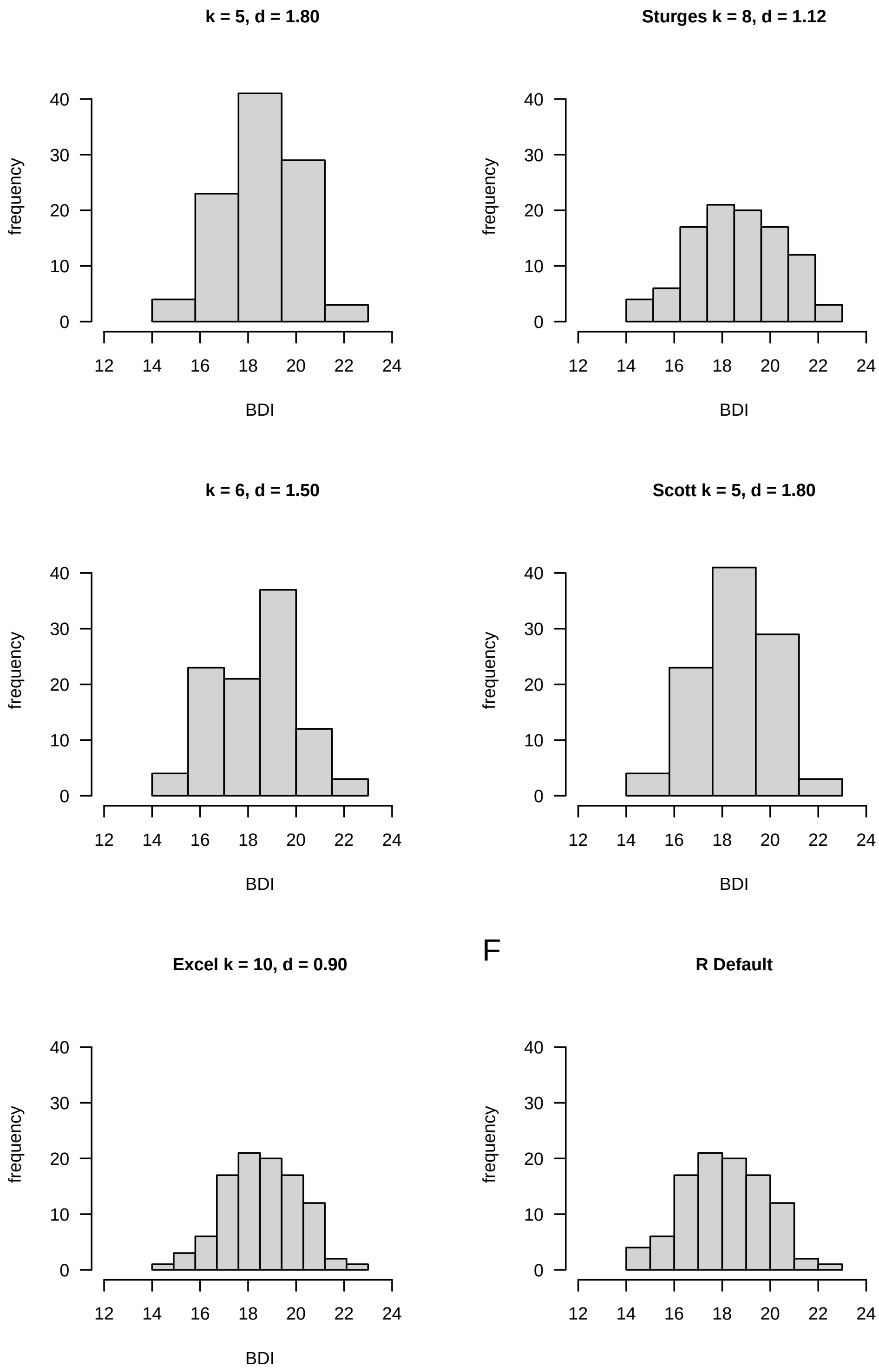

Figure 10.2 Histograms of the absolute frequencies of the pre-intervention BDI-II scores. Note that the qualitative impression that values around 18 occur frequently in the data set and that lower and higher values are rarer is constant across all choices of the number of classes. The exact quantitative appearance, however, depends on the exact choice of the number of classes and the class width.

```
D         = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y         = D$PRE                                                # pre-intervention BDI-II data
n         = length(y)                                            # number of data values
H         = as.data.frame(table(y))                              # absolute frequency distribution as a data frame
names(H)  = c("w", "h")                                          # column naming
H$H       = cumsum(H$h)                                          # cumulative absolute frequency distribution
H$r       = H$h/n                                                # relative frequency distribution
H$R       = cumsum(H$r)                                          # cumulative relative frequency distribution
print(H, digits = 1)
```

```
    w  h   H    r    R
1  14  1   1 0.01 0.01
2  15  3   4 0.03 0.04
3  16  6  10 0.06 0.10
4  17 17  27 0.17 0.27
5  18 21  48 0.21 0.48
6  19 20  68 0.20 0.68
7  20 17  85 0.17 0.85
8  21 12  97 0.12 0.97
9  22  2  99 0.02 0.99
10 23  1 100 0.01 1.00
```

Pay particular attention to the comparison of `h` and `H` as well as `r` and `R`, which illustrate the cumulative totals in Definition 10.3. The following **R** code enables the representation of the cumulative frequency distributions determined as shown in Figure 10.3. Obviously, the absolute and relative frequency of values less than the smallest value in the data set is zero. On the other hand, the absolute and relative frequency of values smaller than the largest value in the data set are n and $n/n = 1$, respectively.

```
pdf(                                                                 # pdf copy function
file       = "./_figures/110-cumulative-frequency-distributions.pdf", # filename
width      = 10,                                                     # width (inches)
height     = 5)                                                      # height (inches)
Hw         = H$H                                                     # h(w) values
Rw         = H$R                                                     # r(w) values
names(Hw)  = H$w                                                     # barplot needs w values as names
names(Rw)  = H$w                                                     # barplot needs w values as names
par(                                                                 # figure parameters
mfcol      = c(1,2),                                                 # rows and columns layout
las        = 1,                                                      # x-tick orientation
cex        = 1)                                                      # annotation enlargement
barplot(                                                             # bar chart
Hw,                                                                  # h(w) values
col        = "gray90",                                               # bar color
xlab       = "w",                                                    # x-axis label
ylab       = "H(w)",                                                 # y-axis label
ylim       = c(0,110),                                               # y-axis limits
las        = 1,                                                      # axis tick label orientation
main       = "Absolute cumulative frequency PRE")                    # title
barplot(                                                             # bar chart
Rw,                                                                  # r(w) values
col        = "gray90",                                               # bar color
xlab       = "w",                                                    # x-axis label
ylab       = "R(w)",                                                 # y-axis label
ylim       = c(0,1),                                                 # y-axis limits
las        = 1,                                                      # axis tick label orientation
main       = "Relative cumulative frequency PRE")                    # title
dev.off()                                                            # end image editing
```

The cumulative analogue of a histogram is the *empirical distribution function*. In contrast to a histogram, no definition of classes is required to determine an empirical distribution function. Instead of the interval division of the real numbers between the minimum and the maximum of a data set, the empirical distribution function simply uses the real numbers, as the following definition makes clear.

Definition 10.4 (Empirical distribution function)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then the function is called

$$F : \mathbb{R} \to [0,1], x \mapsto F(x) := \frac{\text{number of } y_i \text{ in } y \text{ with } y_i \leq x}{n} \tag{10.13}$$

the empirical distribution function (EVF) of y.

•

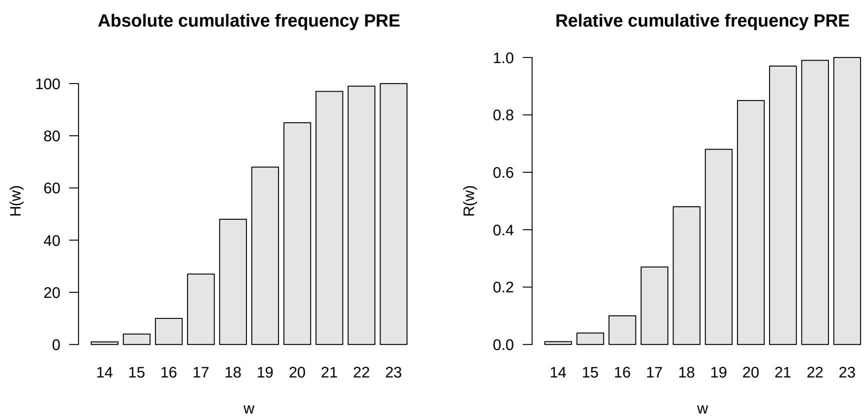

Figure 10.3 Absolute and relative cumulative frequency distributions of pre-intervention BDI-II scores.

The empirical distribution function is sometimes called the *empirical cumulative distribution function*. In general, the empirical distribution function as in Definition 10.4 is limited to specifying the relative frequencies. Of course, not all real numbers appear in data sets. This implies that the relative frequency assigned to real numbers that lie between two adjacent values in the data set is equal to that of the smaller of the two values. Typically, empirical distribution functions are step functions. In **R**, the function `ecdf()` can be used to evaluate and visualize the empirical distribution function, as the following **R** code demonstrates using the example data set and is visualized in Figure 10.4. For example, you can read from Figure 10.4 that the relative frequency of values smaller than 17.9 is approximately 0.3. Values smaller than 17.8 have the same relative frequency.

```
D      = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y      = D$PRE                                                # pre-intervention BDI-II data
evf    = ecdf(y)                                              # evaluation of the EVF
pdf(                                                          # pdf copy function
file   = "./_figures/110-ecdf.pdf",                           # filename
width  = 8,                                                   # width (inches)
height = 5)                                                   # height (inches)
plot(                                                         # plot() knows how to handle ecdf object
evf,                                                          # ecdf object
xlab   = "BDI",                                               # x-axis label
ylab   = "Relative frequency",                                # y-axis label
main   = "Empirical distribution function")                   # title
dev.off()                                                     # end image editing
```

10.4 Quantiles and boxplots

The empirical distribution function provides an answer, at least graphically represented, to the question of how large the relative proportion of values in a data set is that are smaller than a given value. The quantiles introduced in this section can be understood as an inversion of this data set interrogation. Specifically, with quantiles you specify a relative proportion $0 \le p \le 1$ of the values of a data set and ask which value of the data set is such that $p \cdot 100\%$ of the values of the data set are less than or equal to this value. We use the following definition of the term p *quantile*.

Definition 10.5 (p quantile)**.** Let $y = (y_1, ..., y_n)$ be a data set and

$$y_s = \left(y_{(1)}, y_{(2)}, ..., y_{(n)}\right) \text{ with } \min_{1\le i\le n} y_i = y_{(1)} \le y_{(2)} \le \cdots \le y_{(n)} = \max_{1\le i\le n} y_i \qquad (10.14)$$

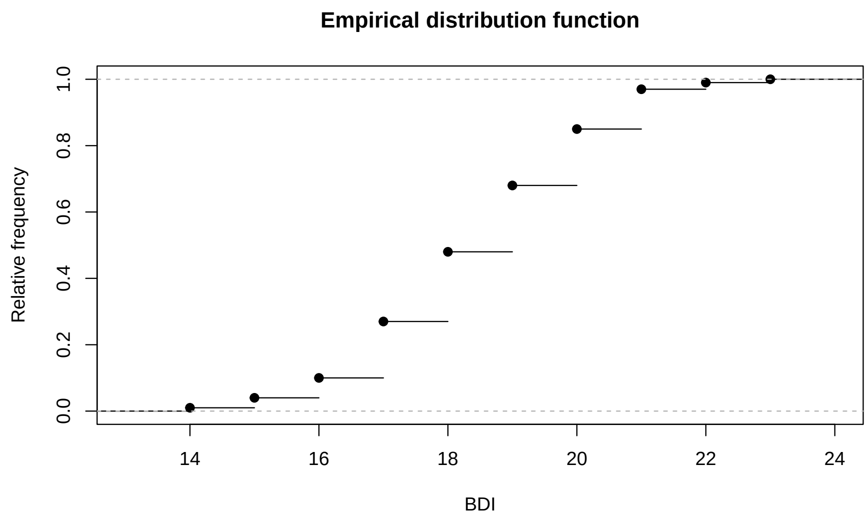

Figure 10.4 Empirical distribution function of the PRE values.

be the corresponding data set sorted in ascending order. Furthermore, let $\lfloor \cdot \rceil$ denote the rounding function. Then for a $p \in [0, 1]$ is the number

$$y_p := \begin{cases} y_{(\lfloor np+1 \rceil)} & \text{if } np \notin \mathbb{N} \\ \frac{1}{2}\left(y_{(np)} + y_{(np+1)}\right) & \text{if } np \in \mathbb{N} \end{cases} \tag{10.15}$$

the p *quantile* of the data set y.

•

Note that the p quantile y_p to Definition 10.5 is either a value of the data set or a value between two values of the data set. After constructing y_p, it applies in particular that at least $p \cdot 100\%$ of all values in the data set are less than or equal to y_p and at least $(1-p) \cdot 100\%$ of all values in the data set are greater than or equal to y_p. The p quantile therefore divides the ordered data set y_s in the ratio p to $(1-p)$. Depending on the choice of p, the p quantiles receive special names:

- $y_{0.25}, y_{0.50}, y_{0.75}$ are called *lower quartile*, *median* and *upper quartile*, respectively,
- $y_{j \cdot 0.10}$ for $j = 1, ..., 9$ are called *deciles* and
- $y_{j \cdot 0.01}$ for $j = 1, ..., 99$ are called *percentiles*.

To clarify the concept of the p quantile, let us consider an example based on Henze (2018), chapter 5. First of all, the data set shown in the table below and its version sorted in ascending order are given.

i	1	2	3	4	5	6	7	8	9	10
y_i	8.5	1.5	75	4.5	6.0	3.0	3.0	2.5	6.0	9.0
$y_{(i)}$	1.5	2.5	3.0	3.0	4.5	6.0	6.0	8.5	9.0	75

We first want to determine the lower quartile, i.e. the 0.25 quantile, of this data set. It's apparently $n = 10$ and we chose $p := 0.25$. Based on Definition 10.5 we find that

$$np = 10 \cdot 0.25 = 2.5 \notin \mathbb{N}. \tag{10.16}$$

So according to Definition 10.5 it applies that

$$y_{0.25} = y_{(\lfloor 2.5+1 \rfloor)} = y_{(3)} = 3.0. \tag{10.17}$$

The lower quartile of the data set under consideration is therefore the value 3.0 of the data set.

Next we want to determine the 0.80 quantile of the data set. Apparently it is still $n = 10$ and we have now chosen $p := 0.80$. In this case we find that

$$np = 10 \cdot 0.80 = 8 \in \mathbb{N}. \tag{10.18}$$

So follows after Definition 10.5

$$y_{0.80} = \frac{1}{2}\left(y_{(8)} + y_{(8+1)}\right) = \frac{1}{2}\left(y_{(8)} + y_{(9)}\right) = \frac{8.5 + 9.0}{2} = 8.75. \tag{10.19}$$

The $y_{0.80}$ quantile of the data set under consideration is with 8.75 the value between the values 8.5 and 9 of the data set. In **R** the quantiles determined in this way are evaluated as follows.

```
y   = c(8.5,1.5,75,4.5,6.0,3.0,3.0,2.5,6.0,9.0)  # sample data
n   = length(y)                                  # number of data values
y_s = sort(y)                                    # sorted data set
p   = 0.25                                       # np \notin \mathbb{N}
y_p = y_s[floor(n*p + 1)]                        # 0.25 quantile
cat("0.25 quantile: ", round(y_p,2))             # output
```

```
0.25 quantile:  3
```

```
p   = 0.80                                       # np \in \mathbb{N}
y_p = (1/2)*(y_s[n*p] + y_s[n*p + 1])            # 0.80 quantile
cat("0.80 quantile: ", round(y_p,2))             # output
```

```
0.80 quantile:  8.75
```

With `quantile()`, **R** also provides a function that automates the quantile determination of a data set. However, it should be noted that the definition of the quantile term given in Definition 10.5 is only one of at least nine different quantile definitions and the exact values can differ from definition to definition. The specific quantile definition used by `quantile()` is selected via the function argument `type`. Hyndman & Fan (1996) provide an overview on the theoretical level, and `?quantile` on the implementation level. As an example, we look at determining the 0.25 quantile of the pre-intervention BDI-II sample data set using `type = 8`.

```
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                               # pre-intervention BDI-II data
y_p = quantile(y, 0.25, type = 8)                         # 0.25 quantile, definition 1
```

```
0.25 quantile (type = 8):  17
```

Finally, a *boxplot* visualizes a quantile-based summary of a data set, typically visualizing the minimum and maximum of a data set as well as the lower quartile, median and upper quartile. In **R** you create a boxplot using the `boxplot()` function, as the following **R** code demonstrates using the pre-intervention BDI-II data set.

```
D          = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
pdf(                                                             # pdf copy function
file       = "./_figures/110-boxplot.pdf",                       # filename
width      = 8,                                                  # width (inches)
height     = 5)                                                  # height (inches)
boxplot(                                                         # box plot
D$PRE,                                                           # dataset
horizontal = T,                                                  # horizontal representation
range      = 0,                                                  # whiskers up to min y and max y
xlab       = "BDI",                                              # x axis label
main       = "Box plot")                                         # title
dev.off()                                                        # end image editing
```

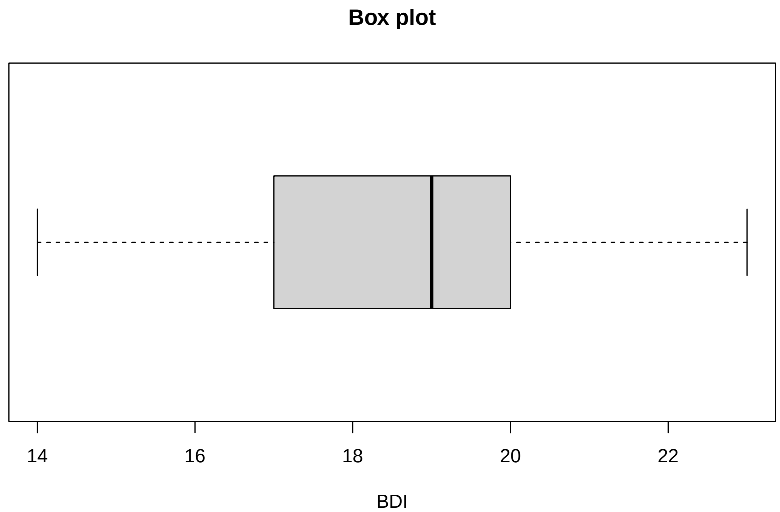

Figure 10.5 Boxplot of the pre-intervention BDI-II scores of the example data set.

In Figure 10.5, $\min y$ and $\max y$ are represented as so-called "whisker endpoints", $y_{0.25}$ and $y_{0.75}$ form the lower and upper boundaries of the central gray box and the median $y_{0.50}$ is represented as a line in the central gray box. Here too, there are a lot of box plot variations, see McGill et al. (1978). A precise explanation of the data set properties displayed graphically is always necessary. Finally, it should be noted that the representation of data sets using boxplots has become somewhat out of fashion in the last twenty years and is only very rarely found in current scientific publications.

10.5 Measures of central tendency

In this section we look at some basic data set metrics that are central to descriptive statistics and provide information about the "average" value of a data set. We start by defining the mean.

10.5.1 Mean

Definition 10.6 (Mean)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then that means

$$\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i \tag{10.20}$$

the *mean* of y.

•

The mean of a data set is the sum of the data values divided by the number of data values. The mean defined here is also referred to as the *arithmetic mean* or *average*. In the context of frequentist inference, we will refer to the mean as the *sample mean* in the context of the random sampling model. In particular, the models of frequentist inference also allow the functional form of the mean to be more deeply substantiated, as we will see at a given point. Using the sum function, you can calculate the mean of a data set in **R** as follows.

```
D     = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y     = D$PRE                                              # pre-intervention BDI-II data
n     = length(y)                                          # number of values
y_bar = (1/n)*sum(y)                                       # average calculation
```

```
Mean of PRE-BDI-II data 18.61
```

With the `mean()` function, **R** of course also provides a function for automated calculation of the mean value.

```
cat("Mean of PRE-BDI-II data", mean(y))  # output
```

```
Mean of PRE-BDI-II data 18.61
```

In the following theorem we first note two properties of the mean value of a data set that make its classification as a measure of the central tendency of a data set plausible.

Theorem 10.1 (Centrality properties of the mean). *Let $y = (y_1, ..., y_n)$ be a data set and $\bar{y}$ be the mean of y. Then apply*

(1) The sum of the deviations from the mean is zero,

$$\sum_{i=1}^{n}(y_i - \bar{y}) = 0. \tag{10.21}$$

(2) The absolute sums of negative and positive deviations from the mean are equal, i.e. if $j = 1, ..., n_j$ denotes the data point indices with $(y_j - \bar{y}) < 0$ and $k = 1, ..., n_k$ denotes the data point indices with $(y_k - \bar{y}) \geq 0$, then with $n = n_j + n_k$ it applies that

$$|\sum_{j=1}^{n_j}(y_j - \bar{y})| = |\sum_{k=1}^{n_k}(y_k - \bar{y})|. \tag{10.22}$$

∘

Proof. (1) It applies

$$\begin{aligned}\sum_{i=1}^{n}(y_i - \bar{y}) &= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \bar{y} \\ &= \sum_{i=1}^{n} y_i - n\bar{y} \\ &= \sum_{i=1}^{n} y_i - \frac{n}{n}\sum_{i=1}^{n} y_i \\ &= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} y_i \\ &= 0.\end{aligned} \tag{10.23}$$

(2) Let $j = 1, \ldots, n_j$ be the indices with $(y_j - \bar{y}) < 0$ and $k = 1, \ldots, n_k$ the indices with $(y_k - \bar{y}) \geq 0$, such that $n = n_j + n_k$. Then applies

$$
\begin{aligned}
\sum_{i=1}^{n}(y_i - \bar{y}) &= 0 \\
\Leftrightarrow \sum_{j=1}^{n_j}(y_j - \bar{y}) + \sum_{k=1}^{n_k}(y_k - \bar{y}) &= 0 \\
\Leftrightarrow \sum_{k=1}^{n_k}(y_k - \bar{y}) &= -\sum_{j=1}^{n_j}(y_j - \bar{y}) \\
\Leftrightarrow |\sum_{j=1}^{n_j}(y_j - \bar{y})| &= |\sum_{k=1}^{n_k}(y_k - \bar{y})|.
\end{aligned}
\tag{10.24}
$$

□

According to Theorem 10.1, the mean value of a data set is just such that the total deviations of the data points from this value equalize to zero and that positive and negative deviations from the mean balance each other out. In **R** you can verify Theorem 10.1 using the example data set as follows.

```
# dataset
D   = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y   = D$PRE                                               # pre-intervention BDI-II data
s   = sum(y - mean(y))                                    # sum of deviations from the mean
s_1     = sum(y[y <= mean(y)] - mean(y))                  # sum of all negative deviations
s_2 = sum(y[y > mean(y)]  - mean(y))                      # sum of all positive deviations
```

```
Sum of deviations from the mean: 0
Sums of positive and negative deviations: -71.28 71.28
```

Theorem 10.2 (Summation properties of the mean). *Given two data sets $y = (y_1, \ldots, y_n)$ and $z = (z_1, \ldots, z_n)$ of the same size and let it be by*

$$
y + z := (y_1 + z_1, \ldots, y_n + z_n) \tag{10.25}
$$

the sum of these data sets is defined. Furthermore, let $\bar{y}$ and $\bar{z}$ be the mean values of the data sets y and z, respectively. Then

$$
\overline{y+z} = \bar{y} + \bar{z} \tag{10.26}
$$

∘

Proof. It applies

$$
\begin{aligned}
\overline{y+z} &:= \frac{1}{n}\sum_{i=1}^{n}(y_i + z_i) \\
&= \frac{1}{n}\sum_{i=1}^{n} y_i + \frac{1}{n}\sum_{i=1}^{n} z_i \\
&=: \bar{y} + \bar{z}.
\end{aligned}
\tag{10.27}
$$

□

If you look at the sum of two data sets, it is irrelevant to determining their mean value whether you first add up the data sets and then determine the mean value of the summed data set or whether you determine the mean value for each of the two data sets and add them up. We verify this result using the pre- and post-intervention BDI-II data from the example data set in **R** as follows.

```
D         = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y         = D$PRE                                               # pre-intervention BDI-II data
y_bar     = mean(y)                                             # mean of pre-intervention BDI-II data
z         = D$POS                                               # post-intervention BDI-II data
z_bar     = mean(z)                                             # mean of post-intervention BDI-II data
yz_bar_1  = mean(y + z)                                         # mean of summed pre and post data
yz_bar_2  = y_bar + z_bar                                       # sum of the means of the pre- and post-data
```

```
Mean of z: 31.68
Sum of the means of y and z :  31.68
```

Theorem 10.3 (Mean under linear-affine transformation). *Let $y = (y_1, ..., y_n)$ be a data set and $\bar{y}$ be the mean of y. Furthermore, be for $a, b \in \mathbb{R}$*

$$ay + b := (ay_1 + b, ..., ay_n + b) \tag{10.28}$$

a linear-affine transformed data set of y is defined. Then that applies

$$\overline{ay + b} = a\bar{y} + b \tag{10.29}$$

∘

Proof. It applies

$$\begin{aligned}
\overline{ay + b} &:= \frac{1}{n}\sum_{i=1}^{n}(ay_i + b) \\
&= \sum_{i=1}^{n}\left(\frac{1}{n}ay_i + \frac{1}{n}b\right) \\
&= \left(\sum_{i=1}^{n}\frac{1}{n}ay_i\right) + \left(\sum_{i=1}^{n}\frac{1}{n}b\right) \\
&= a\left(\frac{1}{n}\sum_{i=1}^{n}y_i\right) + \frac{1}{n}\sum_{i=1}^{n}b \\
&= a\bar{y} + \frac{1}{n}nb \\
&= a\bar{y} + b.
\end{aligned} \tag{10.30}$$

□

A linear-affine transformation of a data set transforms the mean value of the data set in a linear-affine manner. You can take advantage of this if you think about what the mean value should be when rescaling a data set, for example converting from grams to kilograms. Of course, to be on the safe side, you can also simply determine the average of the rescaled data set. We demonstrate this property as an example by rescaling the pre-intervention BDI-II data set with the value $a = 2$ while simultaneously adding the constant $b := 5$.

```
D         = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y         = D$PRE                                               # pre-intervention BDI-II data
y_bar     = mean(y)                                             # mean of pre-intervention BDI-II data
a         = 2                                                   # multiplication constant
b         = 5                                                   # addition constant
z         = a*y + b                                             # linear-affine transformation of the PRE data
z_bar     = mean(z)                                             # mean value of transformed PRE data
ay_bar_b  = a*y_bar + b                                         # pre data mean transformation
```

```
Mean value of the transformed data set: 42.22
Transformation of the mean        : 42.22
```

10.5.2 Median

In the mean value defined above, each value of a data set is included summatively with the same weight $1/n$. This particularly applies to values that are very atypical compared to the other values in the data set. One way to determine the "center" of a data set independent of such atypical values is to determine its median, which we define below. We already learned about the median in section Section 10.4 as the 0.5 quantile, with which it is identical.

Definition 10.7 (Median)**.** Let $y = (y_1, ..., y_n)$ be a data set and $y_s = (y_{(1)}, ..., y_{(n)})$ the corresponding data set sorted in ascending order. Then the median of y is defined as

$$\tilde{y} := \begin{cases} y_{((n+1)/2)} & \text{if } n \text{ ungerade,} \\ \frac{1}{2}\left(y_{(n/2)} + y_{(n/2+1)}\right) & \text{if } n \text{ gerade.} \end{cases} \tag{10.31}$$

•

Definition 10.7 implies that at least 50% of all data values y_i of the data set are less than or equal to $\tilde{y}$ and at the same time at least 50% of all data values y_i of the data set are greater than or equal to $\tilde{y}$. Instead of a formal proof of these statements, we refer to Figure 10.6. As in the definition of the median, we differentiate between the cases where the number of data points is odd (here $n = 5$, Figure 10.6 A) or even ($n = 6$, Figure 10.6 B). In the case of an odd number of data points, the median is the data value with the index number that lies in the middle of the index numbers of the sorted data set, in the example Figure 10.6 A, i.e. the value $y_{(3)}$. In this case, $3/5$, i.e. 60% of the data values (specifically $y_{(1)}$, $y_{(2)}$ and $y_{(3)}$) are less than or equal to the median, but at the same time $3/5$, i.e. 60% of the data values (specifically $y_{(3)}$, $y_{(4)}$ and $y_{(5)}$) greater than or equal to the median. In the case of an even number of data points, the location of the median is somewhat more intuitive: the median lies in the middle of the two middle values of the sorted data set (in Figure 10.6 B specifically in the middle of the values $y_{(3)}$ and $y_{(4)}$). This means that $3/6$, i.e. 50% of the data values (specifically $y_{(1)}$, $y_{(2)}$ and $y_{(3)}$) are smaller than the median and $3/6$, i.e. 50% of the data values (specifically $y_{(4)}$, $y_{(5)}$ and $y_{(6)}$) are larger than the median.

The following **R** code determines the median of the example data set using Definition 10.7.

```
D       = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y       = D$PRE                                               # pre-intervention BDI-II data
n       = length(y)                                           # number of values
y_s     = sort(y)                                             # ascending sorted vector
if(n %% 2 == 1){                                              # n odd, n mod 2 == 1
   y_tilde = y_s[(n+1)/2]                                     # median formula, case n odd
} else {                                                      # n even, n mod 2 == 0
   y_tilde = (y_s[n/2] + y_s[n/2 + 1])/2}                     # median formula, case n even
```

```
Median of PRE-BDI-II data: 19
```

Of course, **R** also provides a function with `median()` for directly determining the median, as the following **R** code demonstrates.

```
cat("Median of PRE-BDI-II data:", median(y))  # output
```

```
Median of PRE-BDI-II data: 19
```

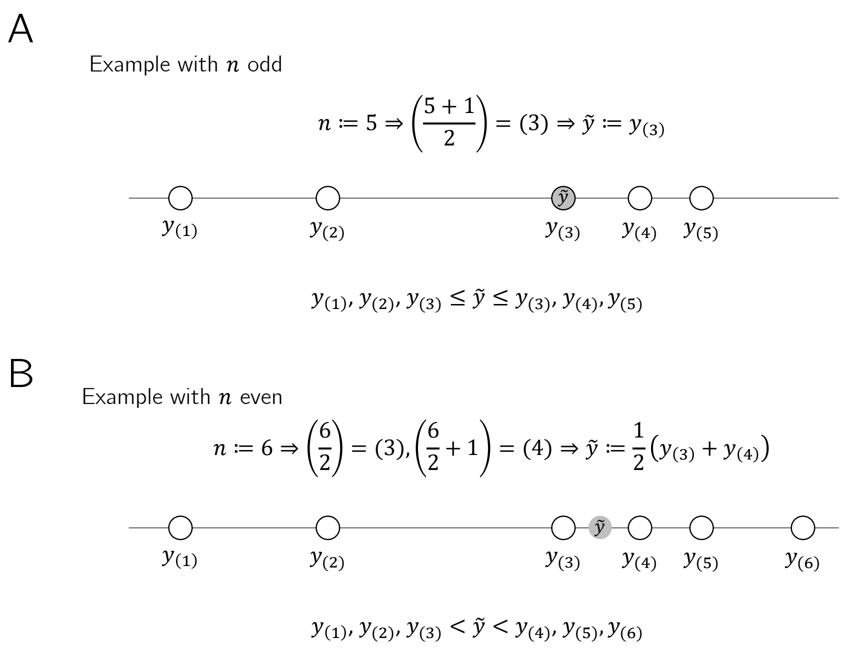

Figure 10.6 Determination and properties of the median.

The following simulation shows as an example that the median, as a measure of the middle of a data set, is less susceptible to outliers than the mean.

```
D = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y = D$PRE                                               # pre-intervention BDI-II data
```

```
Mean and median for data set without outliers: 18.61 19
```

```
z    = y      # new simulated data set
z[1] = 10000  # ... with an extreme value
```

```
Mean and median for data set with outliers: 118.44 19
```

So, although the median appears to be a more robust measure of the central tendency of a data set than the mean, determining averages is certainly the more common approach. On the one hand, this is due to the fact that the mean value is easier to use in an analytical context for designing probabilistic models due to its mathematically less complex definition. On the other hand, the occurrence of extreme values is usually a sign of data collection errors (e.g. a BDI-II value of 10.000 simply should not occur) or large differences between the experimental units under consideration, which are usually noticed by visual inspection of the data set before determining measures of central tendency. However, there is a whole subfield of statistics that deals with - in the broadest sense - measures that are independent of outliers, see e.g. Huber (1981).

10.5.3 Mode

If some values occur repeatedly in a data set, a third measure of central tendency is given with the *mode* as the value that occurs most frequently in a data set. We use the following definition.

Definition 10.8 (Mode). $y := (y_1, ..., y_n)$ with $y_i \in \mathbb{R}$ is a data set, $W := \{w_1, ..., w_k\}$ with $k \leq n$ are the different numerical values occurring in the data set and $h : W \to \mathbb{N}$ is the absolute frequency distribution of the numerical values of y. Then the *mode (or mode)* of y is defined as

$$\text{argmax}_{w \in W} h(w), \tag{10.32}$$

i.e. the value that occurs most frequently in the data set.

•

In **R** you can determine the mode of a corresponding data set as follows.

```
D        = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y        = D$PRE                                                # pre-intervention BDI-II data
n        = length(y)                                            # number of data values (100)
H        = as.data.frame(table(y))                              # absolute frequency distribution (data frame)
names(H) = c("a", "w")                                          # consistent naming
mode     = H$w[which.max(H$w)]                                  # mode
```

```
cat("Mode of PRE-BDI-II data:", as.numeric(as.vector(mode)))  # output as numeric vector, not factor
```

```
Mode of PRE-BDI-II data: 21
```

10.5.4 Data distribution shapes and measures of central tendency

How useful it is to use the mean, median or mode to provide information about the central tendency of a data set depends largely on the nature of the data set. For example, if you have a data set of high-resolution data in which no data value appears multiple times, the mode is certainly not suitable for making a quantitative statement about the central tendency of the data set. The specification of a measure of central tendency should always be viewed in the context of the overall nature of the data set and, if applicable, the inferential statistical model that is used to interrogate the data set. In Figure 10.7 we give four examples of how the measures of central tendency considered above can behave in different data value distribution scenarios. Of course, these four scenarios are not exhaustive and each data set should be viewed critically for a meaningful measure of central tendency.

Figure 10.7 A visualizes the position of the mean, median and mode in a data set that is symmetrically distributed around a value of approximately $y = 50$ and for which values close to this central value occur frequently and values far from this central value occur rarely. As we see later, this corresponds to the typical characteristics of normally distributed data values. In this case, the mean, median and mode are quite close to each other and each of these measures in fact describes the central tendency of the data set.

Figure 10.7 B visualizes the position of the mean, median and mode in a data set in which the data values occur clustered around two values of approximately $y = 25$ and $y = 75$ and rarely occur in the positive and negative directions from these values. For example, there are very few data points in this data set around $y = 50$. The mean and median now take on values around $y = 50$. As such, they do not provide any information about which values occur particularly frequently in this data set, but only about where the average "physical center of gravity" of the data set is. The mode, on the other hand, is at the larger of the two data accumulation points due to the slightly more frequent occurrence of values around $y = 75$. Overall, none of the three measures really satisfactorily describes the central tendency of the data values for such a data set, which is also called *bimodal*.

Figure 10.7 C visualizes the position of the mean, median and mode in a data set in which the data values in the width under consideration occur with approximately the same frequency throughout. Here the mode is relatively arbitrary at around $y = 15$, as there is a minimal accumulation of data values there. In the sense of the "physical center of gravity" of the data set, the mean and median are around $y = 50$, even if the values of this data set have no tendency to occur more frequently there than elsewhere. Here too, the description of the central tendency of the data set by the measures considered is not really satisfactory. Taken together, using Figure 10.7 B and Figure 10.7 C, one might be able to conclude that if a data set does not have a clear central tendency, the measures of central tendency cannot reflect it either.

Finally, Figure 10.7 D shows the location of the mean, median and mode for a data set that is skew-symmetrically distributed around values of approximately $y = 0.5$. The data in this data set only takes positive values, many data values are in fact around $y = 0.5$, but there are also higher data values. We will see later that this appearance is typical for squared normally distributed data values. In a psychological context, reaction times in particular often have a similar distribution, as they cannot be negative, in most cases being in the range of 0.3 to 0.6 seconds, but in some cases can be much longer due to inattention or stimulus variability. In this case, the median describes the actual central tendency of the data values somewhat better than the mean, as this is slightly shifted towards higher values due to the rarely occurring outlier values.

Overall, the specification of one or more measures of central tendency must always be viewed critically against the background of the distribution of the values of a data set and classified qualitatively.

10.6 Measures of variability

In addition to the question of what "average value" a data set assumes, it is often of interest to quantify how much the values in the data set are scattered or, in other words, how "variable" they are. Here we consider the *range*, the *empirical variance* and the *empirical standard deviation* as quantitative measures of the variability of a data set. As with the mean, the full meaning of the latter measures becomes apparent primarily in the context of inferential statistical methods.

10.6.1 Range

The range of a data set is the difference between its largest and smallest values.

Definition 10.9 (Range)**.** Let $y = (y_1, ..., y_n)$ be a data set. Then the *range* of $y_1, ..., y_n$ is defined as

$$r := \max(y_1, ..., y_n) - \min(y_1, ..., y_n). \tag{10.33}$$

•

In **R** you determine the range either by evaluating the maximum and minimum of the data set using the `max()` and `min()` functions or directly using the `range()` function. The following **R** code demonstrates the use of the `max()` and `min()` functions.

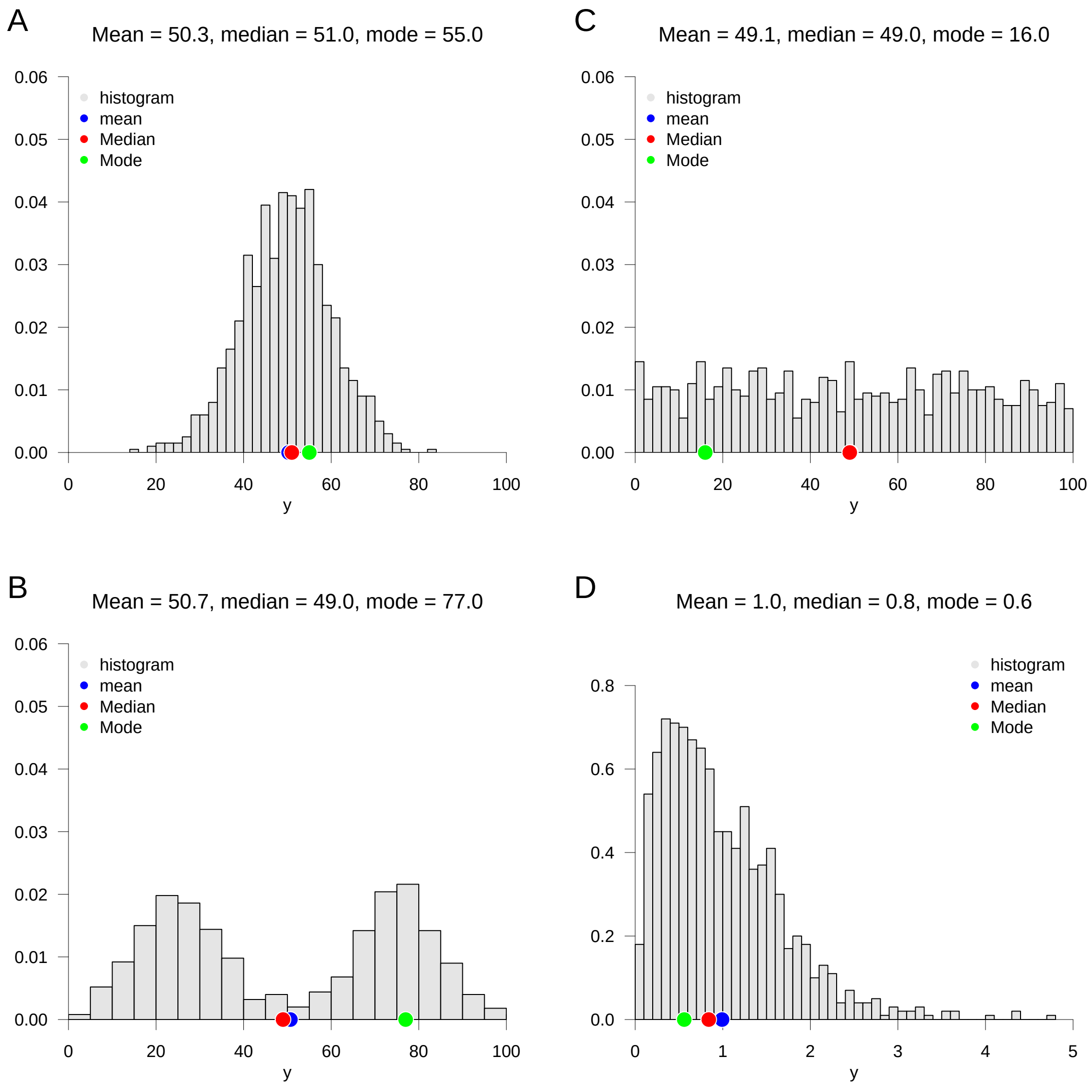

Figure 10.7 Visual intuition on measures of central tendency in different data distribution forms

```
D     = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y     = D$PRE                                               # pre-intervention BDI-II data
y_max = max(y)                                              # maximum of PRE values
y_min = min(y)                                              # minimum of PRE values
r     = y_max - y_min                                       # range
```

```
Range of PRE-BDI-II data: 9
```

The following **R** code demonstrates the use of the `range()` function.

```
MinMax = range(y)              # "automatic" calculation of min(x), max(x)
r      = MinMax[2] - MinMax[1]  # range
```

```
Range of PRE-BDI-II data: 9
```

10.6.2 Empirical variance

A typical statistical measure of the variability of data sets is the standardized sum of the squared differences between the individual data values and their mean. The squared differences between the individual data values and their mean are also referred to as *difference squares*. Depending on the form of standardization, i.e. the formation of an average of the squared deviations, a distinction is made between *uncorrected empirical variance* and *empirical variance*, as the following definition states.

Definition 10.10 (Uncorrected empirical variance and empirical variance)**.** Let $y = (y_1, ..., y_n)$ be a data set with $n > 1$ and $\bar{y}$ be its mean. Then the *uncorrected empirical variance* of y is defined as

$$\tilde{s}^2 := \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \tag{10.34}$$

and the *empirical variance* of y is defined as

$$s^2 := \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{10.35}$$

•

We saw in Theorem 10.1 that the sum of the differences between the data values and their mean is always zero. Squaring the differences

$$y_i - \bar{y} \text{ for } i = 1, ..., n \tag{10.36}$$

in Definition 10.10 now leads to the fact that negative and positive differences between data points do not balance each other out and, depending on the level of the positive and negative differences, with

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 \tag{10.37}$$

a higher or lower measure of the overall deviation of the data values from their mean can be determined. The sum of the squares of the deviations is zero if all data values are identical and therefore identical to their mean. Alternatively, for example, the determination of the absolute amounts $|y_i - \bar{y}|$ would also be conceivable, but the resulting measure of variability has different properties than the measure of the squared deviations in Definition 10.10, which is why this is usually preferred, also in view of its proximity to probabilistic normal distribution models. The uncorrected empirical variance $\tilde{s}^2$ and the (corrected) empirical variance s^2 then differ in terms of the form of the averaging of the squared deviations. For $\tilde{s}^2$ the sum of the squares of the deviations is divided by n, for s^2 by $n-1$. So $\tilde{s}^2$ is always slightly smaller than s^2. However, this difference will be rather small for large n, think for example of the numerical difference between $\frac{1}{3}$ and $\frac{1}{2}$ for small n and between $\frac{1}{1001}$ and $\frac{1}{1000}$ for large n. Note that the empirical variance for $n = 1$ is not defined, but the uncorrected empirical variance can in principle be determined even in this special case. The term *uncorrected* (and implicitly *corrected*) cannot be understood at this point, but only makes sense against the background of a frequentist inference model and will be explained in the later section on parameter estimation.

We summarize the above discussion in the following theorem.

Theorem 10.4 (Ratio of uncorrected empirical variance and empirical variance)**.** *Let $y = (y_1, ..., y_n)$ be a data set with $n > 1$ and let $\tilde{s}^2$ and s^2 be its uncorrected empirical variance and empirical variance, respectively. Then apply*

(1) (conversion)

$$\tilde{s}^2 = \frac{n-1}{n}s^2 \text{ and } s^2 = \frac{n}{n-1}\tilde{s}^2. \tag{10.38}$$

(2) (inequality)

$$0 \le \tilde{s}^2 \le s^2. \tag{10.39}$$

∘

Proof. (1) After Definition 10.10 applies on the one hand

$$\begin{aligned}
\tilde{s}^2 &= \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow \frac{n-1}{n-1}\tilde{s}^2 &= \frac{n-1}{n-1}\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow \tilde{s}^2 &= \frac{n-1}{n}\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow \tilde{s}^2 &= \frac{n-1}{n}s^2
\end{aligned} \tag{10.40}$$

and on the other

$$\begin{aligned}
s^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow \frac{n}{n}s^2 &= \frac{n}{n}\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow s^2 &= \frac{n}{n-1}\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \\
\Leftrightarrow s^2 &= \frac{n}{n-1}\tilde{s}^2.
\end{aligned} \tag{10.41}$$

(2) With the non-negativity of the square function and $\frac{1}{n} > 0$ and $\frac{1}{n-1} > 0$ for $n > 1$, $\tilde{s}^2 \ge 0$ and $s^2 \ge 0$ result directly. Finally applies to $\sum_{i=1}^{n}(y_i - \bar{y})^2 \ne 0$

$$\frac{1}{n} < \frac{1}{n-1} \Leftrightarrow \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 < \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 \Leftrightarrow \tilde{s}^2 < s^2. \tag{10.42}$$

and for $\sum_{i=1}^n (y_i - \bar{y})^2 = 0$ apparently $\tilde{s}^2 = s^2$. So overall:

$$0 \leq \tilde{s}^2 \leq s^2. \tag{10.43}$$

□

The following **R** code demonstrates the determination of the uncorrected empirical variance and the empirical variance using the formulas specified in Definition 10.10.

```
D       = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y       = D$PRE                                               # pre-intervention BDI-II data
n       = length(y)                                           # number of values
s2_tilde = (1/n)*sum((y - mean(y))^2)                         # uncorrected empirical variance
s2      = (1/(n-1))*sum((y - mean(y))^2)                      # empirical variance
```

```
Uncorrected empirical variance of PRE-BDI-II data: 2.998
Empirical variance of PRE-BDI-II data: 3.028
```

The **R** function `var()` determines the empirical variance by default. Using statement (1) in Theorem 10.4, the uncorrected empirical variance can also be determined based on this function.

```
s2      = var(y)            # empirical variance
s2_tilde = ((n-1)/n)*var(y)  # uncorrected empirical variance
```

```
Uncorrected empirical variance of PRE-BDI-II data: 2.998
Empirical variance of PRE-BDI-II data: 3.028
```

The following theorem states how the empirical variance behaves when a data set is transformed with linear affinity.

Theorem 10.5 (Variance in linear-affine transformations)**.** *Let $y = (y_1, ..., y_n)$ be a data set with empirical variance S_y^2 and $z = (ay_1 + b, ..., ay_n + b)$ be the data set with empirical variance S_z^2 transformed linearly with $a, b \in \mathbb{R}$. Then applies*

$$s_z^2 = a^2 s_y^2. \tag{10.44}$$

∘

Proof.

$$\begin{aligned}
s_z^2 &= \frac{1}{n-1}\sum_{i=1}^n (z_i - \bar{z})^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (ay_i + b - (a\bar{y} + b))^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (ay_i + b - a\bar{y} - b)^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n (a(y_i - \bar{y}))^2 \\
&= \frac{1}{n-1}\sum_{i=1}^n a^2(y_i - \bar{y})^2 \\
&= a^2\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\
&= a^2 S_y^2.
\end{aligned} \tag{10.45}$$

□

For example, converting a data set from meters to centimeters changes the variance of the original data set by the factor 100^2. We reproduce Theorem 10.5 using the following **R** codes as an example.

```
y   = D$PRE                                                               # pre-intervention BDI-II data
s2y = var(y)                                                              # empirical variance of y_1,...,y_n
a   = 2                                                                   # multiplication constant
b   = 5                                                                   # addition constant
z   = a*y + b                                                             # z_i = ay_i + b
s2z = var(z)                                                              # empirical variance of z_1,...,z_n
cat("Empirical variance of z_1,...,z_n by direct calculation:", round(s2z,3))  # output
```

```
Empirical variance of z_1,...,z_n by direct calculation: 12.113
```

```
s2z = a^2*s2y                                                             # empirical variance of z_1,...,z_n
cat("Empirical variance of z_1,...,z_n according to theorem:", round(s2z,3))   # output
```

```
Empirical variance of z_1,...,z_n according to theorem: 12.113
```

The following theorem shows a way to determine the uncorrected empirical variance of a data set solely by determining mean values. The theorem finds its analytical equivalent in the variance shift theorem.

Theorem 10.6 (Shift theorem for uncorrected empirical variance). *Let $y = (y_1, ..., y_n)$ be a data set, $y^2 := (y_1^2, ..., y_n^2)$ be its element-wise square, and $\bar{y}$ and $\overline{y^2}$ be the mean values, respectively. Then applies*

$$\tilde{s}^2 = \overline{y^2} - \bar{y}^2 \tag{10.46}$$

∘

Proof.

$$\begin{aligned}
\tilde{s}^2 &:= \frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2 \\
&= \frac{1}{n}\sum_{i=1}^n \left(y_i^2 - 2y_i\bar{y} + \bar{y}^2\right) \\
&= \frac{1}{n}\sum_{i=1}^n y_i^2 - 2\bar{y}\frac{1}{n}\sum_{i=1}^n y_i + \frac{1}{n}\sum_{i=1}^n \bar{y}^2 \\
&= \overline{y^2} - 2\bar{y}\bar{y} + \frac{1}{n}n\bar{y}^2 \\
&= \overline{y^2} - 2\bar{y}^2 + \bar{y}^2 \\
&= \overline{y^2} - \bar{y}^2.
\end{aligned} \tag{10.47}$$

□

We reproduce Theorem 10.6 using the following **R** codes as an example.

```
y        = D$PRE                      # pre-intervention BDI-II data
s2_tilde = mean(y^2) - (mean(y))^2    # \bar{y^2} - \bar{y}^2
cat("Uncorrected empirical variance according to the shift theorem:", round(s2_tilde,3))  # output
```

```
Uncorrected empirical variance according to the shift theorem: 2.998
```

Note that the shift theorem of the uncorrected empirical variance does not apply to the empirical variance, since in this case the multiplication factor $\frac{1}{n-1}$ does not lead to the corresponding mean values. In particular, we have already seen above that for the example data set in question, $s^2 = 3.028$.

10.6.3 Empirical standard deviation

As seen above, the empirical variance is the average squared deviation of a data set value from its mean. If, for example, the unit of the data set in a reaction time task is the millisecond, then subtracting and then squaring results in the unit millisecond^2, a not very intuitive unit. In order to obtain a measure of dispersion whose unit corresponds to the unit of the original data set, the square root function is often applied to the empirical variance. This leads to the concept of *empirical standard deviation*.

Definition 10.11 (Empirical standard deviation)**.** Let $y = (y_1, ..., y_n)$ be a data set. The *empirical standard deviation* of y is defined as

$$s := \sqrt{s^2} \tag{10.48}$$

and the *uncorrected empirical standard deviation* of y is defined as

$$\tilde{s} := \sqrt{\tilde{s}^2}. \tag{10.49}$$

•

In analogy to the properties of the empirical variance and its uncorrected version, the uncorrected empirical standard deviation can also easily be converted into the empirical standard deviation and vice versa, apparently the following apply here

$$\tilde{s} = \sqrt{\frac{n-1}{n}}s \text{ and } s = \sqrt{\frac{n}{n-1}}\tilde{s}. \tag{10.50}$$

The following **R** code demonstrates the determination of the uncorrected empirical standard deviation and the empirical standard deviation using the formulas specified in Definition 10.11.

```
y = D$PRE                                # pre-intervention BDI-II data
n = length(y)                            # number of values
s = sqrt((1/(n-1))*sum((y - mean(y))^2)) # standard deviation
```

```
Empirical standard deviation: 1.74
```

The **R** function `sd()` determines the empirical standard deviation by default.

```
s = sd(y)
```

```
Empirical standard deviation: 1.74
```

If you transform a data set with linear affinity, the multiplication constant of the transformation is only included in the transformation of the empirical standard deviation in terms of its amount. This is the statement of the following theorem.

Theorem 10.7 (Empirical standard deviation for linear-affine transformations)**.** *Let $y = (y_1, ..., y_n)$ be a data set with empirical standard deviation s_y and $z = (ay_1+b, ..., ay_n+b)$ be the data set linearly-affinically transformed with $a, b \in \mathbb{R}$ with empirical standard deviation s_z. Then applies*

$$s_z = |a|s_y. \tag{10.51}$$

∘

Proof.

$$
\begin{aligned}
s_z &:= \left(\frac{1}{n-1}\sum_{i=1}^n (z_i - \bar{z})^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n \left(ay_i + b - (a\bar{y} + b)\right)^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n \left(a(y_i - \bar{y})\right)^2\right)^{1/2} \\
&= \left(\frac{1}{n-1}\sum_{i=1}^n a^2(y_i - \bar{y})^2\right)^{1/2} \\
&= \left(a^2\right)^{1/2}\left(\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2\right)^{1/2}.
\end{aligned}
\tag{10.52}
$$

So $s_z = as_y$ applies when $a \geq 0$ and $s_z = -as_y$ applies when $a < 0$. But this corresponds to $s_z = |a|s_y$. □

We reproduce Theorem 10.7 as an example using the following **R** codes once for the $a \geq 0$ case and once for the $a < 0$ case

```
# a >= 0
D  = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y  = D$PRE                                               # pre-intervention BDI-II data
a  = 2                                                   # multiplication constant
b  = 5                                                   # addition constant
z  = a*y + b                                             # z_i = ay_i + b
sz = sd(z)                                               # empirical standard deviation of z
```

```
Empirical standard deviation after transformation: 3.48
```

```
sz = a*sd(y)  # empirical standard deviation according to theorem
```

```
Empirical standard deviation after transformation: 3.48
```

```
# a < 0
D  = read.csv("./_data/110-descriptive-statistics.csv")  # loading the data set
y  = D$PRE                                               # pre-intervention BDI-II data
a  = -3                                                  # multiplication constant
b  = 10                                                  # addition constant
z  = a*y + b                                             # z_i = ay_i + b
sz = sd(z)                                               # empirical standard deviation of z
```

```
Empirical standard deviation after transformation: 5.221
```

```
sz = -a*sd(y)  # empirical standard deviation according to theorem
```

```
Empirical standard deviation after transformation: 5.221
```

Part II

Imperative programming

On the importance of imperative programming in psychology

Today, scientific data are usually available in digital form. To evaluate these digital data, they are analyzed with the help of a computer. For this purpose, one writes special computer programs that are tailored precisely to the research question at hand. Such programs are called data analysis scripts.

Computer-assisted data analysis follows a typical structure. First, a digital data set is imported and cleaned in order to correct erroneous or incomplete data and prepare them for analysis. Descriptive statistics are then usually computed and visualized in order to obtain an initial overview of the data. This is often followed by a step of probabilistic modeling and inference, in which models based on probability theory are used to test hypotheses or make predictions. Finally, the results are documented and presented clearly, again with the help of the computer.

The central element of computer-assisted data analysis is the *data analysis script*, a text file that documents all steps from the raw data to the final data visualization and thus makes data analysis processes transparent and reproducible. A data analysis script enables third parties to reproduce scientific results, which is a cornerstone of good scientific practice. Data analysis scripts are therefore indispensable tools of everyday scientific work and essential components of scientific publications. In the following sections, starting from a discussion of some basic concepts from computer science and structure-forming elements of imperative programming, we will become familiar with the development of data analysis scripts in the programming environment R.

Various digital tools are used to analyze psychological data. Particularly common are freely available programming languages and platforms such as R, which is widely used in data science, statistics, and psychology, and Python, with its broad range of applications in data science, machine learning, and artificial intelligence. The free programming language Julia, which is in fact optimized for data analysis, still occupies more of a niche and currently has not yet found wide adoption, especially in industrial applications. Free programming languages have the advantage that they are often further developed by a broad community of users and can in this way develop a versatile infrastructure.

In addition, commercial programming environments whose source code cannot be inspected continue to be used in psychology. In neuroimaging and engineering applications, for example, the commercial programming environment MATLAB remains popular even today. Some programs that now seem rather old-fashioned also continue to be used, including SPSS, which was very widespread in the social sciences and psychology around the turn of the millennium, JMP, with areas of application in biology and psychology, and STATA, which was popular in medicine and economics, among other fields.

The popularity of digital tools is subject to constant change. One way to measure the current popularity of different programming languages and environments is to quantify Google searches for programming language tutorials. An overview of current trends is provided by the PYPL Index, which is based on the analysis of such search queries.

In summary, digitalization has a lasting impact on science in particular. Effective research data management poses an acute challenge that researchers must face. Programming is increasingly becoming a central craft of scientific work, and basic knowledge of computer science has become indispensable in the modern working world. This applies not least to psychotherapists, for example in the context of online interventions, digitally supported treatment services, or psychotherapy research.

11 Basic concepts of computer science

In ordinary usage (cf. Wikipedia), computer science is the science of the systematic representation, storage, processing, and transmission of information, with particular emphasis on the automatic processing of information by computers. Computer science is both a basic and formal science and an engineering discipline.

Central components of computer science are, on the one hand, *computers* and, on the other hand, *algorithms and programs*. *Computers* are machines that can store data and carry out simple data operations. They perform these simple operations at extremely high speed. The universality of computers rests on the fact that they can store both data and programs.

Programs are *algorithms* formulated in a programming language. *Algorithms*, in turn, are sequences of *instructions* that prescribe certain *operations*. In algorithms, one distinguishes different levels: the *description level*, as found, for example, in a recipe, assembly instructions, or a data analysis script; the *instructions* of an algorithm, such as "mix flour and water", the pictorial depiction of screwing two parts together, or the instruction `x = c(1, 2, 3)` in **R**; and finally the *execution level*, that is, the actual cooking process, the assembly of a piece of furniture, or the execution of the data analysis script.

Computer science is divided into several subfields that may also be relevant to psychology. *Theoretical computer science* deals with the foundations of computation, such as automata theory, computability theory, and complexity theory. *Applied computer science* concerns the development and use of application software, questions of human-computer interaction, and the social effects of computer science. *Computer engineering* examines the hardware side of computer science, including microprocessor technology, computer architecture, and network technology. Finally, *practical computer science* deals with concrete implementation in the form of programming, the development and analysis of algorithms, and the construction and use of databases. In psychological basic research, practical computer science is the subfield that is applied most often. However, questions from complexity theory, and thus from theoretical computer science, are also of interest in the development of quantitative psychological theories (cf. Van Rooij & Wareham (2012)), and human-computer interaction is one of the topic areas of modern work psychology (cf. Dahm (2005)).

In addition, there are many special areas of computer science that are closely connected with psychology. These include, in particular, the areas of *machine learning* and *artificial intelligence*, which have their origins in quantitative cognitive psychology and today make it possible to analyze large amounts of data automatically and to make predictions. Another important field is *computer graphics and visual computing*, which deals with image recognition and image synthesis as well as the design of virtual reality (VR) and augmented reality (AR). These technologies use findings from perceptual psychology on the one hand and, on the other hand, open up new possibilities for experiments, therapies, and training in psychology. *Computational linguistics* contributes methods of speech recognition and speech synthesis that help analyze, model, and make human communication usable in applications such as chatbots or assistance systems. Finally, *bioinformatics* also plays a major role for psychology, for example by developing methods for evaluating medical imaging procedures that are also used in neuropsychological research and diagnostics.

11.1 Computer architecture

Even though we do not want to go into the technical aspects of computer science in depth here, for imperative programming it is helpful to have at least a rudimentary understanding of the typical physical structure, that is, the hardware, of a computer. A computer consists of several essential hardware components that work together to handle all computation and storage tasks. At the center is the main circuit board, also called the motherboard or mainboard. It is the central printed circuit board on which important components such as the processor, memory, and connectors are linked to one another.

The heart of a computer is the CPU (central processing unit, also called the microprocessor), which combines the arithmetic unit, the control unit, and the registers of the system. It performs all basic computations, controls the flow of data, and coordinates the processes. In addition, it has a small but very fast *cache*, a volatile memory that temporarily stores frequently needed data in order to speed up processing. Typical examples of current CPUs are Intel Core i processors, AMD Ryzen processors, or Apple Silicon M processors.

The CPU is complemented by RAM (random access memory), the temporary, volatile working memory of the system. During the computer's runtime, the programs and data currently needed are stored here to enable fast access. RAM is limited in its capacity, for example to 32 to 64 GB in many modern systems.

Mass storage, the computer's stationary storage, is used for the permanent storage of data and programs. Today, SSDs (solid state drives) are usually used for this purpose; they are much faster and more robust than conventional magnetic hard drives. Cloud storage is also increasingly used as a complement to, or replacement for, local mass storage.

Another important component is the GPU (graphics processing unit), a powerful processor optimized specifically for visualization tasks. The GPU relieves the CPU in graphics-intensive applications such as 3D rendering or video editing and also plays a growing role in computationally intensive tasks such as training neural networks, where it can effectively support the CPU.

The basic functional architecture of modern computers goes back to Von Neumann (1945) and is therefore referred to as the *Von Neumann architecture*. The decisive characteristic of the Von Neumann architecture is that both instructions (program instructions) and data are stored in the same memory, namely in the form of binary numbers. The central processing unit (CPU) reads both the instructions and the data from this shared memory. As a result, programs can be changed dynamically during execution or even generated anew, since in memory they are present merely as data.

Specifically, Von Neumann (1945) describes the following computer structure: the computer is composed of the basic functional units control unit, arithmetic unit, memory, input unit, and output unit. Programs and data are entered into memory together, so that data, programs, and intermediate and final results are stored in the same memory area. The memory itself is divided into equally sized, numbered (addressed) cells whose contents can be retrieved or changed selectively via the respective address. The instructions of a program typically lie in consecutive memory cells, so that the control unit can call the next instruction simply by increasing the instruction address by one. This principle of sequentially processing instructions is the basic principle of imperative programming and is therefore very closely connected with the Von Neumann architecture. Special jump instructions also make it possible to deviate from this fixed order and jump to another point in the program. The basic instructions include arithmetic operations such as addition and multiplication, logical comparisons such as logical AND or OR, and transport instructions by which data are transferred, for example, from the input unit to memory or from memory to the arithmetic unit. All data, that is, both instructions and addresses, are encoded in binary. A suitable circuit technology in the computer handles the binary encoding and decoding of the information.

11.2 Algorithms and programs

A *real-world problem* denotes the problem that is to be solved with the help of a computer. A typical example is the evaluation of questionnaire data from a psychological study. To work on such a problem, one first creates a *problem specification*, that is, the precise verbal formulation of the real-world problem, as one might find it in the methods section of a scientific publication. On this basis, an *algorithm* is designed, that is, a sequence of instructions for solving the problem, for example reading in the data, computing descriptive statistics, and carrying out a statistical test. Finally, the algorithm is implemented in the form of a *program*, that is, as a text file written in a programming language that can be executed by a computer.

Definition 11.1 (Algorithm)**.** An algorithm is a sequence of instructions for deriving certain output data from certain input data, where the following conditions must be satisfied:

- Finiteness. The sequence of instructions is completely described in a finite text.
- Effectiveness. Each instruction must actually be executable.
- Termination. The algorithm ends after finitely many instructions.
- Determinacy. The course of the algorithm is prescribed at every point in time.

If E denotes the set of admissible input data and A denotes the set of admissible output data, then an algorithm is therefore a function:

$$f : E \to A, \quad e \mapsto f(e). \tag{11.1}$$

Conversely, functions that can be described by an algorithm are called *computable functions*.

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11.3 Programming languages

Programming languages specify the rules that a program must obey. They define a *syntax*, that is, the vocabulary and structure of a program, as well as the *semantics*, that is, the meaning of the permitted instructions. In the following, we give some terminological clarifications for categorizing different forms of programming languages. Here one distinguishes *machine language* and so-called *higher-level programming languages*, different *generations of programming languages*, the *imperative programming* that is the focus here from *declarative programming*, and *compiled* from *interpreted* programming languages.

Machine language and higher-level programming languages

Machine language consists of elementary operation instructions such as storing, comparing, or adding, which are encoded as binary numbers. Table 11.1 shows some examples of such instructions:

Table 11.1 Examples of machine language

Instruction	Binary code
Add contents of R1 to contents of R2	1001 0010
Increase contents of R by 1	1001 0110
Transfer contents of R1 to R3	0010 0011

Programs written in machine language are called machine programs. De facto, a computer executes only machine programs, because these are understood directly by the hardware. For humans, however, programming in machine language is very laborious, which is why higher-level programming languages are predominantly used in practice.

Higher-level programming languages use words and sentences modeled on human language to make programming more understandable and efficient. Programs in higher-level programming languages are then translated into machine language by a *compiler* or an *interpreter* so that they can be executed by the computer. Well-known examples of higher-level programming languages are FORTRAN, COBOL, C++, Python, and **R**.

Generations of programming languages

The *first-generation programming languages (1GL)* are the machine languages. Programs are written directly in the form of binary numbers, for example `10110000 01100001`, which corresponds to `B0 61` in hexadecimal notation. Such machine languages were used mainly on the first vacuum-tube and relay computers such as ENIAC or early IBM mainframes of the 1940s and early 1950s.

With the *second generation (2GL)*, assembly languages emerged from the 1950s on as the first form of symbolic programming. Here, human-readable abbreviations are used, although they are still processor-specific. An example of such a processor-specific Intel instruction is `MOV AL, 61H`. This generation of programming languages was used mainly on mainframe computers such as the IBM 704 or early mainframes of the 1950s and 1960s, which already used transistors instead of vacuum tubes.

From the 1970s onward, the *third-generation programming languages (3GL)* became established. These include higher-level programming languages such as FORTRAN, C, C++, and Java. They are much more programming-friendly and, above all, processor-independent, which considerably improves the portability of programs. Such languages were first used widely on minicomputers such as the DEC PDP-11 and later on workstations, PCs, and servers.

Finally, from the 1980s onward, the *fourth-generation programming languages (4GL)* developed; they include languages such as Python, MATLAB, and **R**. These are characterized by minimizing code overhead, high flexibility, support for multiparadigmatic approaches, and a high degree of automation. They are used mainly on modern PCs, high-performance computing clusters (HPC clusters), and cloud-based systems, which became increasingly available from the 1990s onward.

Imperative and declarative programming

In *imperative programming* (from *imperare*, Latin for "to command"), the path to a solution is specified as a sequence of instructions or commands. These commands process data that are addressed via variables. Within imperative programming, two different approaches are distinguished: *procedural* imperative programming and *object-oriented* imperative programming.

In *procedural imperative programming*, data and the commands that manipulate them are treated separately. The central structural concept here is procedures or functions, which encapsulate reusable units of instructions. By contrast, in *object-oriented imperative programming*, data and the commands that act on them are combined as objects. These objects are at the same time data carriers and providers of functions and form the central structural concept of this approach.

In practice, mixed forms of the two paradigms are often used, because the approaches can be combined well depending on the problem. An example of such a mixed form is the newer programming language Julia, which does not offer a classical object-oriented approach but instead a *multiple-dispatch* approach: functions can have different implementations depending on the types of their arguments and can thus combine aspects of object-oriented and procedural programming.

In *declarative programming*, the focus is not on the path to the solution but on the description of the problem itself. The programmer formulates what is to be achieved, and not in detail how the computer is to reach this goal step by step. The path to the solution is determined automatically by the system on the basis of given rules and mechanisms.

A typical feature of declarative approaches is that the focus is on defining relations, conditions, or goals rather than on an explicit sequence of commands. The programmer thus describes the desired result in a formal language, and the computer independently searches for a solution that satisfies these conditions.

Well-known approaches to declarative programming are, for example, logic programming, as implemented in the language Prolog, or functional programming, as in Haskell. In logic programming, facts and rules are formulated from which the system draws conclusions by inference. In functional programming, computations are described by the composition of functions, while the state of the data is not changed explicitly. Declarative programming is particularly suitable for problems in which the path to the solution can be complex and varied, but the desired properties of the result can be clearly specified. Typical application areas include database queries (for example SQL), knowledge bases, constraint solving, and formal proofs.

A modern example of the use of declarative programming is the system Lean, a so-called *theorem prover*. Lean is used to formally prove mathematical theorems. Here the user formulates the axioms, definitions, and statements to be proved in a mixture of declarative and interactive commands. The system automatically checks whether the specified steps are valid and completes the missing parts of the proof according to fixed rules. In this way, Lean combines declarative programming with automated proof and makes it possible to develop large mathematical theories in a formally correct way. For an introduction to Lean in the context of mathematical proof, see, for example, The Mechanics of Proof.

Compiled and interpreted programming languages

Compiled programming languages are characterized by the fact that the entire source code is translated into machine language before execution. A special program, the so-called *compiler*, is used for this translation. The machine code generated by the compiler is executed directly by the processor, which enables the program to run very quickly. Since the executable program is already available in machine language, it no longer has to be translated at runtime. However, whenever the source code is changed, it is necessary to compile the program again. Typical examples of compiled programming languages are C, C++, and Rust.

By contrast, in *interpreted programming languages*, the source code is translated into a machine-like language during execution. This is done by a special program, the so-called *interpreter*. Since translation and execution take place at the same time, execution is usually slower than in compiled programs. One advantage, however, is that after changes no separate compilation step is required; the program can be executed immediately. Typical examples of interpreted programming languages are Python and **R**.

In the sense of the concepts introduced here, **R** is an interpreted multiparadigmatic fourth-generation programming language that supports object-oriented concepts but is often used procedurally and functionally and is tailored to statistical data analysis.

11.4 R

R is a programming language and a software package originally developed by Ross Ihaka (Ihaka & Gentleman (1996)). It is a free dialect of the proprietary software S, introduced by Becker et al. (Becker et al. (1988)). Today, **R** is developed and maintained by the R Core Team and the R Foundation. **R** has a large and active community that has so far contributed more than 20,000 **R** packages, which extend the system with numerous additions. The core of **R** deliberately develops in an evolutionary and conservative way, while the large number of R packages enables consistent and progressive further development. **R** can be easily downloaded and installed from CRAN.

With **R**, one can load, manipulate, and save data sets in different formats. The language makes it possible to carry out a wide variety of computations on different data structures. In addition, **R** offers a broad spectrum of statistical analysis methods that can be applied directly to the data. It is also typical to write reproducible data analysis scripts with which analyses can be automated and results can be documented transparently. Finally, **R** is very well suited for creating figures and visualizations in order to present data clearly.

However, some things that are important in psychological data science can be implemented only to a limited extent with **R**. For example, **R** per se lacks a modern and appealing development environment of its own; embedding **R** in VS Code remedies this. In the area of scientific computing, **R** reaches its limits. Here, languages such as Python, MATLAB, or Julia often offer more powerful and flexible tools. Finally, **R** is hardly suitable for programming psychological experiments, for which Python or MATLAB are also more commonly used.

There are many ways to get help with **R**. Quite classically, a quick search on Google, a request to ChatGPT, or an interaction with Copilot in VS Code often already helps. During programming, and when one already knows the name of a function, one can call help directly in **R**:

```
?mean
help(mean)
```

For more comprehensive, longer tutorials and background information, it is worth taking a look at the so-called *vignettes* of installed packages:

```
browseVignettes()
```

In addition, there are numerous helpful online resources, for example the specialized search engine rseek or the many articles and guides on R-Bloggers, although most information on these platforms is certainly also part of the databases accessible through chatbots such as ChatGPT, Gemini, Claude, or DeepSeek.

11.5 Visual studio code

Visual Studio Code (VS Code) is a free source code editor from Microsoft that has been available for Windows, macOS, and Linux since 2015. Roughly speaking, VS Code is something like the perhaps better-known RStudio environment, except that it is for all programming languages. VS Code can be flexibly extended through extensions and used as an integrated development environment (IDE) for almost any language, for example **R**, Python, Julia, Shell, Quarto, and many more. Since 2018, VS Code has been regarded as the most popular editor overall, as annual surveys show. Remarkably, this means that a Microsoft product in particular is also the most-used editor in the Linux world. VS Code is highly configurable, strongly shaped by the community, and enables the synchronization of settings and extensions across multiple devices via Microsoft's GitHub. Detailed documentation with instructions for use can be found at www.code.visualstudio.com/docs. The corresponding documentation R in Visual Studio Code provides an introduction to working with **R** in VS Code.

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